KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

COLLEGE OF ENGINEERING

DEPARTMENT OF AGRICULTURAL ENGINEERING

(BSC AGRICULTURAL ENGINEERING, THIRD YEAR)

AE 358: RENEWABLE ENERGY SYSTEMS AND MANAGEMENT

[Credit: 3]

F. Kemausuor

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Course Introduction

COURSE OVERVIEW

AE 358: RENEWABLE ENERGY SYSTEMS & MANAGEMENT introduces students to the potentials, fundamental theories, conversion, utilisation and basic economics of renewable energy technologies. It also introduces briefly the application of renewable energy for agricultural activities.

COURSE OBJECTIVES

After going through this course, you should be to:

1. Define and explain the various renewable energy technologies and their potentials
2. Discuss the production processes/conversion of renewable energy technologies
3. Outline the need for and general uses of the various forms of renewable energy
4. Explain how renewable energy can be applied in agriculture

COURSE OUTLINE

The course material has been structures into six (6) units.

- Unit 1: Renewable Energy Systems and their Potential
- Unit 2: Biofuels – Fundamental Theories and Conversion
- Unit 3: Biogas – Fundamental Theories and Conversion
- Unit 4: Solar Energy - Fundamental Theories and Conversion
- Unit 5: Wind Energy - Fundamental Theories and Conversion
- Unit 6: Applications of Renewable Energy to Agriculture in Ghana

GRADING

Continuous assessment: 30%

End of semester examination: 70%

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READING LIST / RECOMMENDED TEXTBOOKS / WEBSITES / CD ROM

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Table Of Contents

Course Writer iii
Course Introduction iv
List Of Tables ix
List Of Figures x
List Of Appendices xii
List Of Abbreviations and Notations xiii
Unit 16
renewable energy systems and their potential 16
SESSION 1-1: INTRODUCTION TO ENERGY SYSTEMS 17
1-1.1 Origin of fossil fuels 17
1-1.1.1 Biogenic theory 18
1-1.2 Advantages and disadvantages of fossil fuel 20
SESSION 2-1: Renewable Energy Sources 21
2-1.1 Hydropower 24
2-1.2 Solar power 24
2-1.3 Wind power 25
2-1.4 Geothermal power 26
2-1.5 Tidal power 26
2-1.6 Biomass and biofuels 27
Unit 31
BIOFUELS – FUNDAMENTAL THEORIES AND CONVERSION 31
SESSION 1-2: BIODIESEL 32
1-2.1 Feedstocks for biodiesel production 33
1-2.2 Technical aspects of biodiesel production 35
1-2.3 Uses of biodiesel 40
1-2.4 Economics of biodiesel 40
SESSION 2-2: BIOETHANOL 42
2-2.1 Bioethanol production 43
2-2.1.1 Ethanol from sugar feedstock 43
2-2.1.2 Ethanol from starch feedstock 45
2-2.2 Step by step ethanol production process 46
2-2.3 Uses of Bioethanol 48
2-2.4 Economics of bioethanol 50
Unit 54
BIOgas – FUNDAMENTAL THEORIES AND CONVERSION 54
SESSION 1-3: BIOGAS PRODUCTION 55

1-3.1 Introduction 55
1-3.2 Stages in biogas production 56
1-3.3 Types of biogas plants 59
1-3.4 Construction of biogas plants 61
SESSION 2-3: APPLICATIONS AND ECONOMICS OF BIOGAS 69
2-3.1 Biogas application technologies 69
2-3.2 Biogas production costs 70
Unit 74
SOLAR ENERGY - FUNDAMENTAL THEORIES AND CONVERSION 74
SESSION 1-4: OVERVIEW OF SOLAR PHOTOVOLTAICS 75
1-4.1 Introduction to solar energy 75
1-4.2 Photovoltaic (PV) systems 75
SESSION 2-4: SOLAR THERMAL APPLICATIONS 84
Unit 89
Wind Energy - Fundamental theories and conversion 89
SESSION 1-5: OVERVIEW OF WIND ENERGY 90
1-5.1 Introduction 90
1-5.2 Power in the wind 92
SESSION 2-5: WIND TURBINES AND WIND RESOURCE ASSESSMENT 94
2-5.1 Description of wind turbines 94
2-5.2 Wind resource assessment 100
2-5.3 Potential environmental and social impacts of wind power development 102
Unit 106
applications of renewable energy to agriculture in ghana 106
SESSION 1-6: overview of energy use in agriculture 107
1-6.1 Introduction 107
1-6.2 Agriculture in Ghana 107
1-6.3 Using energy to modernise agriculture in Ghana 108
SESSION 2-6: renewable energyoptions for agriculture 111
2-6.1 Introduction 111
2-6.2 Solar dryers and pumps 112
2-6.3 Windmills 114
2-6.4 Biofuels and biogas 114

List Of Tables

Table 1: Status of renewable energy technologies – characteristics and costs 23

Table 2: Vegetable oil yields for some crops..... 34

Table 3: Ethanol production step by feedstock and conversion techniques 44

Table 4: Principal differences between wet and dry mill ethanol plants 45

Table 5: Properties of fuel ethanol..... 49

Table 6: Parameters for bioethanol in comparison with gasoline..... 50

Table 7: Potentials of Biogas Production from some types of Dung..... 57

Table 8: Data for determining biogas plant capacity 62

Table 9: Biogas production temperatures 63

Table 10: Parameters for sizing a biogas plant 64

Table 11: Fixed Capital Investment Cost for Biogas Installations in Some African Countries ... 71

Table 12: Conventional and renewable energy options for modernizing agriculture..... 109

Table 13: Renewable Energy Options for Agriculture 111

List Of Figures

Figure 1: Crude oil prices (1998 - 2009) 17

Figure 2: Formation of oil and gas..... 18

Figure 3: Biogenic origin of oil and gas 19

Figure 4: Scenario of World oil production and “peak oil”..... 20

Figure 5: Renewable energy Share of global final energy consumption, 2009..... 23

Figure 6: Solar PV, Existing world capacity, 1995-2010..... 25

Figure 7: Wind turbine..... 26

Figure 8: Tidal Energy Generation 27

Figure 9: Examples of sources of feedstocks for biodiesel production 35

Figure 10: Process Flow Scheme for Biodiesel Production..... 39

Figure 11: Schematic outline of the biomass-to-ethanol process 47

Figure 12: Benefits, applications and uses of biogas..... 56

Figure 13: Schematic diagram of feedstock and unit operations in a typical biogas plant..... 57

Figure 14: Fixed dome digester 60

Figure 15: Different models of Fixed dome plants..... 60

Figure 16: Floating drum digester..... 61

Figure 17: Application Areas of Biogas 69

Figure 18: Cell, Module and Array..... 76

Figure 19: Solar panel..... 76

Figure 20: Application of Solar Panels..... 77

Figure 21: Block diagram of a PV system..... 77

Figure 22: Solar water heating (Passive system) 85

Figure 23: solar cooking 86

Figure 24: Global wind..... 90

Figure 25: Sea breeze 91
Figure 26: Land breeze 92
Figure 27:Vertical axis wind turbines 95
Figure 28: Schematic of a Horizontal Axis Wind Turbine 97
Figure 29: Components of a Horizontal Axis Wind Turbine 98
Figure 30: Vestas V47 – 650 kW Turbine Power Curve 99
Figure 31: Cup Anemometer 101
Figure 32: Features of a basic solar dryer 112
Figure 33: Typical arrangement of the components of a solar-powered pumping system 113
Figure 34: Schematic and photo of a windmill application for water pumping 114

List Of Appendices

Appendix 1: Biodiesel feedstock classification 118

Appendix 2: Global Solar Radiation Map of Ghana 121

Appendix 3: Monthly Averages of Solar Irradiation (kWh/m2-day) at 9 Synoptic Stations 121

Appendix 4: Wind Resource Map for Ghana - 50 m Wind Power 122

List Of Abbreviations and Notations

Abbreviation	Definition
PV	Photovoltaic
CSP	Concentrating Solar Power
US	United States
MW	megawatt
kWh	kilowatt-hour
kW	kilowatt
m	metres
n/a	Not applicable
m^3	Cubic metres
TWh	Terawatt-hour
GW	Gigawatt
EU	European Union
i.e.	that is
SVOs	Straight Vegetable Oils
PPOs	Pure Plant Oils
e.g.	Example
FFA	Free Fatty Acids
ha	Hectare
ºC	Degree Celcius
%	Percent
~	Approximately
>	Greater than
L/yr	Litres per year
CSTR	Continuous Stirred-Tank Reactors
COD	Chemical Oxygen Demand
BOD	Biochemical Oxygen Demand
HFCS	High-Fructose Corn Syrup
DDGS	Dried Distillers Grains with Soluble
CGF	corn gluten feed
CGM	corn gluten meal
FFVs	Flexible fuel vehicles
kg	Kilogramme
l	Litres
mm	millimetres
s	second
MJ	Megajoule
<	Less than
RON	Research Octane Number
AD	Anaerobic digestion
ppm	parts per million
HRT	Hydraulic retention time
V_d	digester volume

S_d	Volume of substrate fed per unit time
d	Day
B	Biomass
W	Water
G	Amount of biogas generated
VS	Volatile solids content
Gy	specific gas yield
Gp	Specific gas production
TS/d	daily total solids input
L_d	Digester loading
LSU	livestock unit
Vg	gasholder volume
V	Volt
Wp	Peak watt
W	Watt
DoD	Depth of Discharge
Ah	ampere hours
Aut	Days of autonomy
BCU	Battery Control Unit
DC	Direct current
AC	Alternating current
E_T	Energy of battery
E_D	Energy removed during discharging
E_R	Remaining Battery Energy
$C_{BAT,R}$	Remaining Battery Capacity
A	Cross Sectional Area
I	Current
L	Length
$\triangle P$	Power loss
A	Ampere
TV	Television
E	Energy requirement
Q	Daily water demand
h	Water level
η	Efficiency
G_{DESIGN}	Design insolation
E_{LOAD}	energy consumption
η_{BAT}	Battery Efficiency
η_{INV}	Mean Operating Efficiency of the Inverter
E_{GEN}	Energy supplied by generator
P_{GEN}	Generator power
C_{BAT}	Battery capacity
U_{BAT}	EMF of battery
P	Density
V	Volume

VAWT	Vertical Axis Wind Turbine
HAWT	Horizontal Axis Wind Turbine
C_p	Capacity factor
A	friction coefficient
DEC	Dedicated Energy Crop
SFA	Saturated Fatty Acids
WFOs	Waste Frying Oils

Unit 1

RENEWABLE ENERGY SYSTEMS AND THEIR POTENTIAL

Introduction

Unit 1 presents an overview of energy systems in general and the potentials for renewable energy across the globe but also narrowing it down to Africa. A generation introduction to fossil fuels, including the challenges associated with it is first presented. The various renewable energy technology options are then briefly introduced before an in-depth discussion is presented in later Units.



Learning Objectives

After reading this unit you should be able to:

1. Explain why renewable energy is the energy of the future
2. Define the various sources of renewable energy and their potentials

Unit content

Session 1-1: Introduction to Energy Systems

- 1-1.1 Origin of fossil fuels
- 1-1.2 Advantages and disadvantages of fossil fuel

Session 2-1: Renewable Energy Sources

- 2-1.1 Hydropower
- 2-1.2 Solar power
- 2-1.3 Wind power
- 2-1.4 Geothermal power
- 2-1.5 Tidal power
- 2-1.6 Biomass and biofuels

SESSION 1-1: INTRODUCTION TO ENERGY SYSTEMS

Energy is defined as the ability to perform work. Energy is used in our daily activities. The standard of living of the people of any country is considered to be proportional to the energy consumption by the people of that country. The use of energy worldwide can be grouped into four main sectors. These include industry, transportation, residential/commercial and agriculture. Generally, the percentage utilization of energy by the agriculture sector is small (2 to 5% of total energy use). The world reserves of primary energy and raw materials are, obviously, limited. According to an estimate, the reserves will last another 218 years for coal, 41 years for oil, and 63 years for natural gas, under a business- as-usual scenario. Oil has no equal as an energy source for its intrinsic qualities of extractability, transportability, versatility, and cost. Concerns have arisen in recent years about the relationship between the growing consumption of oil and the availability of oil reserves, as well as the impact of the potentially dwindling supplies and rising prices on the world's economy and social welfare. Oil accounts for approximately one-third of all the energy used in the world. Future world demand for oil is uncertain because it depends on economic growth and government policies throughout the world. Rapid economic growth in China and India could significantly increase world demand for oil, while environmental concerns, including oil's contribution to global warming, may spur conservation or the adoption of alternative fuels that would reduce future demand for oil. It is also worth noting that crude oil have risen significantly over the last two decades or so (Figure 1).

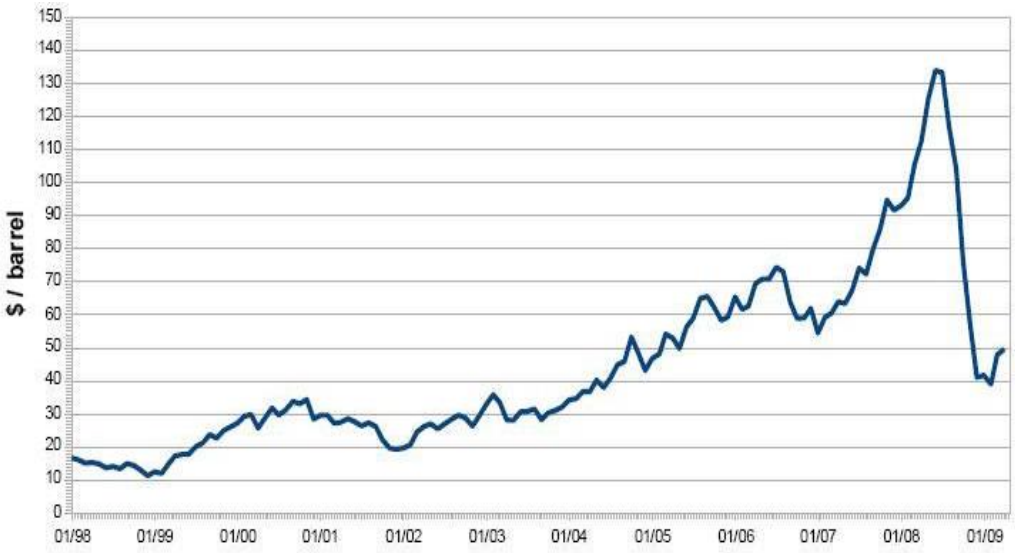


Figure 1: Crude oil prices (1998 - 2009)

1-1.1 Origin of fossil fuels

Fossil fuels are sources of energy that were formed by the death, decay, and transformation, or diagenesis, of life (Figure 2). The term diagenesis encompasses physical and chemical

changes that are associated with lithification and compaction. Sediment can be lithified, or made rocklike, by the movement of minerals into sedimentary pore spaces. The minerals can form cement that binds grains of sediments together into a rocklike structure that has less porosity than the original sediment in a process called cementation. Compaction is the process of applying pressure to compress the rocklike structure.

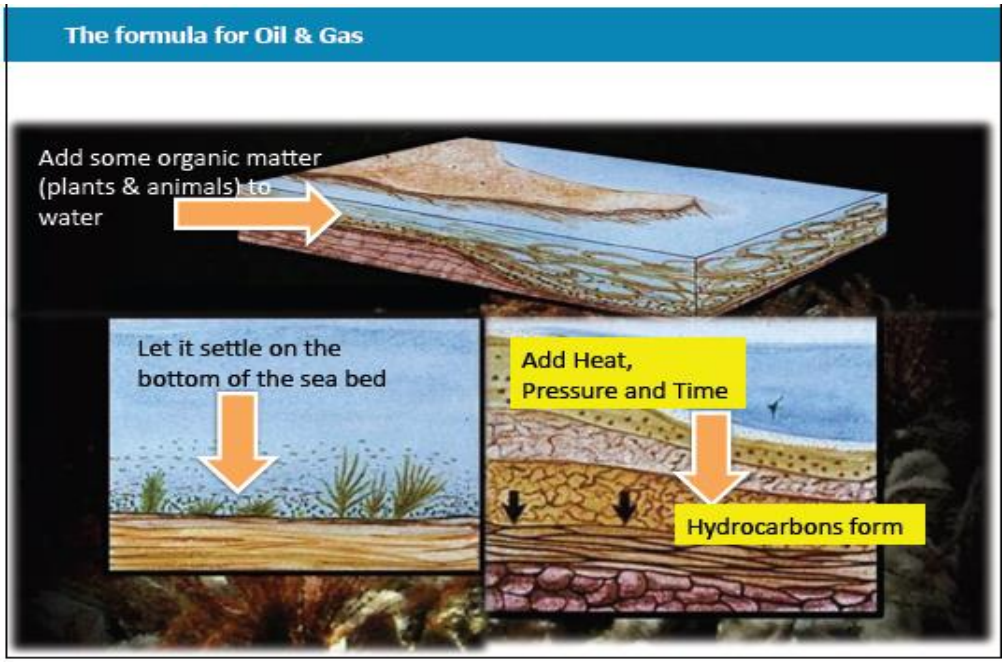


Figure 2: Formation of oil and gas

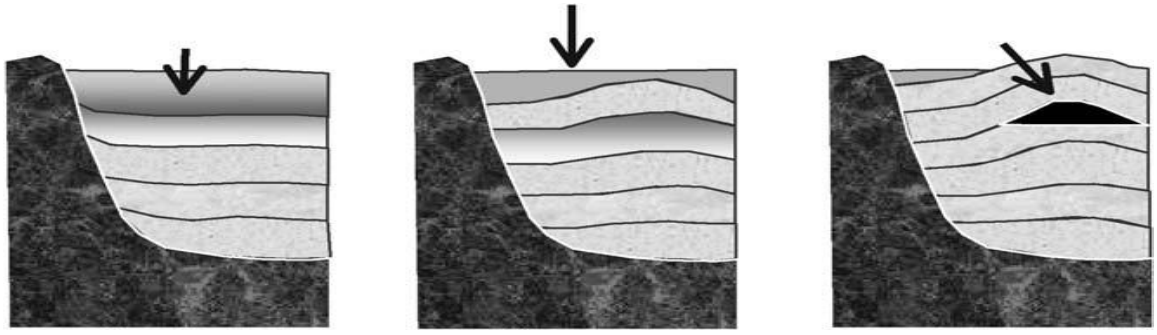
1-1.1.1 Biogenic theory

The biogenic theory is the mainstream scientific view of the origin of fossil fuels. In the biogenic theory, a type of biochemical precipitation called organic sedimentation forms coal, oil, and gas. When vegetation dies and decays in aqueous environments such as swamps, it can form a carbon-rich organic material called peat. If peat is buried by subsequent geological activity, the buried peat is subjected to increasing temperature and pressure. Peat can eventually be transformed into coal by the process of diagenesis. A similar diagenetic process is thought to be the origin of oil and gas.

Oil and gas are petroleum fluids. A petroleum fluid is a mixture of hydrocarbon molecules and inorganic impurities, such as nitrogen, carbon dioxide, and hydrogen sulfide. Petroleum can exist in solid, liquid, or gas form, depending on its composition and the temperature and pressure of its surroundings. Natural gas is typically methane with lesser amounts of heavier hydrocarbon molecules such as ethane and propane. The elemental mass content of petroleum fluids ranges from approximately 84% to 87% carbon and 11% to 14% hydrogen, which is comparable to the carbon and hydrogen content of life. This is one piece of evidence for the origin of petroleum from biological sources.

The biochemical process for the formation of petroleum is illustrated in Figure 3. It begins with the death of microscopic organisms such as algae and bacteria. The remains of the

organisms settle into the sediments at the base of an aqueous environment as organic debris. Lakebeds and seabeds are examples of favourable sedimentary environments. Subsequent sedimentation buries the organic debris. As burial continues, the organic material is subjected to increasing temperature and pressure, and is transformed by bacterial action into oil and gas. Petroleum fluids are usually less dense than water and will migrate upwards until they encounter impermeable barriers and are collected in traps. The accumulation of hydrocarbon in a geologic trap becomes a petroleum reservoir.



A. Death and Burial B. Bacterial and Geologic Action C.Migration and Trapping

Figure 3: Biogenic origin of oil and gas

1-1.1.2 The Concept of Peak Oil

The current global energy supply is highly dependent on fossil sources (crude oil, lignite, hard coal, natural gas). These are fossilised remains of dead plants and animals, which have been exposed to heat and pressure in the Earth's crust over hundreds of millions of years. For this reason, fossil fuels are non-renewable resources which reserves are being depleted much faster than new ones are being formed The World’s economies are dependent today of crude oil. There is some disagreement among scientists on how long this fossil resource will last but according to researchers, the “peak oil production” has already occurred or it is expected to occur within the next period of time as illustrated in Figure 4.

The peak oil production is defined as “the point in time at which the maximum rate of global production of crude oil is reached, after which the rate of production enters its terminal decline. In view of the fact that fossil oil is peaking, renewable energy sources like wind, solar, geothermal, hydrogen, and biomass will play an important role in the future of our energy demand.

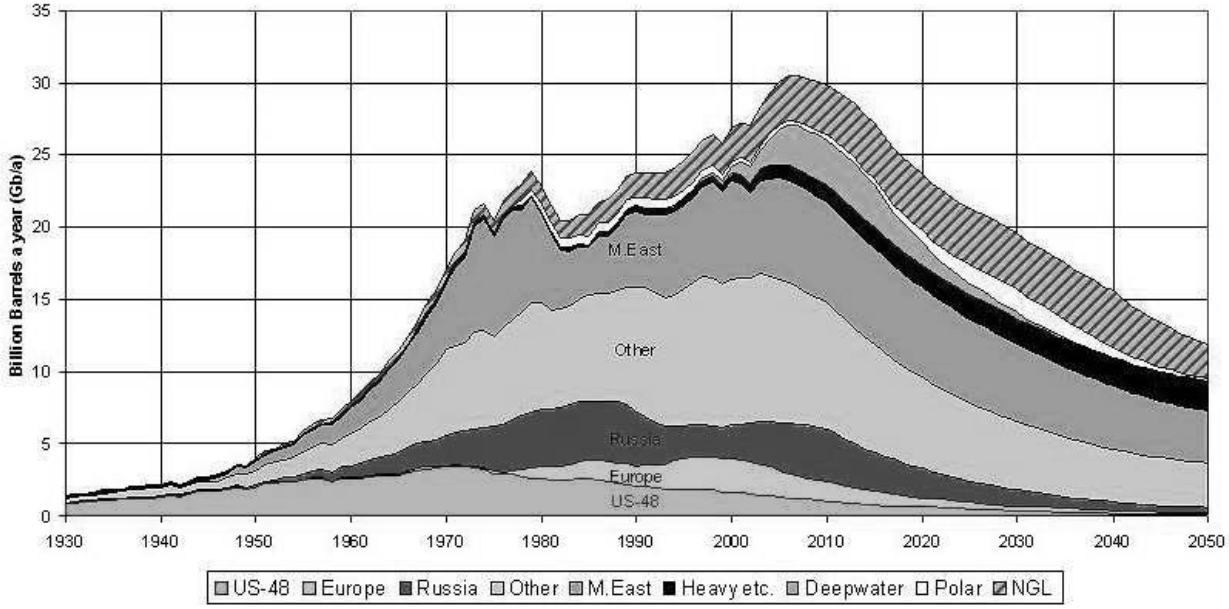


Figure 4: Scenario of World oil production and "peak oil"

1-1.2 Advantages and disadvantages of fossil fuel

There are both advantages and disadvantages to the use of fossil fuels. A major advantage of fossil fuels is their capacity to generate huge amounts of electricity in just a single location. Coal is in abundant supply and when used in power plants, they are very cost effective. Transporting oil and gas to the power stations can be made through the use of pipes making it an easy task. Power stations that make use of fossil fuel can be constructed in almost any location. This is possible as long as large quantities of fuel can be easily brought to the power plants.

The disadvantages of fossil fuels include the following.

- Fossil fuels are a non-renewable energy source that takes millions of years to form and therefore once existing and any new reserves are depleted there is no way to obtain more. Locating additional oil reserves is an extremely costly ongoing process with no guarantee of success.
- While oil is relatively easy to transport, one of the more common transportation methods is shipping oil from port to port by 'supertanker' – massive purpose-built tanker ships capable of transporting enormous quantities of oil by sea. While spills are rare they are by no means unheard of and the resulting inevitably catastrophic effect on marine life, birds and coastlines is evident for many years. The 2010 oil spill in the Gulf of Mexico have had devastating effects on marine life in the Gulf.
- Pollution is a major disadvantage of fossil fuels. This is because they give off carbon dioxide when burned thereby causing a greenhouse effect. This is also the main contributory factor to the global warming experienced by the earth today.

- Oil contains sulphur which when burnt forms sulphur dioxide and sulphur trioxide – these compounds combine with atmospheric moisture to form sulphuric acid, leading to 'acid rain'. This can lead to destruction of forests and the progressive erosion of rock and masonry structures, both natural and man-made. The use of low-sulphur fuels can help to reduce the impact of this.
- Many of the countries where oil can be found are politically unstable – around 70% of the world's oil reserves are believed to be located in the Middle East. Developed nations with an oil-dependency have a vested interest in such countries and may seek to influence the politics of such countries to their own advantage.
- Environmentally, the mining of coal results in the destruction of wide areas of land. Mining this fossil fuel is also difficult and may endanger the lives of miners. Coal mining is considered one of the most dangerous jobs in the world.



Self Assessment 1-1

1. How is oil and gas formed in nature?

2. What do you understand by the term 'peak oil'?

3. What in your opinion is accounting for the increasing cost of crude oil?

4. What are some of the advantages and disadvantages of crude oil?



Answer tips

1. Read session 1-1.1
2. Read session 1-1.1.2
3. Read introduction to session 1-1
4. Read session 1-1.2

SESSION 2-1: RENEWABLE ENERGY SOURCES

Renewable energy is energy which comes from natural resources such as sunlight, biomass, wind, rain, tides, and geothermal heat, which are renewable (naturally replenished). The fact that renewable energy is derived from resources that are regenerative means that, for all practical purposes, they cannot be depleted. For this reason, renewable energy sources are

fundamentally different from fossil fuels, and do not produce as many greenhouse gases and other pollutants as fossil fuel combustion. With the depletion of oil resources as well as the negative environmental impact associated with the use of fossil fuels, there is a renewed interest in renewable energy sources. The drivers for development and use of renewable energy include:

1. Diminishing reserves of readily recoverable oil
2. Concern over global climate change
3. Increasing fuel prices
4. The desire for energy independence and security.

According to the 2011 report of the global status of renewable energy, renewable energy supplied an estimated 16% of global final energy consumption in the year 2009 – counting traditional biomass, hydropower, wind, solar, geothermal, modern biomass, and biofuels (See Figure 5). Traditional biomass, which is used primarily for cooking and heating in rural areas of developing countries, accounted for approximately 10% of the total renewable energy share. Hydropower represented 3.4% and is growing modestly but from a large base. All other renewables accounted for approximately 2.8% in 2009 and are growing very rapidly in many developed countries as well as in some developing countries. Renewable energy replaces fossil and nuclear fuels in four distinct markets: power generation, heating and cooling, transport fuels, and rural/off-grid energy services.

During the period from the end of 2005 through 2010, total global capacity of many renewable energy technologies – including solar photovoltaics (PV), wind, concentrating solar power (CSP), solar water heating systems, and biofuels – grew at average rates ranging from around 15% to nearly 50% annually. Solar PV increased the fastest of all renewables technologies during this period, followed by biodiesel and wind. Hydropower, biomass power and heat, and geothermal heat and power are growing at more ordinary rates of 3–9% per year, making them more comparable with global growth rates for fossil fuels (1–4%, although higher in some developing countries). In several countries, however, the growth in these renewable technologies far exceeds the global average. (See Table 1 for a summary of the main renewable energy technologies and their characteristics and costs.)

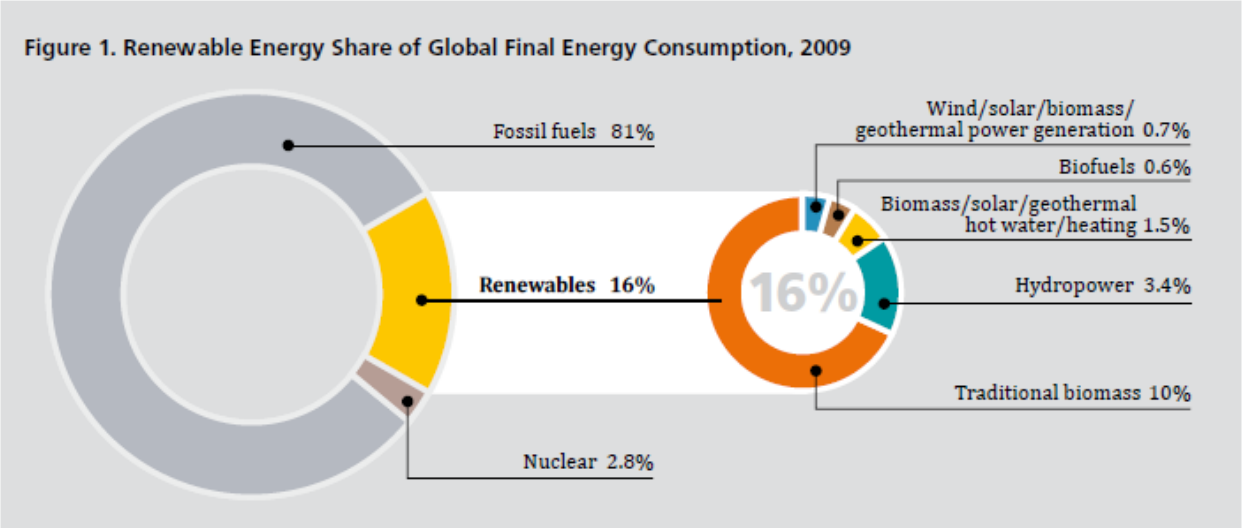


Figure 5: Renewable energy Share of global final energy consumption, 2009

Table 1: Status of renewable energy technologies – characteristics and costs

Technology	Typical characteristics		Typical energy costs (US cents/kWh)
Large hydro	Plant size:	10 – 18,000MW	3-5
Small hydro	Plant size:	1 – 10MW	5-12
Biomass power	Plant size:	1 – 20 MW	5-12
Geothermal power	Plant size:	1 – 100MW	5-12
Rooftop solar PV	Peak capacity:	2 – 5 kW_{peak}	17-34
Utility-scale solar PV	Peak capacity:	200kW – 100MW	15-30
On-shore wind	Turbine size	1.5-3.5MW; rotor diameter 60-100m	5-9
Off-shore wind	Turbine size	1.5-5MW; rotor diameter 70-125m	10-20
Ethanol	Feedstocks	Sugarcane, sugar beets, corn, cassava, sorghum, wheat	30-50 cents/litre (sugar) 60-80 cents/litre (corn) (gasoline equivalent)
Biodiesel	Feedstocks:	Soy, rapeseed, mustard seed, palm, jatropha, waste vegetable oils and animal fats	40-80 cents/litre (corn) (diesel equivalent)
Biogas digester	Digester size:	6-8 m^3	n/a

2-1.1 Hydropower

Hydropower is power that is derived from the force or energy of moving water, which may be harnessed for useful purposes. Hydroelectricity is the term referring to electricity generated by hydropower; the production of electrical power through the use of the gravitational force of falling or flowing water. It is the world's leading form of renewable energy, assuming traditional biomass was not counted. The long-term technical potential is believed to be nine to twelve times the current hydropower production, but environmental concerns increasingly block new dam construction. There are also growing interests in mini-hydro projects, which avoid many of the problems of the larger dams. The top countries for hydro capacity are China, Brazil, the United States, Canada, and Russia, which account for 52% of total installed capacity. It is estimated that Africa has 1,750 TWh potential of hydropower but only 5% of the continent's potential of hydropower has been exploited.

2-1.2 Solar power

Solar power is the technology of obtaining usable energy from sunlight. It has been used in many traditional technologies for centuries, and has come into widespread use where other sources of power are absent, such as in remote locations and in space. Commercial solar cells can presently convert about 20% of the energy of incident sunlight to electrical energy. Solar energy is currently used in a number of applications, such as for the generation of heat for boiling water, heating rooms, cooking, electricity generation in photovoltaic cells and heat engines, and for desalination of seawater.

Solar photovoltaic (PV) capacity was added in more than 100 countries during 2010, ensuring that PV remained the world's fastest growing power-generation technology. An estimated 17 GW of Photovoltaic capacity was added worldwide in 2010 (compared with just under 7.3 GW in 2009), bringing the global total to about 40 GW – more than seven times the capacity in operation five years (See Figure 6.) The EU dominated the global PV market, accounting for 80% of the world total with about 13.2 GW newly installed – enough to meet the electricity consumption of some 10 million European households. Africa receives abundant solar radiation throughout the year but this resource has not been exploited for electricity generation to the extent at which it should.

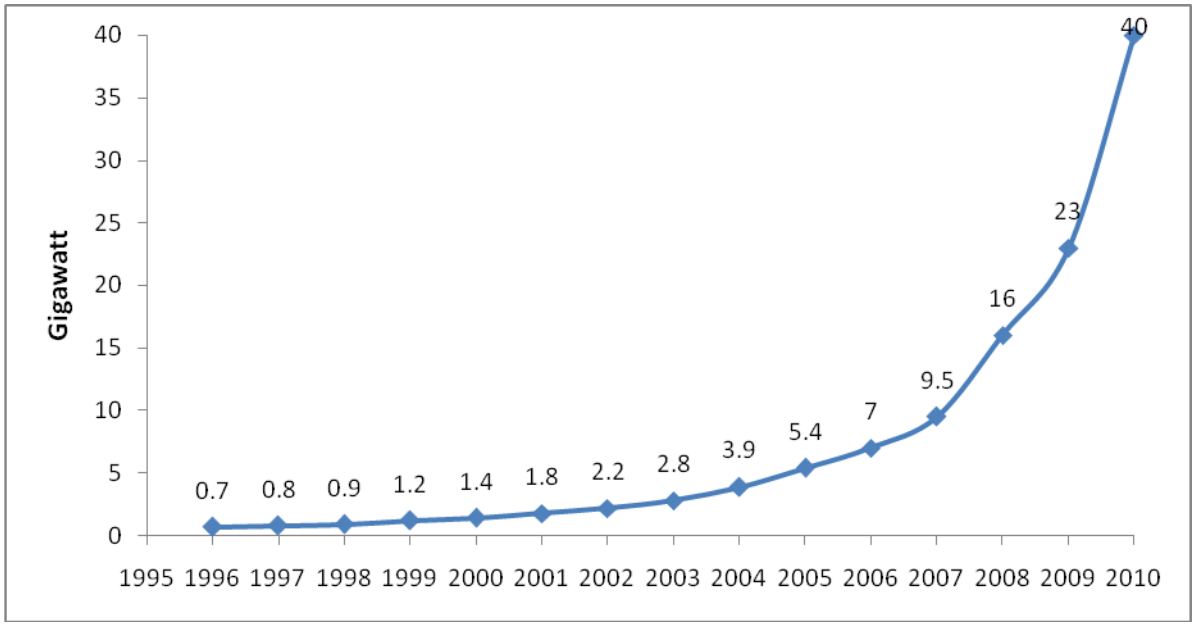


Figure 6: Solar PV, Existing world capacity, 1995-2010

2-1.3 Wind power

Wind power is the conversion of wind energy into a useful form of energy, such as using wind turbines (Figure 7) to make electricity, wind mills for mechanical power, wind pumps for pumping water or drainage, or sails to propel ships Large-scale wind farms are connected to the electric power transmission network; smaller facilities are used to provide electricity to isolated locations. It is one of the most cost-competitive renewable energy sources today. Its long-term technical potential is believed to be five times the current global energy consumption, or 40 times current electricity demand. New wind power capacity added during 2010 reached 39 GW, more than any other renewable technology and over three times the 11.5 GW of wind added worldwide just five years earlier. This growth was driven primarily by China, which accounted for 50% of global capacity additions in 2010, up from 4.4% in 2005. China added 18.9 GW of new wind capacity, a 37% increase over the 2009 market, bringing the country into the global lead with a total of 44.7 GW

The wind energy is ample, renewable, widely distributed, clean, and reduces toxic atmospheric and greenhouse gas emissions if used to replace fossil-fuel-derived electricity. The intermittency of wind seldom creates problems when using wind power at low to moderate penetration levels.

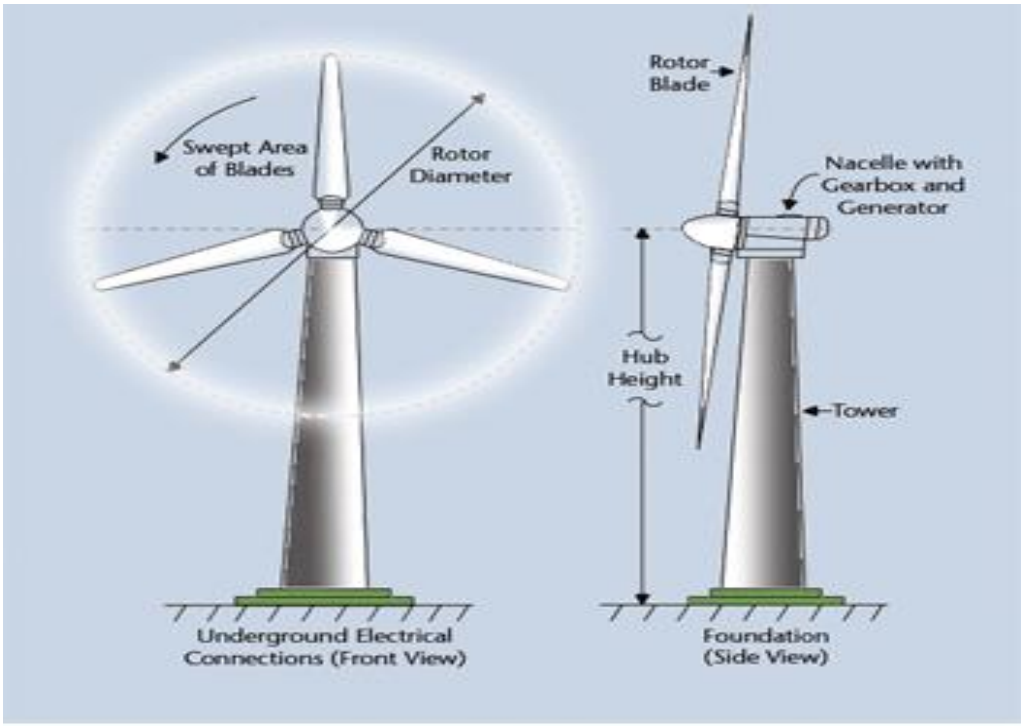


Figure 7: Wind turbine

2-1.4 Geothermal power

Geothermal power is the use of geothermal heat to generate electricity. Geothermal energy is thermal energy generated and stored in the Earth. Thermal energy is energy that determines the temperature of matter. Earth's geothermal energy originates from the original formation of the planet, from radioactive decay of minerals, from volcanic activity, and from solar energy absorbed at the surface. The geothermal gradient, which is the difference in temperature between the core of the planet and its surface, drives a continuous conduction of thermal energy in the form of heat from the core to the surface.

Geothermal power is not dependent on the sun but is limited to special locations. Since 2005, significant additions of electric capacity have occurred in Iceland, Indonesia, New Zealand, the United States, and Turkey, and global electricity production from geothermal has increased more than 20%. By the end of 2010, total global installations came to just over 11 GW, up an estimated 240 MW from 2009, and geothermal plants generated about 67.2 TWh of electricity during the year. The three largest plants commissioned in 2010 were in New Zealand (the largest single-shaft turbine project ever developed, at 0.1 GW), Italy (0.04 GW), and Kenya (just under 0.04 GW). The addition in Kenya increased the plant's capacity to 0.1 GW, making it the largest in Africa, and brought the country's total installations above 0.2 GW. It is estimated that Africa has 14,000 MW of geothermal potential but only 0.6% has been exploited.

2-1.5 Tidal power

Tidal power is the power achieved by capturing the energy contained in the moving water in tides and ocean currents (Figure 8). It is classified as a renewable energy source, because

tides are caused by the orbital mechanics of the solar system and are considered inexhaustible. The root source of the energy is the orbital kinetic energy of the earth-moon system, and also the earth-sun system. The tidal power has great potential for future power and electricity generation because of the essentially inexhaustible amount of energy contained in these rotational systems. All the available tidal energy is equivalent to one-fourth of the total human energy consumption today. Tidal power is reliably predictable (unlike wind energy and solar power). In Europe, tide mills have been used for nearly a thousand years, mainly for grinding grains. Several smaller tidal power plants have recently started generating electricity in Canada and Norway. They all exploit the strong periodic tidal currents in narrow fiords using subsurface water turbines.

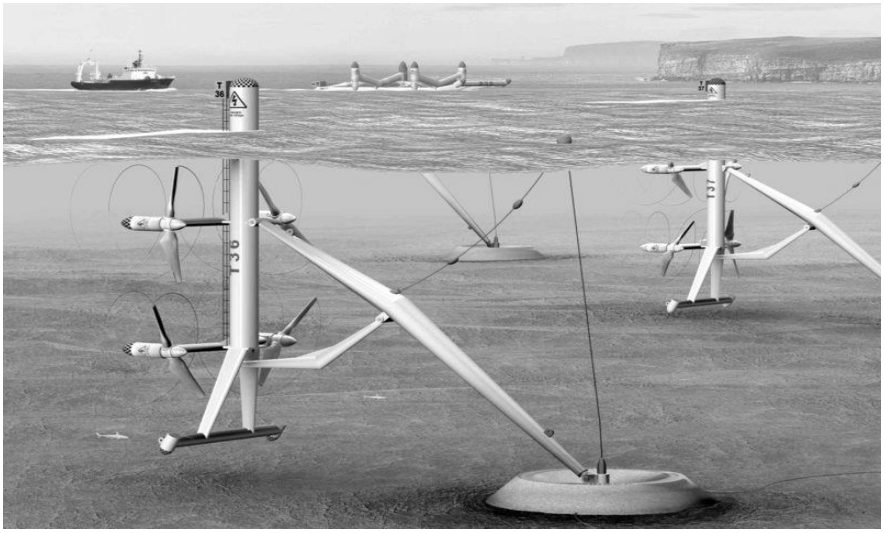


Figure 8: Tidal Energy Generation

2-1.6 Biomass and biofuels

The term *biomass* refers to any plant derived organic matter available on a renewable basis, including dedicated energy crops and trees, agricultural food and feed crops, agricultural crop wastes and residues, wood wastes and residues, aquatic plants, animal wastes, municipal wastes, and other waste materials. Biomass can be used to produce electricity, transportation fuels, or chemicals. The use of biomass for any of these purposes is called biomass energy. The most conventional way in which biomass is used, however, still relies on direct incineration. Biomass (produced by burning biological materials to generate heat), biofuels (produced by processing biological materials to generate fuels such as biodiesel and bioethanol), and biogas (using anaerobic digestion to generate methane from biodegradable materials and wastes) are other renewable sources of energy to produce an array of energy-related products, including electricity, liquid, solid, and gaseous fuels, heat, chemicals, and other materials.

Biofuels are a wide range of fuels which are in some way derived from biomass and include bioethanol, biodiesel and biogas. Bioethanol is an alcohol made by fermenting the sugar components of plant materials and it is made mostly from sugar and starch crops. Biodiesel is

made from vegetable oils, animal fats or recycled greases and biogas is methane produced by the process of anaerobic digestion of organic material by anaerobes.

There are several reasons for biofuels to be considered as relevant technologies by both developing and industrialized countries. These include energy security, environmental concerns, foreign exchange savings, and socioeconomic issues related to the rural sector.

The United States has since 2006 been the largest producer of bioethanol whereas Germany is the global leader in the production of biodiesel. Other European countries such as France, Italy and Sweden, are also major producers of biodiesel, giving Europe a massive lead over all other regions of the world. Apart from Brazil, developing countries that have been active in the production of biofuels include India, Malaysia, Indonesia, Argentina, China and Thailand. Several African countries have been making plans and preparations for the production of biofuels. Except for a few industrial scale ethanol plants that produce ethanol mainly for the chemical industry, large scale biofuel processing plants are yet to come on board in most African countries. In most instances, such as in Ghana, Ethiopia, Mali, Mozambique, Nigeria and Senegal, large scale plantations of biofuel crops are being planned, with a few thousand hectares of jatropha and other crops already in place.

Africa is endowed with biomass resources which can be used sustainably to produce biofuels to solve some of its energy problems. The biofuels industry in Africa can follow the footsteps of some of the traditional biofuels producers and create an alternative energy source that is produced ‘in-house’. African countries can target job creation, enhanced economic growth and reduced oil import bill using biofuels, as Brazil and other countries have done. There is great potential for job creation through biofuels production as the case has been in Brazil, US, Europe and other countries/regions. This can help to improve household incomes of especially rural farmers and feedstock outgrowers.



Self Assessment 2-1

1. List the various renewable energy sources that are available and discuss the drivers for renewable energy technology development

2. Discuss the differences between renewable and non-renewable energy

3. Define the renewable energy technologies that were addressed in this unit.

Answer tips

1. Read introduction to session 2-1
2. Read introduction to session 2-1
3. Read session 2-1.1 to 2-1.6

Learning Track Activities

Unit Summary

1. In this unit, we have introduced fossil fuels as the predominant energy source in the world today. We have also shown that fossil fuels may not be the energy of the future because of rising costs, environmental impacts and the peak oil concept. In view of this fact, renewable energy sources may be the way to go due the advantages that they offer.

2. We have discussed the various sources of renewable energy and current contribution to global energy consumption.

Key terms/ New Words in Unit

1. *Fossil fuels*: sources of energy that were formed by the death, decay, and transformation, or diagenesis, of life.

2. *Renewable energy*: energy which comes from natural resources such as sunlight, biomass, wind, rain, tides, and geothermal heat, which are renewable.

3. *Hydropower*: power that is derived from the force or energy of moving water, which may be harnessed for useful purposes.

4. *Hydroelectricity*: term referring to electricity generated by hydropower; the production of electrical power through the use of the gravitational force of falling or flowing water.

5. *Solar power*: the technology of obtaining usable energy from sunlight.

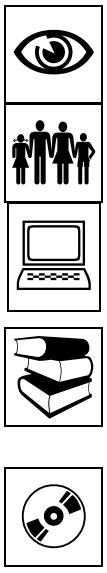
6. *Wind power*: the conversion of wind energy into a useful form of energy.

7. *Geothermal energy*: is thermal energy generated and stored in the earth.

8. *Geothermal power*: is the use of geothermal heat to generate electricity.

9. *Tidal power*: is the power achieved by capturing the energy contained in the moving water in tides and ocean currents.

10. *Biomass*: any plant derived organic matter available on a renewable basis, including dedicated energy crops and trees, agricultural food and feed crops, agricultural crop wastes and residues, wood wastes and residues, aquatic plants, animal wastes, municipal wastes, and other waste materials.



(choose an appropriate activity and give detail instructions)

- **Discussion Question:** Having read unit 1, discuss what in your opinion would be viable renewable energy technologies that Ghana could adopt based on the resources available within the country. At this point, you may ignore the economics of the technology and base your discussion mainly on the resources.



Unit Assignments 1

1. Name two advantages of generating electricity from renewable energy sources.
2. Describe three main advantages that renewable energy resources have over non-renewable resources for the generation of electricity
3. You are an engineer who wants to promote more renewable energy. Suggest how you would persuade people to want more renewable energy sources

Unit 2

BIOFUELS – FUNDAMENTAL THEORIES AND CONVERSION

Introduction

Biofuels are fuels produced from biological materials. In this Unit, only the liquid biofuels, i.e. biodiesel and bioethanol, will be discussed. Biogas, which is also a biofuel, is treated as a stand alone topic in the next unit. Areas covered include production processes, uses and some basic economics.



Learning Objectives

After reading this unit you should be able to:

1. Identify feedstock sources for biodiesel and bioethanol
2. Explain the production process for biodiesel and bioethanol
3. Discuss the uses of biodiesel and bioethanol

Unit content

Session 1-2: Biodiesel

- 1-2.1 Feedstocks for biodiesel production
- 1-2.2 Technical aspects of biodiesel production
- 1-2.3 Uses of biodiesel
- 1-2.4 Economics of biodiesel

Session 2-2: Bioethanol

- 2-2.1 Bioethanol production
- 2-2.2 Step by step ethanol production process
- 2-2.3 Uses of bioethanol
- 2-2.4 Economics of bioethanol

SESSION 1-2: BIODIESEL

Biodiesel is a mixture of mono-alkyl esters produced by the transesterification of vegetable oils/fats and animal fats (which are triglyceride esters) with alcohols in the presence of a catalyst. It is important to distinguish between biodiesel and straight vegetable oils – SVOs (also called pure plant oils – PPOs), which were the original form of this fuel. Even though the latter can be used in place of petro-diesel to run machinery, including vehicles, due to technical difficulties (especially high viscosities and presence of impurities such as free fatty acids, phospholipids, sterols, water and odourants) associated with the use of SVOs, they are becoming less important. The idea of using vegetable oils to power diesel engines dates back to the advent of the development of the diesel engine itself a little over one hundred years ago. Dr Rudolf Diesel conducted engine tests on plant oils. One prototype of his engines was tested on peanut oils at the World's Exhibition in Paris in 1900. Dr. Diesel was a visionary. In 1911 he stated that ''the diesel engine can be fed with vegetable oils and would help considerably in the development of agriculture of the countries, which use it''. In 1912, Diesel again said that, ''the use of vegetable oils for engine fuels may seem insignificant today. But such oils may become in course of time as important as petroleum and the coal tar products of the present time''.

The historical records indicate that the idea of testing peanut oils as fuel in diesel engines originated from the then French government's desire to find a means of domestic fuel production for her African colonies. Other European countries accepted and utilised the idea. With the advent of cheap petroleum, appropriate fractions of crude oil were refined to be used in the diesel engine, with the appropriate modification of the engine. Vegetable oils were also used as emergency fuels and for other purposes during the Second World War. The oil crisis of the early 1970s, however, brought renewed interest in this source of energy.

One of main technical problems associated with the use of SVO as fuel for diesel engines is its high kinematic viscosity which is about eleven to seventeen times higher than that of petro-diesel. The high viscosity causes poor atomization of the fuel in the engine's combustion chambers and ultimately results in operational problems, such as engine. The viscosity problems can be solved either by adapting the engine to be suitable for the fuel or by adapting the fuel to the use of the engine. Both approaches were tried but the second approach became more commercially relevant.

Since the renewal of interest in vegetable oil-derived fuels during the late 1970s, four possible approaches to adapting the fuel to the use of the diesel engine in order to overcome the problem of high viscosity were investigated. These were transesterification, pyrolysis, dilution with conventional petroleum-derived diesel fuel, and microemulsification. Pyrolysis is the thermal decomposition of a molecule, brought about in the absence of air or oxygen and optionally in the presence of a catalyst. Microemulsification involves the formation of a thermodynamically stable dispersion (colloidal in nature) of two otherwise immiscible liquids in the presence of emulsifiers, usually surfactants. Dilution or blending simply involves the mixing of the vegetable oils with light petro-diesel in appropriate proportions in order to attain the desirable viscosity. Transesterification of the plant oils with lower alcohols turned

out to be the ideal approach to modifying the plant oils and hence the term biodiesel is now used to denote products obtained through this technology. The production of biodiesel will, therefore, be described in terms of this technology.

1-2.1 Feedstocks for biodiesel production

Three main raw materials are necessary for the production of biodiesel. These are: the vegetable oil/fat or animal fat, the alcohol and the catalyst. All these raw materials are available in different forms. The selection of the right kind and quality of raw materials is very essential in achieving good yields and obtaining a product of the right quality and right price.

The kind of vegetable oils/fats or animal fats used for biodiesel production should satisfy some requirements in order to make the use of such feedstock economically feasible. Firstly, the source of the feedstock must be sustainable. For feedstocks from plant sources, the yield per hectare of land must be high and the cost of production should be reasonable. Table 2 shows the oil yields for some plant seeds. Many of the high yielding crops, e.g., oil palm and coconut are grown in the tropics and this should give many African countries a comparative advantage. However, other factors such as production costs, use of the feedstock for other purposes e.g., for food and for other industrial applications and environmental degradation can determine whether a high yielding crop is used or not. Secondly, for feedstocks from both plant and animal sources, the oil yield from the source must be high and the process for obtaining the oil must be technically easy and economically cost effective. Thirdly, the levels of impurities such as free fatty acids (FFA) and water must be low. Feedstocks containing significant levels of FFA require different processing to biodiesel than do refined oils and fats. It is vital that the FFA be removed because they can be detrimental to fuel systems and engines. All current biodiesel quality specifications impose strict limits on the levels of FFA allowed. The FFA composition should be typically less than 0.5% (w/w) for a high yield of biodiesel to be obtained. Examples of lipid sources that can be used for biodiesel production are given in Figure 9.

The types of oils/fats used for biodiesel production can be classified into:

- edible oils
- non-edible oils
- animal fats
- waste or recycled oils and
- microalgae.

A detailed description of the classes of feedstocks is presented in Appendix 1.

Table 2: Vegetable oil yields for some crops

Crop	Yield (litres oil/ha)	Crop	Yield (litres oil/ha)
Oil Palm	5950	Coriander	596
Coconut	2689	Camelina	583
Avocado	2638	Mustard	572
Brazil nuts	2392	Pumpkin	524
Macadamia	2246	Euphorbia	524
Jatropha	1892	Hazel nuts	482
Jojoba	1818	Linseed	478
Pecan Nuts	1791	Coffee	459
Castor Beans	1413	Soya	446
Olives	1212	Hemp	363
Rapeseed	1190	Cotton	325
Groundnuts	1059	Calendula	305
Sunflower	952	Kenaf	273
Tung Oil Tree	940	Lupine	232
Rice	828	Oats	217
Safflower	779	Cashew	176
Sesame	696	Maize	172

Biodiesel yield = oil yield x 0.8

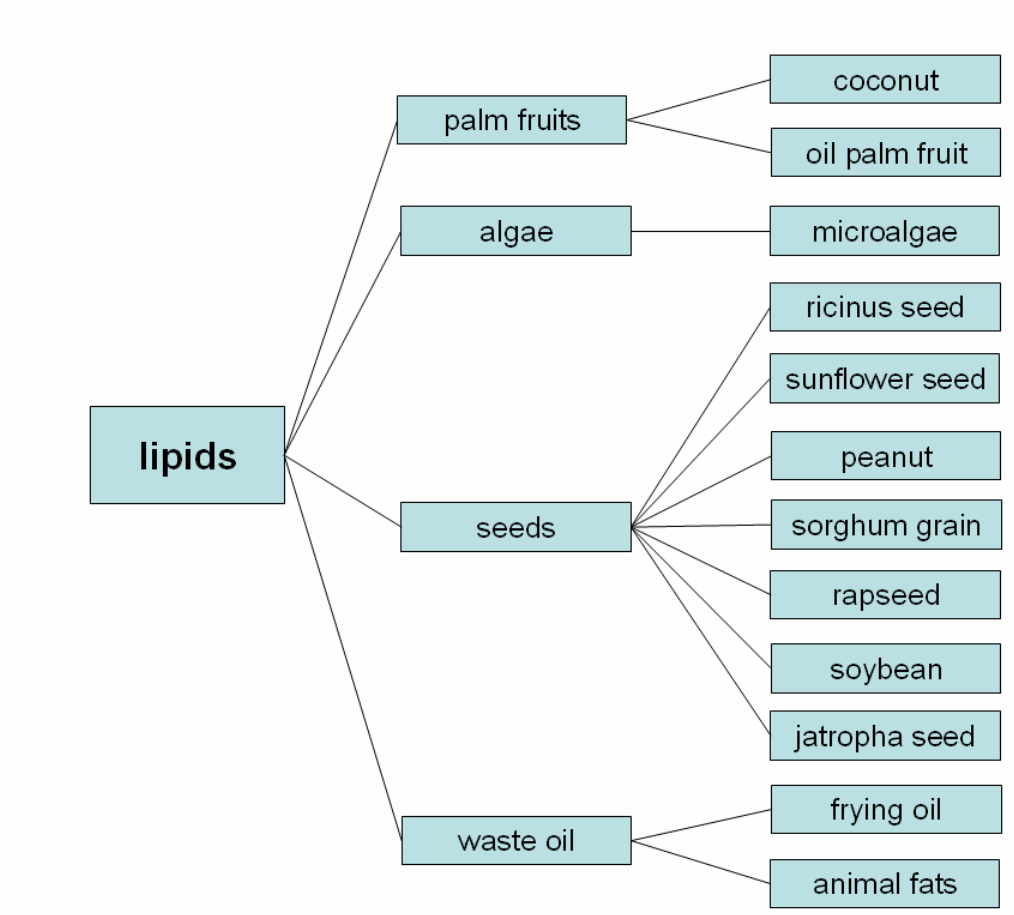


Figure 9: Examples of sources of feedstocks for biodiesel production

1-2.2 Technical aspects of biodiesel production

There are three major steps from the raw feedstocks (seeds) to the finished products (biodiesel and glycerol by-product). These are: (a) extraction of oil, (b) refining of the oil and (c) the transesterification of the oil to obtain the biodiesel.

1-2.2.1 Extraction of oil

There are two fundamental production process types for vegetable oils:

- Industrial: this is based on centralized production by refining in large industrial plants
- Small scale pressing: this is a decentralized cold pressing process directly on farms or in cooperatives

In small scale cold pressing facilities, the cleaned oil seeds are exclusively mechanically pressed at maximum temperatures of 40 ºC. Suspended solids are removed by filtration or sedimentation. As a co-product, the press cake is left with a remaining oil content of usually over 10 %, which is used as a protein-rich fodder. Due to higher production costs, the decentralized oil production by farmers is not widely applied today in developed countries, although the chance of additional income for farmers is given. Furthermore, the co-product could be directly used for feeding the animals. Until large scale industrial oil extraction plants

are in place, many African communities can depend on small scale cold pressing facilities for their oil extraction. This will have an added advantage of social cohesion among the farmers.

The common way in oil extraction is the treatment of feedstock in centralized industrial large scale plants. First, the feedstock has to be pre-treated. Within the pre-treatment the seeds have to be dried first, then cleaned and peeled if necessary. After the treatment the seeds are crushed, and temperature and moisture content are conditioned. The adjustment of the specific moisture content is important as too high contents make penetration of the solvent difficult, whereas too low contents enhances compactness and consequently also make solvent penetration difficult. The conditioning of temperature above 80 °C is necessary to deactivate microorganisms and to avoid smearing of the press through coagulated proteins. Additionally the crushed seeds can be better penetrated by the solvent and the oil flow is better as it is more liquid.

After conditioning, the oil seeds are pressed at relatively higher temperatures than used in small-scale cold pressing. Thereby approximately 75 % of the total oil content can be extracted. This pressed raw oil then is filtered and dehydrated and the final pure oil can be used for further refining into PPO or for biodiesel production.

1-2.2.2 Oil refining

The refining process is an important treatment of creating PPO and of preparing vegetable oil for the transesterification process of biodiesel. It is important in order to remove undesirable substances, such as phosphatides, free fatty acids, waxes, tocopherols and colorants. These substances can alter oil storage life and hamper further processing. During this first refining step the oil mass (4 to 8 %) and the solvent contents are reduced. Since the refining process depends on the vegetable oil quality, the refining steps depend on the feedstock source. There also exist alternatives of refining and some refining steps are merging.

The first purification step of oil refining is the removal of phosphatides, also known as degumming process. This is necessary as phosphatides make the oil become turbid during storage and as they promote the accumulation of water. Phosphatides can be removed by two different ways: water degumming and acid degumming. Water soluble phosphatides can be removed by water degumming. In this case water is added to the oil at 60-90°C and the mixture is separated by centrifugal separation of the water phase and the oil phase. Acid degumming is applied to phosphatides which cannot be hydrated. Acid substances like citric or phosphoric acid are added.

The second refining step is the deacidification. It is an important step for edible oils as the development of rancid flavours FFA is prevented. The content of these FFA's in unrefined pure oil is between 0.3 and 6 %. In this step also phenol, oxidized fatty compounds, heavy metals and phosphatides are removed. The purification of all these substances is not only important to edible oils, but also to fuel production as these compounds alter the storage life and influence transesterification in the biodiesel process. Several methods of deacidification are in operation:

- Neutralization with alkali: This is the most applied method. FFA's are saponified with alkaline solutions and the resulting soap is separated.
- Distillation: For this alternative more energy is needed.
- Deacidification by esterification: This is done by esterification of FFA's with Glycerine

- Deacidification and extraction of colorants and odours with various solvents: (e.g.,. ethanol, furfural, propane)

In the third step which is bleaching, colorants are removed. This process step enhances storage life of the biofuel. Bleaching is mainly conducted by adsorbing substances, such as bleaching earth, silica gel or activated carbon. Oxygen, ozone, hydrogen peroxide and heat (200°C) can also be used for bleaching.

In the deodorization step odorous substances (ketone, aldehyde) are removed by steam distillation. Finally a dehydration step has to be conducted, as traces of water may decrease conversion in the transesterification process of biodiesel production. The removal of water is either accomplished by distillation under reduced pressure or by passing a stream of nitrogen through the fatty material.

1-2.2.3 Transesterification

Transesterification is simply a chemical reaction in which a shorter chain alcohol replaces a longer chain one in an ester of an organic acid in order to produce a lower molar mass ester of the acid. This reaction is reversible and usually takes place in the presence of a catalyst. Since transesterification has been extensively used for the production of biodiesel, it has often been defined in terms of biodiesel production but transesterification is not necessarily synonymous to biodiesel production. The transesterification reaction is sometimes called alcoholysis because an alcohol is used to convert the oil or fat into the ester (biodiesel). Hence the process is sometimes called methanolysis or ethanolysis depending on whether methanol or ethanol is used as the alcohol. The transesterification reaction can be generally represented as follows:

$$\mathbf{RCOOR^1} + \mathbf{R^2OH} \underset{}{\overset{\text{Catalyst}}{\rightleftharpoons}} \mathbf{RCOOR^2} + \mathbf{R^1OH} \qquad (1)$$

(Original Ester) (Original Alcohol) (New Ester) (New Alcohol)

In the production of biodiesel, the triglyceride (oil or fat) is the original ester and the mixture of new esters is the biodiesel. The new alcohol is the glycerol by-product whilst the original alcohol is the methanol or ethanol used for the transesterification. The detailed chemistry will not be discussed in this text. Students who are interested in the chemistry should refer to some of the books cited in the reference list.

The presence of the catalyst allows the transesterification reaction to be carried out under mild conditions. Hasher conditions, e.g., high temperatures and pressures are required for the reaction in the absence of a catalyst. In the absence of the catalyst, the reaction will take a much longer time to complete. Technically, the presence of the catalyst should not affect the yield. It only shortens the time within which a specific amount of the biodiesel can be obtained under specified conditions of temperature and pressure. Selecting the right kind of catalysts and the right amount required for the transesterification is essential for obtaining a high yield of the product within a relatively short time. Since the reaction is reversible, removing the products as they are formed allows the process to be carried out more efficiently.

The transesterification process consists of four principal steps: pre-treatment of the lipid feedstock, biodiesel production through transesterification, biodiesel purification and glycerol purification.

1. Pretreatment of the lipid feedstock

This step depends on the nature and quality of the lipid feedstock. Different feedstocks require different types of pretreatment. The description given here is for tallow (animal fats) or vegetable oil. Pretreatment of tallow or oil feedstock is meant to remove components that will be detrimental to subsequent processing steps. The process is somewhat different for tallow and vegetable oils, the former removing free fatty acids and the latter gummy materials. Acid-catalyzed pretreatment is applied to oil or fat containing significant amounts of FFA. Used cooking oils typically contain 2–7% FFA, and animal fats contain 5–30% FFA. Some very low-quality feedstocks, such as trap grease, can approach 100% FFA. When an alkali catalyst is added to these feedstocks, the FFA react with the catalyst to form soap and water as shown in the reaction below:



(2)

Fatty acid Potassium hydroxide Potassium soap Water

Up to ~5% FFA, the reaction can still be catalyzed with an alkali catalyst, but additional catalyst must be added to compensate for that lost to soap. The soap created during the reaction is either removed with the glycerol or washed out during the water wash. When the FFA level is >5%, the soap inhibits separation of the glycerol from the methyl esters and contributes to emulsion formation during the water wash. For these cases, an acid catalyst such as sulphuric acid can be used to esterify the FFA to methyl esters as shown in the following reaction:



(3)

Fatty acid Methanol Methyl ester Water

This process can be used as a pretreatment to convert the FFA to methyl esters, thereby reducing the FFA level. Then, the low-FFA pretreated oil can be transesterified with an alkali catalyst to convert the triglycerides to methyl esters. As shown in the reaction scheme, water is formed and, if it accumulates, it can stop the reaction well before completion. It has been proposed that the alcohol should be separated from the pretreated oil or fat after the reaction. Removal of this alcohol also removes the water formed by the esterification reaction and allows for a second step of esterification; alternatively, one may proceed directly to alkali-catalyzed transesterification. The methanol-water mixture will also contain some dissolved oil and FFA that should be recovered and reprocessed.

2. Biodiesel production through transesterification

The processes described here are based on the use of methanol as the alcohol, sodium hydroxide or methoxide as the catalyst. In the actual transesterification the pretreated triglycerides are reacted with methanol to form the raw methyl esters and glycerol. There are two basic steps: the reaction process followed by separation of the methyl ester and glycerol streams. In most technologies, these two steps are undertaken twice in order to push the transesterification closer to completion by reducing the concentration of glycerol in the second stage.

For alkali-catalyzed transesterification, Figure 10 shows a basic schematic diagram of the processes involved in biodiesel production from feedstocks containing low levels of FFA. These include soybean oil, canola (rapeseed) oil, and the higher grades of waste restaurant oils. Alcohol, catalyst, and oil are combined in a reactor and agitated for ~1 h at 60°C. Smaller plants often use batch reactors but most of the larger plants (>4 million L/yr) use continuous flow processes involving continuous stirred-tank reactors (CSTR). The reaction is sometimes done in two steps in which ~80% of the alcohol and catalyst is added to the oil in a first-stage CSTR. Then, the product stream from this reactor goes through a glycerol removal step before entering a second CSTR. The remaining 20% of the alcohol and catalyst is added in this second reactor. This system provides a very complete reaction with the potential of using less alcohol than single-step systems.

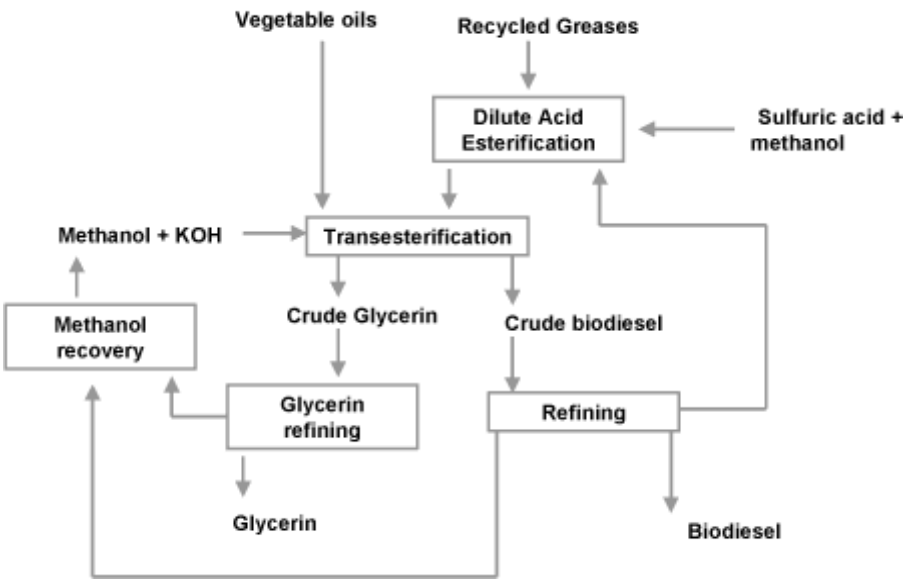


Figure 10: Process Flow Scheme for Biodiesel Production

After the reaction, glycerol is removed from the methyl esters. Due to the low solubility of glycerol in the esters, this separation generally occurs quickly and can be accomplished with either a settling tank or a centrifuge. The excess methanol tends to act as a solubilizer and can slow the separation. However, this excess methanol is usually not removed from the reaction stream until after the glycerol and methyl esters are separated due to concern about reversing the transesterification reaction. Water may be added to the reaction mixture after the transesterification is complete to improve the separation of glycerol.

3. Biodiesel purification

After separation from the glycerol, the methyl esters enter a neutralization step and then pass through a methanol stripper, usually a vacuum flash process or a falling film evaporator, before water washing. Acid is added to the biodiesel product to neutralize any residual catalyst and to split any soap that may have formed during the reaction. Soaps will react with the acid to form water-soluble salts and FFA according to the following equation:

R-COONa + HAc ⟶ R-COOH + NaAc (4)

Sodium soap Acid Fatty acid Salt

The salts will be removed during the water washing step and the FFA will stay in the biodiesel. The water washing step is intended to remove any remaining catalyst, soap, salts, methanol, or free glycerol from the biodiesel. Neutralization before washing reduces the amount of water required and minimizes the potential for emulsions to form when the wash water is added to the biodiesel. After the wash process, any remaining water is removed from the biodiesel by a vacuum flash process.

4. Glycerol purification

The glycerol stream leaving the separator is only ~50% glycerol. It contains some of the excess methanol and most of the catalyst and soap. In this form, the glycerol has little value and disposal may be difficult. The methanol content requires the glycerol to be treated as hazardous waste. The first step in refining the glycerol is usually to add acid to split the soaps into FFA and salts. The FFA is not soluble in the glycerol and will rise to the top where they can be removed and recycled. The salts remain with the glycerol, although depending on the chemical compounds present, some may precipitate out. After acidulation and separation of the FFA, the methanol in the glycerol is removed by a vacuum flash process, or another type of evaporator. At this point, the glycerol should have a purity of ~85% and is typically sold to a glycerol refiner. The glycerol refining process takes the purity up to 99.5–99.7% using vacuum distillation or ion exchange processes. Methanol that is removed from the methyl ester and glycerol streams will tend to collect any water that may have entered the process. This water should be removed in a distillation column before the methanol is returned to the process.

1-2.3 Uses of biodiesel

Biodiesel has similar characteristics as petro-diesel and hence can generally be applied in machinery that make use of petro-diesel. The applicability of biodiesel in such machinery is based on the benefits listed above. Currently, it is used either in the neat form (100% biodiesel – B100) or as a blend in powering vehicles with diesel engines with little or no engine modifications. It is suitable for buses and heavy duty machinery. Due to its high flash point, biodiesel is very suitable for use in underground mining operations. Since mining operations create marginal lands and some of the energy crops that are used for the production of biodiesel can be grown on marginal lands, biodiesel programmes in mining communities can serve a dual purpose –land reclamation and provision of safe energy for mining operations.

Biodiesel can be used to run agricultural machinery. In rural agricultural communities, biodiesel programmes can help provide the energy needs for their farming activities if they mobilize themselves into cooperatives to own the farm machinery. Such an arrangement can also help to increase their income. Biodiesel can also be used in lamps in place of kerosene.

1-2.4 Economics of biodiesel

The success of replacing petroleum diesel with biodiesel depends to a large extent on the economics of the process. At present, the cost of producing biodiesel is more expensive than the cost of producing petroleum based diesel. In some instances, the cost of producing biodiesel is about 1.5 times higher than the cost of producing diesel depending on feedstock.

Reasons for the high cost of producing biodiesel can be grouped into three main areas for the purpose of this material. The areas are:

- Cost associated with the cultivation of the oil seed (feedstock)
- Cost associated with producing the biodiesel
- Other factors that may affect the cost of production

The cost of producing biodiesel is reported to depend to a large extent on the cost of feedstock. The cost of feedstock is country specific and depends on:

- Cost of land
- Labour
- Farm machinery
- Cost of fertilizer
- Cost of planting materials (seeds), herbicides, irrigation, etc.

These factors combined constitute the total cost of cultivating the oil seed. Availability and cost of land is country specific. Cost of labour, planting materials and irrigation are also country specific. For instance in African countries, land, labour and planting materials may be lower than in developed countries.

Cost associated with producing the biodiesel can be grouped into capital and operational cost. The capital cost is basically the cost of setting up the plant. This includes the cost of land, clearing, equipment and installation of equipment. All these cost are dependent on the size of the plant. Operational cost includes labour, utilities, and plant overhead cost which are also country specific.

Other factors that may affect the cost of biodiesel positively when considered and put in their proper contest include government subsidies and additional income from the sale of by-products of the biodiesel production process. Government subsidies are essential tools needed to make the price of biodiesel comparable to diesel. So far most of the economics carried out on biodiesel do not include the potential additional revenue that can accrue from the sale of the by-product. Glycerine which is a by product of biodiesel production is a potential raw material for the production of soap.



Self Assessment 1-2

1. What is biodiesel?

2. Which of the oil crops for producing biodiesel have been successfully cultivated in Ghana?

3. What are the five main classifications that biodiesel feedstocks can be grouped into, after having read this unit?

4. What are the principal components that contribute to the cost of producing biodiesel?



Answer tips

1. Read introduction to session 1-2.
2. Read session 1-2.1
3. Read sessions 1-2.1 to 1-2.5
4. Read session 1-2.4

SESSION 2-2: BIOETHANOL

Ethanol (CH3CH2OH) belongs to the group of chemical compounds whose molecules contain a hydroxyl group, −OH, bonded to a carbon atom. It is a clear, colourless liquid with a characteristic, agreeable odour. In dilute aqueous solution, it has a somewhat sweet flavour, but in more concentrated solutions it has a burning taste. The terms ‘fuel ethanol’ and ‘ethanol fuel’ are usually used in literature to refer to ethanol used as a transportation fuel. ‘Bioethanol’ is ethanol that is obtained from the conversion of carbon-based feedstocks that are considered renewable. Bioethanol, as a renewable fuel, is manufactured through the fermentation of sugars; it may be produced from sugar-, starch- and cellulose-based feedstocks (including forest trimmings and agricultural residues). For first generation bioethanol, the feedstocks include sugarcane, corn, sugar beet, sweet sorghum, cassava and wheat. Process descriptions in this chapter will dwell on sugar and starch crops.

It is worth noting that sugar feedstocks, such as sugarcane and sweet sorghum are voluminous crops and their transportation is costly. Meanwhile, they must be processed within forty-eight hours after harvesting, otherwise, energy yields start to decrease significantly. For these reasons, such feedstocks should be produced, as much as possible, in the vicinity of the ethanol plant. On the other hand, corn and other grains are far less perishable and can be transported over longer distances at reasonable costs, a great advantage over the sugar feedstock. Therefore, a corn-fed ethanol plant can rely on a much more decentralized area of supply and it is easier to include smallholders into the logistical processes required.

As a biofuel crop, cassava has the advantage that the root can remain in the ground for months without deterioration, allowing for carefully planned and continuous harvest schemes. Once it is harvested, cassava needs to be processed within two to three days and makes it possible for cassava to be grown at considerable distance from the ethanol plant. The roots are not as voluminous as sugarcane or sorghum stalks and cassava is therefore easier and cheaper to transport.

2-2.1 Bioethanol production

Ethanol could be produced through chemical reactions or by conversion of biomass materials by fermentation processes following three major steps: the formation of a solution of fermentable sugars; the fermentation of these sugars to ethanol; and the separation and purification of the ethanol, usually by distillation. The production processes depend on whether the feedstock is a sugar crop (sugar cane and sweet sorghum), or starch crop (grain and tuber crops). The principal differences have to do with the initial processes the feedstock goes through before the fermentation stage. But no matter the feedstock used, the final stage of ethanol production is fermentation. Sugar crops follow the same process line whereas starch crops can be converted to ethanol using the dry-milling or wet-milling production route.

2-2.1.1 Ethanol from sugar feedstock

Sugarcane consists of a main stalk, a green top, and a significant amount of leaves. In most cane processing operations worldwide, only the stalk (without the tops and leaves) is delivered to the mill for processing. The production of ethanol from sugar crops is the simplest way to produce ethanol. Raw cane (from the sugarcane or sweet sorghum plant) is washed, chopped, and crushed in rolling mills to separate the sugar-laden juice (sucrose) from the fibre in the cane, called bagasse. The juice, which contains over 90 percent of the sucrose in the cane, is filtered to remove impurities. Filtered juice is heated and fermented and then fed to a distillery. In the fermentation stage, yeasts and other microbes are added to produce a dilute alcohol product with ethanol concentration of 10 – 15%. The fermented solution is then distilled to remove most of the water from the mixture to give "hydrous ethanol" with about 95% ethanol content and 5% water. This mixture of ethanol and water is azeotropic, so that the compounds cannot be separated by simple addition of heat, as is done in the distillation process. Hydrous ethanol can be commercially used, but cannot be blended with gasoline. A final dehydration step is required to remove the remaining water to produce "anhydrous ethanol" that can be blended with gasoline. An additional reactant, e.g. cyclohexane, is needed in order to dehydrate the ethanol An ethanol production pathway is presented in Figure 11 and ethanol production routes by feedstock and production techniques is shown in Table 3.

Co-products from processing of sugar feedstock

The sugar mill that is used for crushing the raw sugarcane stalks produces considerable volumes of cane residue called bagasse which is often used in combustion furnaces in the plant to produce heat and electricity. Bagasse can be fired in a Combined Heat and Power (CHP) plant where heat and electricity are generated simultaneously. The electricity can then be supplied to the grid whereas the heat can be used in the plant as process heat. Bagasse cogeneration is very popular in sugar and 'ethanol from sugar feedstock' producing plants. Another co-product from the process is stillage, a potassium-rich liquid, drained from the bottom of the distillation columns. Stillage is usually treated in closed circuit, anaerobic digesters before being released in order to reduce high Chemical Oxygen Demand (COD) and Biochemical Oxygen Demand (BOD) levels. The treated stillage can then be used to fertilise and irrigate the sugarcane plantation.

Table 3: Ethanol production step by feedstock and conversion techniques

Feedstock type	Feedstock	Harvest technique	Feedstock conversion to sugar	Process heat	Sugar conversion to alcohol	Co-products
Sugar crops	**Sugarcane and sweet sorghum**	Cane stalk cut, mostly taken from field	Sugars extracted through bagasse crushing, soaking, chemical treatment	Primarily bagasse	Fermentation and distillation of alcohol	Heat, electricity and molasses
Starch crops	**Corn**	Starchy parts of plants harvested, stalks mostly left on the field	Starch separation, milling, conversion to sugars via enzyme application	Typically from fossil fuel	Fermentation and distillation of alcohol	Animal feed and sweetener
	Cassava	Starchy root tuber harvested. Sticks serve as planting material	Peeling, drying, milling, conversion of sugars via enzyme application	Typically from fossil fuel	Fermentation and distillation of alcohol	Animal feed, manure or raw material for biogas plant

2-2.1.2 Ethanol from starch feedstock

Ethanol from starch crops is produced by one of two processes: wet milling and dry milling. The principal differences between the two processes is summarised in Table 4. The wet mill is more versatile, though, because the starch stream, being nearly pure, can be converted into other products (for instance, high-fructose corn syrup (HFCS)). Co-product output from the wet mill is also more valuable. The market for co-products adds tremendous economic viability to the ethanol industry. Ethanol-from-cassava is similar to the ethanol-from-grain process. Both wet milling and dry milling processes can be used. But again, like grain, the dry milling of cassava is popular in Thailand, a country with great expertise in cassava-to-ethanol conversion processes. In the dry milling process, the cassava is peeled, sliced into chips and dried. The production of starch and its subsequent conversion to ethanol follows the corn to ethanol process description in the next section.

The dry milling process involves cleaning and breaking down the cereals kernel into fine particles using a hammer milling process. This creates a powder with a coarse flour type consistency. The powder contains the cereals germ, starch and fibre. In order to produce a sugar solution the mixture is hydrolysed or broken down into sucrose sugars using enzymes or a dilute acid. The mixture is then cooled and yeast is added in order to ferment the mixture into ethanol. The dry milling process is normally used in factories producing less than 50 million gallons of ethanol every year. In the wet milling process, corn kernel is steeped in a solution of water and sulphur dioxide (SO_2); this helps to break down the proteins and release the starch present in the corn and helps to soften the kernel for the milling process. The corn is then milled to produce germ, fibre and starch products. The germ is extracted to produce corn oil and the starch fraction undergoes centrifugation and saccharification to produce gluten wet cake. The ethanol is then extracted by the distillation process. The wet milling process is normally used in factories producing several hundred million gallons of ethanol every year. The 30- to 40-hour extra soaking of corn in the wet milling process requires additional tanks that contribute to the higher construction costs of wet milling plants.

Table 4: Principal differences between wet and dry mill ethanol plants

Parameter	Dry milling	Wet milling
Cost of construction	Lower	Higher
Versatility of products	Less versatile	More versatile
Co-products	Dried distillers grain and corn meal	Sweeteners, corn oil, gluten feed and gluten meal
Value of co-products	Less valuable	More valuable
Yield per feedstock	Higher	Slightly Less

2-2.2 Step by step ethanol production process

Processes 1-3 is initial processing of starch feedstock

1. **Milling:** The feedstock (corn) is passed through a hammer mill and is grinded into a fine meal to free the starchy material from the grain kernel and to allow water access to all the starch in the grains. It is much like the grinding process that agricultural residues go through so that the particles have uniform size and to make the production process more efficient.

2. **Liquefaction:** The meal is mixed with warm water to form a mash. An enzyme (alpha-amylase) is added and heat is applied to enable liquefaction. The mixture is cooked at a high temperature of between 120 – 150 °C. The temperature is then lowered to about 95°C. The initial higher temperature enables the elimination of possible bacteria and the lower holding temperature ensures that the mixture is still in the liquid state.

3. **Saccharification:** The mash from the preceding process is cooled to 60 °C and a secondary enzyme (gluco-amylase) is added. The purpose of this enzyme is to convert the starch to fermentable sugars, also known as dextrose. The temperature keeps the reaction progressing fast enough while at the same time avoiding unwanted side reactions. Saccharification in a wet mill may take up to 48 hours, though it usually requires less time, depending on the amount of enzyme used. In modern dry mills, saccharification has been combined with the fermentation step in a process called simultaneous saccharification and fermentation, to save cost and time.

Processes 4-7 is similar for juice from sugar feedstock

4. **Fermentation:** The mash is then further cooled to a fermentation temperature of about 32 °C. Yeast is added to ferment the sugars (in mash or juice depending on feedstock) to ethanol and carbon dioxide (CO_2) and small quantities of other organic compounds. The yeast contains an enzyme called invertase, which acts as a catalyst and helps to convert the sucrose sugars into glucose and fructose (both $C_6H_{12}O_6$). The chemical reaction is shown below in equation 5:

$$\underset{\text{Sucrose}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{\text{Invertase}} \underset{\text{Fructose}}{C_6H_{12}O_6} + \underset{\text{Glucose}}{C_6H_{12}O_6} \quad (5)$$

The fructose and glucose sugars then react with another enzyme called zymase, which is also contained in the yeast to produce ethanol and carbon dioxide. The chemical reaction is shown in equation 6:

$$\underset{\text{Fructose/Glucose}}{C_6H_{12}O_6} \xrightarrow{\text{Zymase}} \underset{\text{Ethanol}}{C_2H_5OH} + 2CO_2 \quad (6)$$

The fermenting mash is allowed to flow or cascade using a continuous process, through several fermenters until the mash is fully fermented and then leaves the final tank. In a batch fermentation process, the mash stays in one fermenter for about 48 hours before the

distillation process is started. The fermenting mash is agitated and also circulated through external exchangers to remove the unwanted heat generated by the fermentation process. The yeast, which produces almost as much CO_2 as ethanol, ceases fermenting when the concentration of alcohol is between 12 and 18 percent by volume, with the average being about 15 %.

5. **Distillation**: The fermented mash (beer) contains significant quantity of water, as well as all the non fermentable solids from the feedstock and the yeast cells which must be removed. This is achieved by using the fractional distillation process. The mash is distilled in two different stages, to increase purification and separation. The distillation process works by boiling the water and ethanol mixture. Since ethanol has a lower boiling point (78.3 °C) compared to that of water (100 °C), the ethanol turns into the vapour state before the water and can be condensed and separated. The ethanol leaves the top of the final column at about 95 – 96% strength (ethanol and water form a 95 – 96% azeotrope so it is not possible to reach 100% by simple distillation), and the residue mash, called stillage, is transferred from the base of the column to the co-product processing area.

6. **Dehydration:** The ethanol from the top of the column is passed through an absorber containing a molecular sieve (dehydration system) to remove the remaining water. The ethanol at this stage is called anhydrous (pure, without water) ethanol and is approximately 200 proof. Three absorbers may be used to enhance efficiency. A drink that's 100 proof is 50% alcohol, while a 200 proof drink is 100% pure alcohol.

7. **Denaturing:** Fuel ethanol is at this stage denatured with a small amount (2 – 5%) of some product such as gasoline, to make it unfit for human consumption.

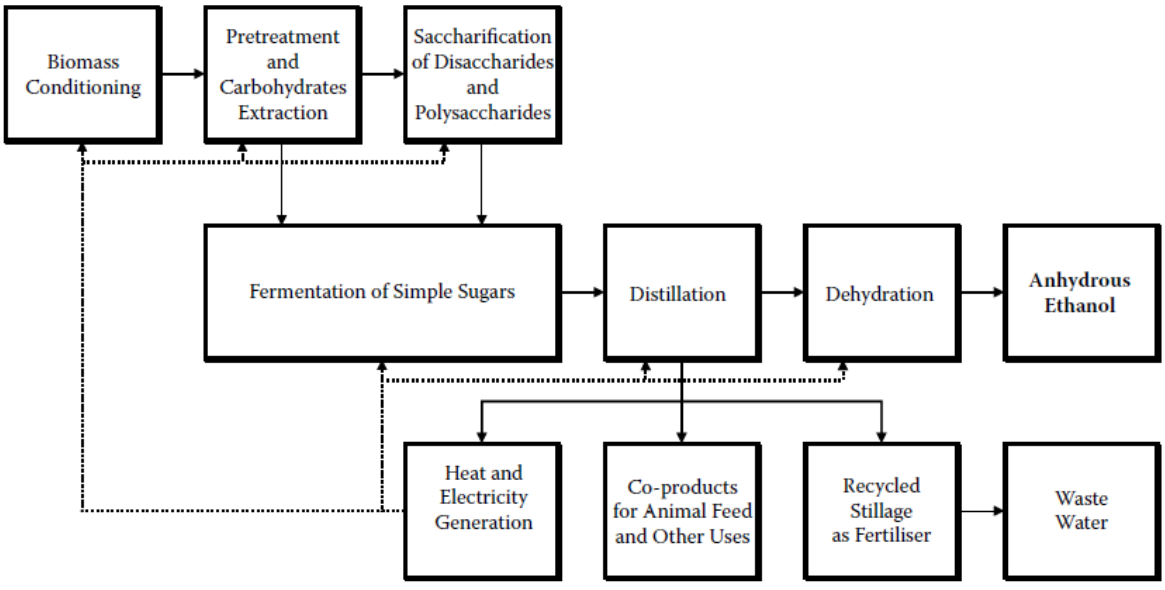


Figure 11: Schematic outline of the biomass-to-ethanol process

Co-product from starch crops

In the dry milling process of ethanol production, there are two main co-products created: CO_2 and Dried Distillers Grains with Soluble (DDGS). CO_2 is given off in great quantities during fermentation. The CO_2 can be collected and cleaned of any residual alcohol, compressed, and sold for use in carbonate beverages or to flash freeze meat. CO_2 can also be released to the atmosphere. DDGS is high in protein and other nutrients and a highly valued livestock feed ingredients. For every tonne of cereals used for ethanol production on average one third will enter the animal feed stream as DDGS. Because of its high protein level it is very much favoured as replacement of soy cake and could be sold to users.

In the wet milling process, the germ is removed from the kernel, and corn oil is extracted from the germ. The remaining germ meal is added to the hulls and fibre to form the corn gluten feed (CGF) stream. Gluten, a high-protein portion of the kernel, is also separated and becomes corn gluten meal (CGM), a high-value, high-protein (60 percent) animal feed.

Also during the chemical processing, streams of wastewater containing biodegradable impurities are generated. For effective wastewater management, a wastewater treatment plant may be necessary, with the aim of treating the wastewater effectively so that a large volume of it can be recycled and reused within the plant. The remaining treated water may be discharged into rivers with only a small volume of wastewater being discharged to sewers to minimise cost. The cellulosic parts of the biomass feedstock – like the bagasse of sugar cane – are utilized in useful ways during ethanol processing, to create heat and electricity to power milling plants.

Storage and distribution

Just like gasoline, bioethanol must be stored appropriately within the period just after production and consumption. There is the need for long term storage tanks and short term storage tanks, depending on the market and feedstock situations. Road tankers have been recommended for the distribution of bioethanol as they are by far more flexible and can deliver from the factory itself straight to the destination. Pipelines are frequently contaminated with water and rust and therefore not recommended for the shipping of bioethanol. In some cases, feedstock may have to be stored at the production site before they are processed. If this is the case, then storage containers are necessary to protect the feedstock from harsh weather conditions. The properties of fuel ethanol are presented in Table 5.

2-2.3 Uses of Bioethanol

The principal application of bioethanol is in the spark ignition engine as a transportation fuel. Bioethanol is best applied in spark ignition engines because of its high octane rating, implying very good anti-knock characteristics. The energy content is however lower than that of petrol (for pure ethanol, about two-third that of petrol) as shown in Table 6. Bioethanol is also corrosive when used as a pure fuel, but low blends with petrol helps avoid this problem. Bioethanol can be used in blends of between 10 and 20% with fossil gasoline without necessary engine modifications. In Brazil, all brands of automotive gasoline contain anhydrous ethanol in the range of 20−25 %. Foreign vehicles have been adapted by using ethanol-compatible materials

in the fuel system and by tuning the engines for a mid-range point, usually at the 22 % ethanol level (E22). This customization has resulted in good drivability and performance, with fuel consumption comparable to gasoline operation. An international classification for ethanol blends has been adopted by most bioethanol users. A blend of ethanol in gasoline is usually classified as EX where X is the percentage of ethanol in the blend. For example, E10 denotes a fuel blend with 10% ethanol and 90% petrol. E85 is fuel blend with 85% ethanol and 15% petrol.

Table 5: Properties of fuel ethanol

Property	Comment
Vapour density	Ethanol vapour, like gasoline vapour, is denser than air and tends to settle in low areas. However, ethanol vapour disperses rapidly
Solubility in water	Fuel ethanol will mix with water, but at high enough concentrations of water, the ethanol will separate from the water
Energy content	For identical volumes, ethanol contains less energy than gasoline. On an energy basis, 1.0 gallon of E85 is equivalent to 0.72 gallons of gasoline
Flame visibility	A fuel ethanol flame is less bright than a gasoline flame but is easily visible in daylight
Specific gravity	Pure ethanol and ethanol blends are heavier than gasoline
Conductivity	Ethanol and ethanol blends conduct electricity. Gasoline, by contrast, is an electrical insulator
Stoichiometric fuel-to-air ratio	E85 needs more fuel per pound of air than gasoline; therefore, E85 cannot be used in a conventional vehicle
Toxicity	Ethanol is less toxic than gasoline or methanol. Carcinogenic compounds are not present in pure ethanol; however, because gasoline is used in the blend, E85 is considered to be potentially carcinogenic
Flammability	At low temperatures, the vapour concentration in an E85 tank can fall into the flammable range The temperature for flammable E85 vapours depends on the E85 volatility. Although less likely than with E85, gasoline tanks can also contain flammable vapours at extremely low temperature

New vehicle technologies, Flexible Fuel Vehicles, have been developed that can use blends having higher concentrations of ethanol in gasoline as fuels while achieving reliable, low-emission operation. Flexible fuel vehicles (FFVs) are capable of operating on gasoline, E85, or a mixture of both. Unlike natural gas and propane driven biofuel vehicles, flexible fuel vehicles contain one fuelling system, which is made up of ethanol compatible components and is set to accommodate the higher oxygen content of E85. Other than their fuelling capability and ethanol

compatible components, FFVs are similar to their conventional gasoline counterparts. Their power, acceleration, payload, and cruise speed are comparable whether running on ethanol or gasoline. Sensors of these flexible fuel vehicles (E85 FFV) can automatically detect the type of the fuel and adapt engine running. They adjust the air/fuel ratio and the ignition timing to compensate the different octane levels of the fuel in the engine cylinders. The main reason to limit ethanol content to 85% is to enhance volatility conditions for cold start, particularly in cold climates. Therefore the technology does not need any cold start ancillary system.

FFVs have specially developed engines that can withstand the high octane rating of higher blends of bioethanol. FFVs are similar to gasoline vehicles; their main differences are the materials used in the fuel management system and modifications to the engine calibration system. There are no switches to flip, no additional fuelling tanks or fuel dispensing systems required, and E85 can be handled in the same manner as gasoline. Usually in order to adapt an engine to run on higher bioethanol blends, changes to the compression ratio and spark timing are needed to obtain maximum benefits. Also, to refit an engine which will be fuelled with pure ethanol, larger carburettor jets, which are about 30-40% larger by area, have to be installed. Additionally, below temperatures of 13 °C, ethanol engines also need a cold-starting system to maximize combustion and minimise uncombusted non-vaporized ethanol.

A high-proportion bioethanol blend E85 (85% ethanol and 15% gasoline) can be used in FFVs with modified engines that are able to run on either E85 or gasoline, or any mixture of the bioethanol and gasoline. The gasoline in E85 makes the vehicle easier to start in cold weather and increases the vehicle's range. FFVs have been adopted in countries which have widely adopted an ethanol fuel economy. Brazil, Sweden and the US have a lot of FFVs ruining on E85. Many of the world's largest car producers, including General Motors, Ford, Peugeot, Volkswagen, Fiat and Renault now have a presence in the flex-fuel car market.

Table 6: Parameters for bioethanol in comparison with gasoline

	Density (kg/l)	**Viscosity (mm^2/s)**	**Flashpoint (°C)**	**Calorific value (@ 20°C MJ/kg)**	**Calorific value (MJ/l)**	**Octane number (RON)**	**Fuel equivalence (l)**
Gasoline	0.76	0.6	<21	42.7	32.45	92	1
Bioethanol	0.79	1.5	<21	26.8	21.17	>100	0.65

2-2.4 Economics of bioethanol

The cost of developing first generation bioethanol is dependent on factors that may vary with local conditions. The primary influences include the capital and operating costs of the plant, including the processing plant, services, feedstock and product storage, enzymes for the conversion process and buildings. The cost of feedstock production or purchasing varies among

countries and relies on land availability, agricultural productivity, labour costs, etc. The size and technology of the plant also vary greatly among countries. The by-product, which provides a secondary revenue stream to the bioethanol produced acts as an offset against the unit cost of bioethanol production. Biomass feedstock cost represents around 40% of the ethanol production cost. An important factor for reducing the cost of bioethanol production is to use larger industrial facilities rather than smaller ones and also to integrate into already existing industries, such as the sugar industry. By increasing the plant size, the investment per unit output of product falls off, reducing unit capital cost charges and conversion cost which often reduce profitability.



Self Assessment 2-2

1. Differentiate between sugar and starch feedstocks used for producing bioethanol.

2. Briefly discuss the bioethanol production process to a fellow student in the social sciences.

3. Following the ethanol production process described in sessions 2-2.1 and 2-2.2, sketch your own diagram to depict the bioethanol production route.

4. Given the same amount of petrol and bioethanol, which of these would you prefer to use in your engine, if the costs were the same and environmental implications were to be overlooked. Why would you make that preference?



Answer tips

1. Read session 2-1.1
2. Read session 2-2.2
3. Read the sessions mentioned and also study Figure 11.
4. Read session 2-2.3



Learning Track Activities

Unit Summary

1. In this unit, we have identified feedstock sources for the production of biodiesel and bioethanol. For biodiesel production, we have shown that the feedstock sources can be classified into five different groups and for bioethanol into two different groups.

2. We have discussed the detailed production process of biodiesel and bioethanol. For biodiesel, we have shown that the state of the feedstock is critical to the production process and could also determine pre-treatment methods and production costs. We have also shown that depending on the feedstock category that is used in producing bioethanol, the earlier stages in the production process differs. But once the fermentation stage is reached, the rest of the process follows the same path, irrespective of the feedstock category used.

3. We have discussed the uses and presented the major items that contribute to the cost of production of the two biofuels.

Key terms/ New Words in Unit

1. Bagasse

2. Free Fatty Acids

3. Microalgae

4. Degumming

5. Deacidification

6. Transesterification

(choose an appropriate activity and give detail instructions)

- **Review Question:** List all the feedstocks for the production of biodiesel and bioethanol that you have found in this unit which you are aware can be

cultivated in Ghana.

- **Discussion Question:** Given one (1) tonne each of sunflower send and sugarcane, discus the processes you would follow in order to produce biodiesel and bioethanol respectively.



Unit Assignments 2

1. Differentiate between the five feedstock classifications for the production of biodiesel.
2. What is distillation? Why is it an important step in the ethanol production process?
3. What does a catalyst contribute to the transesterification process?
4. What might call for pre-treatment in the transesterification process?

Unit 3

BIOGAS – FUNDAMENTAL THEORIES AND CONVERSION

Introduction

This unit discusses the basics in biogas production. The fundamental topics covered include feedstock sources, stages in the production process, types of biogas plants, application of biogas and basic economics.



Learning Objectives

After reading this unit you should be able to:

1. Identify feedstock sources that are suitable for the production of biogas
2. Discuss the stages in biogas production and identify digester types for biogas production
3. Discuss the applications of biogas
4. Discuss the components that contribute to the capital and operational cost of a biogas plant

Unit content

Session 1-3: Biogas Production

- 1-3.1 Introduction
- 1-3.2 Stages in biogas production
- 1-3.3 Types of biogas plants
- 1-3.4 Construction of biogas plant

Session 2-3 Application and Economics of Biogas

- 2-3.1 Biogas application technologies
- 2-3.2 Biogas production costs

SESSION 1-3: BIOGAS PRODUCTION

1-3.1 Introduction

Biogas can be defined as a gas composed principally of a mixture of methane (CH_4) and carbon dioxide (CO_2) produced by anaerobic digestion of biomass with a heating value of about 55%–60% that of natural gas. Each year, some 590 - 880 million tons of methane are released worldwide into the atmosphere through microbial activity. About 90 percent of the emitted methane is derived from biogenic sources, i.e., from the decomposition of biomass. The remainder is of fossil origin (e.g., petrochemical processes).

Biogas technology is generally accepted to be simple and readily available for use in rural, peri-urban and urban areas. It can be utilised for many purposes including lighting, heating and power generation at the individual and community levels. Biogas technology offers many benefits including:

- provision of sanitation option
- agriculture via manure (bio-solid fraction) and effluent (liquid fraction)
- job creation (employment of artisans for construction and appliance manufacturing).

Figure 12 illustrates the various applications, uses and benefits to be derived from biogas. Africa has suitable climatic characteristics (mean temperatures of 24 and 28 ^{o}C with little seasonal variation and annual rainfall above 1,500 mm) that are ideal for biogas production.

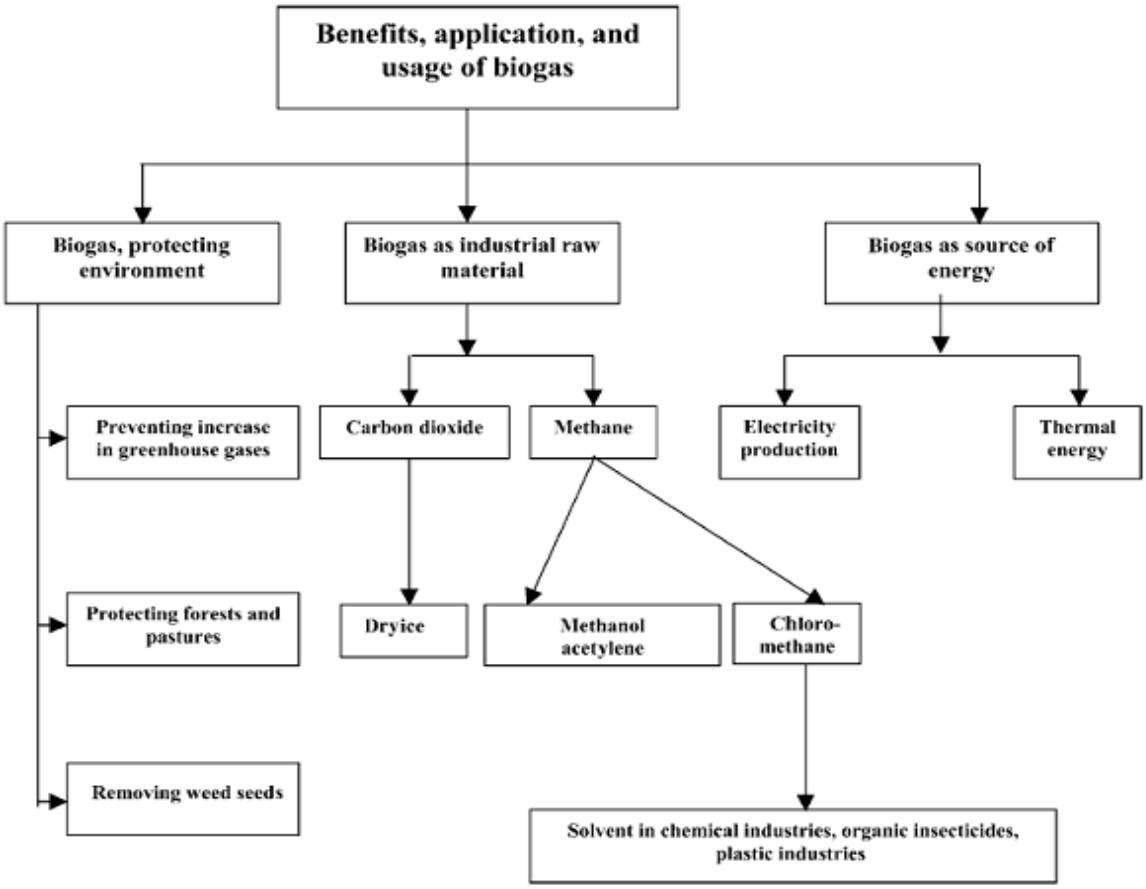


Figure 12: Benefits, applications and uses of biogas

1-3.2 Stages in biogas production

Biogas is produced by bacteria during digestion or fermentation of organic matter under airless condition (anaerobic process). The gas consists mainly of CH_4 and CO_2. Typically, biogas consists of about 60 % methane (CH_4) and 40 % carbon dioxide (CO_2). It also contains small proportions of other substances, including up to 1% hydrogen sulphide (H_2S). The water content is usually removed through water drains.

1-3.2.1 Biogas Feedstock

Many kinds of organic waste can be used in a biogas plant as feedstock as shown in Figure 13. A wide range of agricultural, industrial, and urban activities generate residue streams with high organic loadings that are suitable for anaerobic treatment. They include: animal dung, chicken droppings, human excreta or solid waste (from agricultural residues, domestic or municipal sources). The design of the biogas plant can be greatly influenced by the feedstock available. Table 7 shows biogas potentials of some types of feedstock. The state of the feedstock determines whether premixing has to be carried out to generate slurry for feeding the digesters.

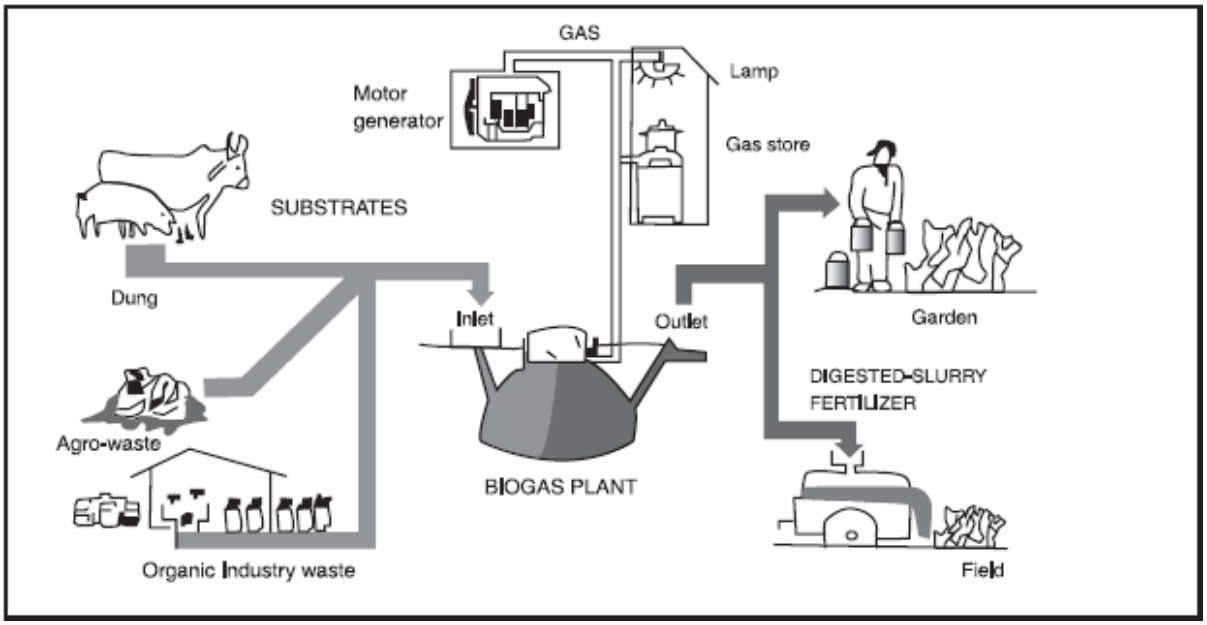


Figure 13: Schematic diagram of feedstock and unit operations in a typical biogas plant

1-3.2.2 Anaerobic digestion (AD)

Anaerobic digestion is a biochemical process during which complex organic matter is decomposed in absence of oxygen, by various types of anaerobic microorganisms. This process is common to many natural environments such as the stomach of ruminants.

In a biogas installation, the result of the anaerobic digestion process is the *biogas and the digestate.* If the substrate is a homogenous mixture of two or more feedstock types (e.g. animal slurries and organic wastes from food industries), the process is called "co–digestion" and is common to most biogas applications today.

Table 7: Potentials of Biogas Production from some types of Dung

Dung type	Potential Biogas Yield
Cattle (cows and buffalos)	0.023 - 0.040
Pig	0.040 - 0.059
Poultry (Chicken)	0.065-0.116
Human	0.020- 0.028
Pre-treated Crop Waste	0.037
Water Hyacinth	0.045

1-3.2.3 Stages in the biogas production

In a typical biogas plant, there are *three main stages* in the production of biogas. These stages are hydrolysis, acidogenesis and methanogenesis.

Hydrolysis

First of all, complex biomass material is decomposed by a heterogeneous set of microorganisms, not necessarily confined to anaerobic environments. These decompositions comprise hydrolysis of cellulosic material to simple glucose, using enzymes provided by the microorganisms as catalysts. Similarly, proteins are decomposed to amino acids and lipids to long-chain acids. The significant result of this first phase is that most of the biomass is now water soluble, and in a simpler chemical form, suited for the next process step. The chemical equation for the phase one process is shown in equation 7.

$$\mathbf{(C_6H_{10}O_5) + nH_2O \rightarrow nC_6H_{12}O_6} \quad \mathbf{(7)}$$

Acidogenesis

The second stage involves dehydrogenation (removing hydrogen atoms from the biomass material), such as changing glucose into acetic acid, carboxylation (removing carboxyl groups) of the amino acids, and breaking down the long-chain fatty acids into short-chain acids, again obtaining acetic acid as the final product. These reactions are fermentation reactions accomplished by a range of acidophilic (acid-forming) bacteria. Their optimum performance requires a pH environment in the range of 6−7 (slightly acid), but because the acids already formed will lower the pH of the solution, it is sometimes necessary to adjust the pH, for example, by adding lime. The net result of the phase two reaction is shown in equation 8.

$$nC_6H_{12}O_6 \rightarrow 3nCH_3COOH \quad \mathbf{(8)}$$

Methanogenesis

Finally, the third phase is the production of biogas (a mixture of methane and carbon dioxide) from acetic acid by a second set of fermentation reactions performed by methanogenic bacteria. These bacteria require a strictly anaerobic (oxygen-free) environment. Often all processes are made to take place in a single container, but separation of the processes into stages will allow greater efficiencies to be reached. The third phase takes of the order of weeks, the preceding phases of the order of hours or days, depending on the nature of the feedstock. The third phase reaction is represented in equation 9.

$$3nCH_3COOH \rightarrow 3nCO_2 + 3nCH_4 \quad \mathbf{(9)}$$

1-3.3 Types of biogas plants

The structure in which biogas is produced is referred to as a biogas digester. There are two main types of biogas digsters, the fixed-dome and floating-drum types. Like the names suggest, the main difference is in the gas holder (drum), whether it is fixed or floating.

1-3.3.1 Fixed-Dome plants

The fixed-dome plant consists of a digester with a fixed, non-movable gas holder, which sits on top of the digester (Figure 14). When the production of gas starts, the slurry is displaced into the compensation tank. Many models of this type exist, four of them schematically shown in Figure 15. Its advantages include:

- Relatively low construction costs
- Absence of moving parts and rusting steel parts
- If well-constructed, fixed-dome plants have a long life span
- The underground construction saves space and protects the digester from temperature changes (Figure 14)
- The construction provides opportunities for skilled local employment.

The disadvantages are:

- Frequent problems with the gas-tightness of the brickwork gas holder, where even a small crack in the upper brickwork can cause a large loss of biogas.
    - Fixed-dome plants are therefore recommended only where construction can be supervised by experienced biogas technicians.
- The gas pressure fluctuates substantially depending on the volume of the stored gas.
- Even though the underground construction protects against temperature extremes, digester temperatures are low.

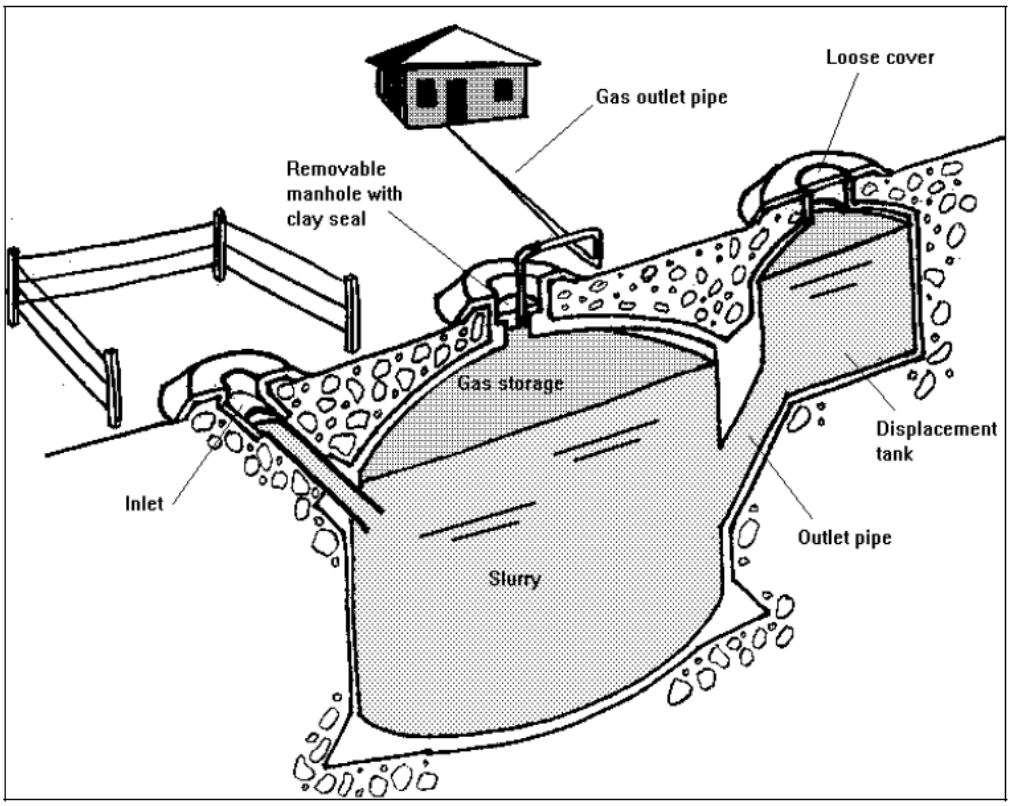
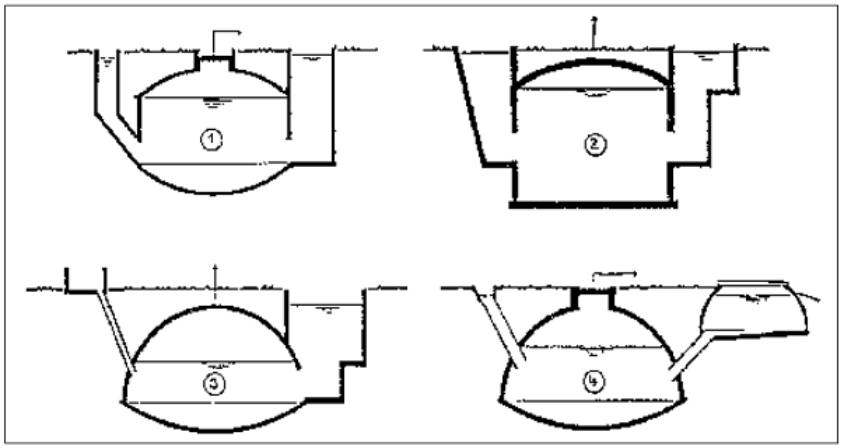


Figure 14: Fixed dome digester



Where 1= Biogas plant from Chengdou/China, 2 = Janata plant from India, 3 = Dheenbandhu plant of AFPRO from India and 4 = Modified BORDA plant from Cankuso in Burundi.

Figure 15: Different models of Fixed dome plants

1-3.3.1 Floating Drum plants

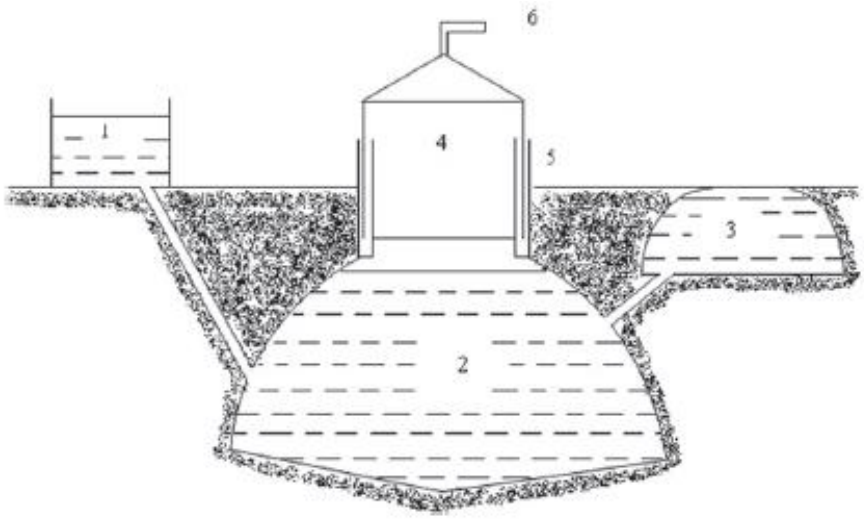
Floating-drum plants (Figure 16) consist of an underground or a stationary digester and a moving gasholder. The gasholder floats either directly on the fermentation slurry or in a water jacket of its own. The gas is collected in the gas drum, which rises or moves down, according to the amount of gas stored. Its advantages include:

- simplicity and ease of operation, as the volume of stored gas is directly visible to the user.
- The construction is relatively easy and mistakes do not lead to major problems in operation or gas yield.

The disadvantages are:

- high material costs of the steel drum
- susceptibility of steel parts to corrosion
- shorter life span than fixed-dome plants
- regular maintenance costs for the painting of the drum.

The conventional material for digester construction in Africa is masonry. However, these can be made of plastic materials and are appreciated for the benefits of being inexpensive to acquire compared to other biogas models.



1. Mixing tank with inlet pipe. 2. Digester. 3. Compensation tank. 4. Gasholder. 5. Water jacket. 6. Gas pipe.

Figure 16: Floating drum digester

1-3.4 Construction of biogas plants

The success or failure of any biogas plant mainly depends upon the quality of construction. The capacity of the plant is dependent on the availability of feedstock. Table 8 gives some relevant data about these six different sizes of biogas plants.

Table 8: Data for determining biogas plant capacity

Size of Plant(m^3)*	Daily Fresh Dung (kg)	Daily Water (Litres)	Approx. No. of Cattle Required
6	36	36	3 – 4
8	48	48	4 – 6
10	60	60	6 – 9
15	90	90	9 – 14

* Plant size is the sum of digester volume and gas storage and is based on a hydraulic retention time of 70 days

1-3.4.1 Design of biogas plant

- Factors such as the specifications of the site, including the nature and type of soil, must be taken into account when considering structural stability.
- The type of feedstock to be used in the digester determines the quantity of biogas produced and therefore affects the design capacities of both the digester and the storage facility.
- The availability and choice of the construction material affect both the capital cost and the operating cost
- The availability of highly trained workers with the requisite skills affects the longevity of the plant and minimizes the need to import labour

Construction

During the construction of digesters the following engineering issues must be taken into consideration:

a) The shape of the digester affects the susceptibility of the structure to external loading effects, especially for fixed-dome digesters, and therefore influences the structural integrity of the digester;
b) Feedstock loading affects the efficiency of the system;
c) Inlet and outlet configurations can affect the pressure dynamics of the digester by virtue of their locations, thereby affecting the operational efficiency.

1-3.4.2 Operational issues

Slurry mixing is an essential stage or unit of the biogas system that ensures uniform feedstock material and the attainment of the appropriate water content of the slurry. Otherwise, flow of the feed into the digester may be severely hampered and thereby reduce the production of biogas significantly. Conversely, too much of water content will lead to reduction of digestion efficiency due to decreased retention time and therefore lower gas production. Digesters can be fed on batch or continuous regime. The feedstock state, which is dependent on the distance

between feedstock source and digester, influences the feedstock stock significantly and the storage condition is equally influential in this regard.

1-3.4.3 Parameters and process optimization

The efficiency of the biogas production is affected by the following factors:

- suitable substrate temperature
- suitable pH value
- Hydraulic retention time
- substrate (feedstock) solid content and mixing
- air tightness
- Carbon : Nitrogen-ratio (C:N recommended to be between 30:1 and 10:1)
- necessary nutrients
- water content
- loading rate
- toxicity

Substrate temperature

The anaerobic digestion process can take place at different temperatures, divided into three temperature ranges shown in Table 9.

Table 9: Biogas production temperatures

Thermal stage	Process temperature	Minimum retention time
Psychrophilic	< 20 ^{o}C	70 to 80 days
Mesophilic	30 to 42 ^{o}C	30 to 40 days
Thermophilic	43 to 55 ^{o}C	15 to 20 days

NOTE: The operation temperature is chosen with consideration to the feedstock used and the necessary process temperature is usually provided by floor or wall heating systems, inside the digester.

pH Value

The pH-value is the measure of acidity/alkalinity of a solution (respectively of substrate mixture, in the case of AD), expressed in ppm (parts per million). The pH value of the AD substrate influences the growth of methanogenic microorganisms. Methane formation takes place within a relatively narrow pH interval, from about 5.5 to 8.5, with an optimum interval between 7.0 to 8.0. Optimum pH values differ for different thermal stages.

Hydraulic retention time (HRT)

The HRT is the average time interval when the substrate is kept inside the digester tank. HRT is correlated to the digester volume and the volume of substrate fed per time unit, according to the following equation:

$$HRT = \frac{V_d}{S_d}$$

HRT is the Hydraulic retention time

V_d – digester volume (m^3)

S_d – volume of substrate fed per unit time (m^3/d)

1-3.4.4 Digester Conditions

Digester conditions govern the type and rate of degradation of the feed material. However, the microbial content of digesters depends on feedstock composition. Rumen tends to have higher amylolytic microorganisms whilst dung has proteolytic organisms as predominant species. The bacterial action is inhibited by the presence of metal salts, penicillin, soluble sulphides, or ammonia in high concentrations. Hence these elements are avoided in digesters. Since the microbial activities are also pH dependent, generally, digesters are operated at pH range of pH 6.6 to 7.6.

1-3.4.5 Sizing a biogas plant and establishing plant parameters

The size of the biogas plant depends on the quantity, quality and kind of available biomass and on the digesting temperature. Important relations that are necessary to size a biogas plant and establish other plant parameters are presented in Table 10.

Table 10: Parameters for sizing a biogas plant

Parameter	**Relation**	**Unit**	**Definition of terms**
Digester volume, **V_d**	**V_d** = **S_d** × **HRT**	(m^3 = m^3/day × number of days)	S_d = Substrate input, HRT = retention time
Substrate input, S_d	**S_d** = biomass (**B**) + water (**W**)	(m^3/d)	B = biomass, W = water
Amount of biogas generated, G	$G = VS * Gy$	[m3/d = kg × m3/(d×kg)]	VS = Volatile solids content, Gy = specific gas yield

Specific gas production, Gp	$Gp = \frac{G}{V_d}$	[$(m^3/d) / m^3$]	See terms above
Digester loading, L_d (based on **Total** solid input)	$LdT = \frac{TS/d}{V_d}$	[$kg/(m^3\ d)$]	TS/d = daily total solids input
Digester loading, L_d (based on **Volatile** solid input)	$LdV = \frac{VS/d}{V_d}$	[$kg/(m^3\ d)$]	VS/d = daily volatile solids input

Sizing the digester

The size of the digester, i.e., is determined on the basis of the chosen retention time **HRT** and the daily substrate input quantity, $\mathbf{S_d}$. The retention time, in turn, is determined by the chosen/given digesting temperature whilst the substrate input depends on how much water has to be added to the substrate in order to arrive at a solids content of 4-8%. In most agricultural biogas plants, the mixing ratio for dung (cattle and / or pigs) and water (**B**:**W**) amounts to between 1:3 and 2:1.

Example 3.1

Calculate the size of a digester required to generate biogas if the daily input quantity is $7m^3$ and a retention time of 40 days.

Solution

Given parameters

Substrate input quantity, $S_d = 70\ m^3/day$

Retention time, RT = 40days

Digester Volume, $Vd = Sd * HRT$

$= 7 * 40$

$\rightarrow Vd = \mathbf{280m^3}$

Example 3.2

What size of a digester is required to generate a biogas if the daily input quantity is obtained from a mixture of 9 m^3 of water and 5 m^3 cow dung? Take the retention time to be 60 days.

Solution

Given parameters

Quantity of biomass, B=5 m^3/day

Quantity of water, W= 9 m^3/day

Retention time, HRT=60 days

Daily input quantity, $S_d = B + W$

$= 5 + 9$

$= 14m^3/day$

Digester Volume, $V_d = S_d * HRT$

$= 14 * 60$

$= \mathbf{840}\boldsymbol{m^3}$

Calculating the daily gas production, G

The amount of biogas generated each day **G (**m^3 gas/d), is calculated on the basis of the specific gas yield, **Gy** of the substrate and the daily substrate input $\mathbf{S_d}$. See Table 10 for formular.

Example 3.3

Estimate the amount of biogas generated each day for a substrate with a volatile solids content of 17kg and a specific gas yield of 20m^3/(d.kg).

Solution

Volatile solids content, VS = 17kg

Specific gas yield, Gy = 20m^3/(d.kg)

$Amount\ of\ biogas, G = VS * Gy$

= 17*20

= 340 m^3/d

Example 3.4

Calculate the volatile solids content of a substrate with a specific gas yield of 5.8m^3/(d.kg) if the daily biogas production is 69.6m^3.

Solution

Amount of biogas, G = 69.6 m^3/d

Specific gas yield, Gy= 5.8 m^3/(d.kg)

$$Volatile\ solids\ content, VS = \frac{G}{Gy}$$

$$= \frac{69.6}{5.7}$$

$$= \mathbf{12kg}$$

1. the weight of the moist mass, B

G = **B** × **Gy** (moist mass) [m3/d = kg × m^3/(d.kg)]

2. Standard gas-yield values per livestock unit, LSU

G = number of **LSU** × **Gy** (species) [m^3/d = number× m^3/(d×number)]

Establishing the plant parameters

The degree of safe-sizing certainty can be increased by defining a number of plant parameters:

Specific gas production Gp

This is the daily gas generation rate per m^3 digester volume, **Vd.** It is calculated according to the equation below:

$$Gp = \frac{G}{V_d} \qquad [\ (m^3/d)\ /\ m^3\ ]$$

Example 3.5

Estimate the specific gas production, Gp if the daily gas production is 432m^3 and a digester volume of 16m^3.

Solution

Amount of gas, G = 432m^3/day

Digester volume, V_d = 16m^3

$$Specific\ gas\ production, Gp = \frac{G}{V_d}$$

$$= \frac{432}{16}$$

$$= \mathbf{27\ (m^3/d)\ /\ m^3}$$

Digester loading, L_d

The digester loading **L_d** is calculated from the daily total solids input, **TS/d** or the daily volatile solids input **VS/d** and the digester volume **V_d**.

Sizing the gasholder

The size of the gasholder, i.e. the gasholder volume **Vg**, depends on the relative rates of gas generation and gas consumption. The gasholder must be designed to:

- cover the peak consumption rate and
- hold the gas produced during the longest zero-consumption period



Self Assessment 1-3

1. A biogas digester of size 82m^3 has a mixture of rice straw and water as its feedstock. The ratio of the rice straw to water is 1:2. If 1.5m^3/day of rice straw is used,
a). find the amount of water in the mixture
b). estimate the retention time.

2. Calculate the daily biogas production of a substrate whose weight of the moist mass is 33kg with a specific gas yield of 7m^3/(d.kg).

3. What weight of moist mass of substrate is required to generate a gas of 336m^3/day if the specific gas yield is 8m^3/(d.kg)?



Answer tips

1. a). 3 m^3/day b). 18 days
2. 231 m^3
3. 42 kg

SESSION 2-3: APPLICATIONS AND ECONOMICS OF BIOGAS

2-3.1 Biogas application technologies

Biogas usage has proven to be good in replacing the other forms of biomass usage as heat and power source in cooking, lighting, operating dual fuel engines as in vehicles, refrigerators and boilers. It is used in replacing conventional power sources such as electricity by burning as fuel source in generators and also when upgraded, it can be used as a transportation fuel. A schematic diagram of the areas of application is shown in Figure 17.

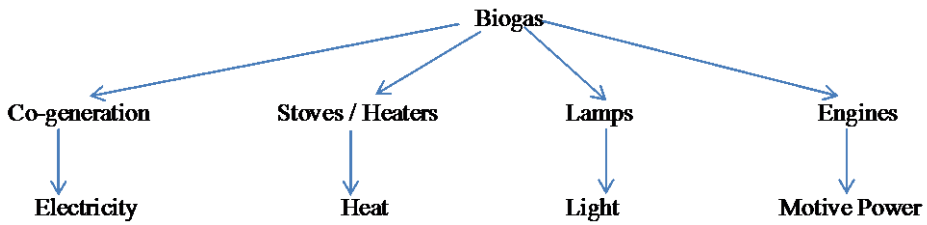


Figure 17: Application Areas of Biogas

2-3.1.1 Biogas Appliances

Biogas is a lean gas that can, in principle, be used like other fuel gases for households and industrial purposes such as:

- Gas cookers/stoves
- Industrial Burners
- Lamps
- Incubators
- Refrigerators
- Engines

Gas cookers/stoves

A cooker is more than just a burner. Biogas cookers and stoves must satisfy certain aesthetic and utility requirements, which can vary widely from region to region. Various basic requirements include:

- Simple and easy to operate
- Versatility e.g. for pots of various size, for cooking and boiling
- Easy to clean, acceptable cost and easy to repair
- Good burning properties, i.e., self-stabilizing flame and high efficiency

- Agreeable in appearance

Biogas-fuelled engines

A minimum of 10 m^3/day of biogas output is needed for it to be used for fuelling engines. For example, to generate 1 kWh of electricity with a generator, about 1.0 m^3 biogas is required. Small-scale systems are therefore not suitable as energy suppliers for engines and are treated elsewhere.

The following types of engines are, in principle, well-suited for operating on biogas:

- Four-stroke diesel engines
- Four-stroke spark-ignition engines
- Converting diesel engines
- Converting spark-ignition engines

2-3.2 Biogas production costs

The costs of establishing and running a methane digester are dependent on the specific type and size of the digester. These costs are:

a) Capital Cost
b) Operational Cost

2-3.2.1 Capital Cost

Capital cost of establishing a biogas plant includes interest and financing the plant, interest rate, loans, equity, etc., which are dependent on the size of the operation. To reduce capital cost, digesters may be built with local construction materials to local specifications. The capital costs include the land, excavation-work, construction of the digester and gasholder, the piping system, the gas utilization system, the dung storage system and other buildings, materials and . The construction costs comprise wages and material.

The capital costs of biogas plants are determined by the following factors:

- purchasing costs for land which is needed for the biogas plant and slurry storage;
- model of the biogas plant;
- size and dimensioning of the biogas unit
- amount and prices of material
- labor input and wages

Even though current production costs for local biogas plants are not readily available, the capital investment costs in Table 11 will serve as a guide and provide an idea as to the costs of biogas plants.

Table 11: Fixed Capital Investment Cost for Biogas Installations in Some African Countries

Plant location	Capacity (m^3)	Year built	Original cost	Original cost (normalised to 2004 US$)
Ethiopia	4	2000	554 US$	618
Nigeria	10	2005	429 100 Naira	3565
South Africa	10	2001	20 000 Rand	2541
Rwanda	16	2004	2, 500 US$	2500
Zimbabwe	16	2004	2, 212, 804 Zim $	3173
Ghana	20	2000	7, 974 US$	8901

2-3.2.1 Operating Cost

Acquisition of raw materials, water for mixing materials, feeding and operations of the plant, preventative and on-going maintenance, supervision, storage and disposal of the slurry, gas distribution and utilization, and administration are all different operating costs associated with running a biogas plant.

The operation and maintenance costs consist of wage and material cost for:

- acquisition (purchase, collection and transportation) of the substrate
- water supply for cleaning the stable and mixing the substrate
- feeding and operating of the plant
- supervision, maintenance and repair of the plant
- storage and application of the biofertilizer
- gas distribution and utilization

The running costs of a biogas plant are just as important as the construction costs, and normally amount to no more than 4% of initial biogas plant cost per year.



Self Assessment 2-3

1. In which ways can we apply biogas?

2. What are the components that constitute capital cost and operating costs of a biogas plant?

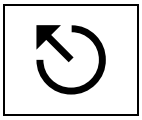


Answer tips

1. Read session 2-3.1
2. Read session 2-3.2



Learning Track Activities



Unit Summary

1. In this unit, we have identified the feedstock sources that are ideal for the production of biogas, including potential biogas yield from some of the feedstocks.

2. We have discussed the stages in biogas production and identified two main types of biogas digesters, the fixed dome type and the floating drum type.

3. We have discussed the applications of biogas.

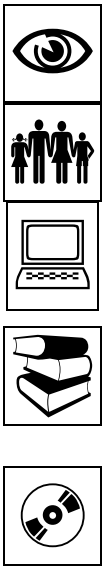
4. We also identified and discussed briefly, the components that contribute to the cost of establishing and managing a biogas plant.



Key terms/ New Words in Unit

1. Methane
2. Anaerobic digestion
3. Digestate
4. Hydrolysis

5. Acidogenesis
6. Methanogenesis
7. Substrate
8. Hydraulic retention time
9. Psychrophilic
10. Mesophilic
11. Thermophilic



(choose an appropriate activity and give detail instructions)

- **Review Question:** [insert details of review question]
- **Discussion Question:** Discuss the production of biogas, identifying the important stages and parameters in the production process.
- **Web Activity:** [insert details of web Activity or online v-classroom]
- **Reading:** [insert details of literature for reading]
- **Interactive CD:** [insert details of interactive CD]



Unit Assignments 3

[insert here details of Assignments]

Unit 4

SOLAR ENERGY - FUNDAMENTAL THEORIES AND CONVERSION

Introduction

This unit covers solar energy, which is basically energy produced directly by the sun and collected elsewhere, normally the Earth. A step by step description of generating energy from the sun is discussed, including the equipment needs for each of the stages.



Learning Objectives

After reading this unit you should be able to:

1. Explain how energy from the sun is converted to electricity and heat using solar photovoltaic and solar thermal applications
2. Discuss the applications of solar photovoltaics

Unit content

Session 1-4: Overview of solar photovoltaics

1-4.1 Introduction to solar energy

1-4.2 Photovoltaic (PV) systems

Session 2-4: Solar thermal applications

SESSION 1-4: OVERVIEW OF SOLAR PHOTOVOLTAICS

1-4.1 Introduction to solar energy

Solar energy is quite simply the energy produced directly by the sun and collected elsewhere, normally the Earth. Solar energy is an inexhaustible resource. The sun produces vast amounts of renewable solar energy that can be collected and converted into heat and electricity. It is important to note that solar energy is the most readily available source of energy and also the most important of the non-conventional sources of energy because it is non-polluting.

Solar energy has many uses. It can be used to provide/generate:

(i) Heat
(ii) Light
(iii) Electricity.

Solar energy received at the Earth's surface can be separated into two basic components: direct solar energy and diffuse solar energy. The radiation coming directly from the sun and received at the earth's surface on a unit area normal to the solar radiation is called direct solar radiation. Diffuse solar radiation is solar radiation reaching the Earth's surface after having been scattered from the direct solar beam by molecules or suspensoids in the atmosphere. The direct component of sunlight and the diffuse component falling together on a horizontal surface make up Global Solar Radiation. Insolation is a measure of solar radiation energy received on a given surface area in a given time. The measurements are taken using either a pyranometer (measuring global radiation) and/or a pyrheliometer (measuring direct radiation) and the unit is kW/m^2 or kWh/m^2 per day. Solar radiation varies throughout the day and is strongly dependant on location and local weather. A solar radiation map of Ghana is shown in Appendix 2. Appendix 3 shows Monthly Averages of Solar Irradiation (kWh/m^2-day) at 9 Synoptic Stations in Ghana.

For solar energy systems, if the insolation is absorbed and utilised without significant mechanical pumping and blowing, the solar system is said to be passive. If the solar heat is collected in a fluid, usually water or air, which is then moved by pumps or fans for use, the solar system is said to be active.

Solar energy technologies can be divided into two categories: solar thermal systems and solar photovoltaic systems. Solar thermal systems use the sunlight to produce directly usable thermal energy or heat for water or space heating, drying, cooking, cooling or electricity generation while solar photovoltaic is the direct conversion of solar radiation into electricity.

1-4.2 Photovoltaic (PV) systems

Photovoltaic is the field of technology and research related to the devices which directly convert sunlight into electricity. Photovoltaic comes from the words photo meaning light and volt, a

measurement of electricity. Sometimes photovoltaic cells are called PV cells or solar cells for short. Solar-powered toys and calculators all use solar cells to convert sunlight into electricity.

Solar cells are made of semiconductor materials, such as silicon. One of the properties of semiconductors that makes them most useful is that their conductivity may easily be modified by introducing impurities into their crystal lattice. Silicon is the second most common substance on Earth. Solar cells can supply energy to anything that is powered by batteries or electrical power. The solar cell is the elementary building block of the photovoltaic technology. A number of solar cells electrically connected to each other and mounted in a single support structure or frame is called a 'photovoltaic module'. Modules are designed to supply electricity at a certain voltage, such as a common 12 volt system. The current produced is directly dependent on the intensity of light reaching the module. Several modules can be wired together to form an array as shown in Figure 18. A typical solar panel or photovoltaic module (Figure 19) produces direct-current electricity. They can be connected in both series and parallel electrical arrangements to produce any required voltage and current combination. The solar panel can be used as a component of a larger photovoltaic system to generate and supply electricity in commercial and residential applications.

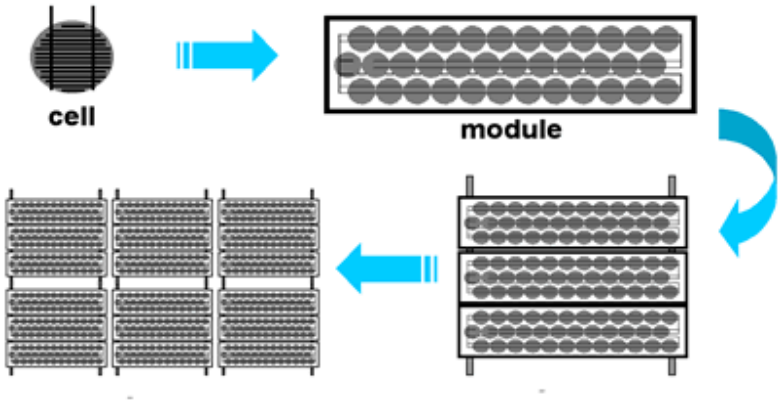


Figure 18: Cell, Module and Array



Figure 19: Solar panel

Because a single solar panel can produce only a limited amount of power, many installations contain several panels. A photovoltaic system typically includes an array of solar panels, an inverter, and sometimes a battery and interconnection wiring. Electricity is produced when sunlight strikes the solar cell, causing the electrons to move around. The action of the electrons starts an electric current. The conversion of sunlight into electricity takes place silently and instantly. There are no mechanical parts to wear out. Figure 20 shows a simple solar system set-up and application.

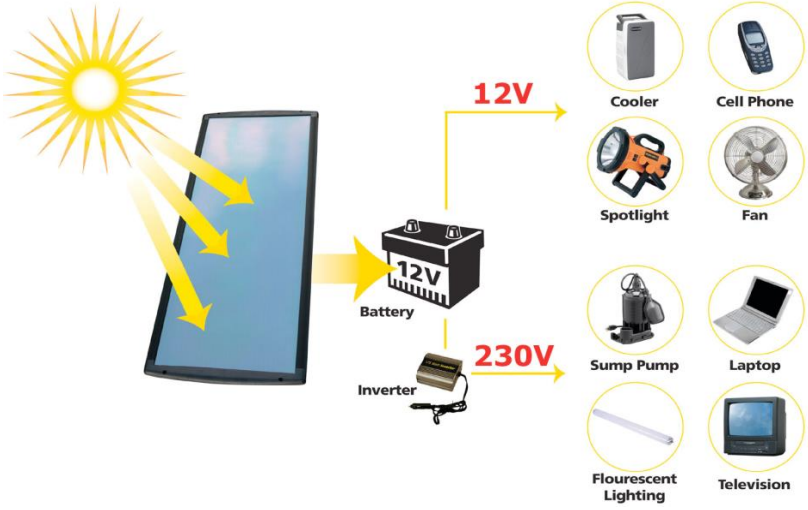


Figure 20: Application of Solar Panels

1-4.2.1 Systems and Components

A typical solar system for power supply to buildings consists of a solar generator for power generation, a battery to store this energy, a charge controller to operate the battery in a suitable way, the appliances and sometimes a power conditioning unit such as an inverter as shown in Figure 21. Besides these technical components, the users always represent an important interactive part of the system which should not be forgotten.

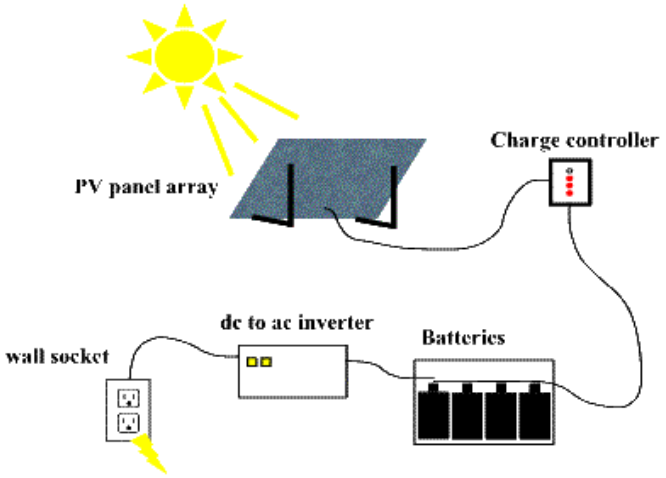


Figure 21: Block diagram of a PV system

The generator

The heart of a photovoltaic system is the generator itself. The generator is the connection of any number of solar modules (also called panels). A standard module consists of 36 solar cells. Modules have a lifetime of more than 20 years. In the field, modules with mono-crystalline and poly-crystalline cells are applied almost exclusively. Solar modules are rated in peak watts (Wp) according to their output under optimal outdoor conditions. Thus, a 50 Wp module can be expected to supply 50 W of power in full sunshine. The performance is reduced by high temperatures. The modules are connected in series and/or in parallel depending on the system requirements. A serial connection increases the voltage; a parallel connection increases the current.

Example 4.1

An Engineer in a solar company is required to produce a 24 V system with a nominal power rating of 300Wp. How should the modules be connected if they are available in 50Wp? Each module also has a voltage of 12V.

Solution

The 24V system can be obtained from the combination of two modules connected in series.

$$\rightarrow 2 \times 12V = 24V$$

To obtain a 300Wp from the two modules, the modules should be connected in three parallel strings.

$$\rightarrow P = 3 \times 2 \times 50$$

$$\rightarrow P = 300W$$

The battery

Batteries are applied to store energy for times of little or no sunlight. They are the most sensitive component of any photovoltaic system and require a certain amount of care. They are not easygoing like batteries in cars. Only lead-acid batteries are used in the field. They come in different types which include:

- Automotive batteries
- Solar batteries

The service life of batteries, which is expressed in cycles of charging and discharging periods, increases with:

- the quality of the battery,
- the quality of the charge controller,
- low temperature,
- a low depth of discharge and

- careful operation and maintenance.

The depth of discharge (DoD) is that portion of the battery capacity which is removed during discharging. The capacity is indicated in ampere hours (Ah) or sometimes in days of autonomy. This is the number of days a fully charged battery can satisfy the energy demand. As a rule of thumb, reducing the DoD by half will double the number of cycles. As with modules, batteries are connected in series for higher voltage levels or in parallel to increase the capacity. Batteries that are connected should be identical in type, capacity, age and electrolyte density. Parallel connection should be avoided if possible.

Example 4.2

For a 12V, 60Ah battery at a 75% depth of discharge, calculate:

i. the remaining battery capacity
ii. the days of autonomy, Aut if the daily energy demand is 270Wh.

Solution

$Energy\ of\ battery, E_T = 12V \times 60Ah$

$= 720Wh$

At 75% DoD,

Energy removed during discharging, $E_D = 0.75 \times 720Wh$

$$E_D = 540Wh$$

Therefore, Remaining Energy, $E_R = 720 - 540$

$$= 180Wh$$

i. Remaining Capacity, $C_{BAT,R} = \frac{180Wh}{12V} = \mathbf{15Ah}$

ii. Days of Autonomy, Aut $= \frac{Energy\ removed\ during\ discharging}{Daily\ Energy\ Demand}$

$$= \frac{540Wh}{270Wh} = \mathbf{2\ days}$$

The controller

Batteries have to be protected against overcharging and deep discharging by a charge controller. Charge controllers are also referred to as charge regulators and battery control units (BCU). During charging, the battery voltage increases. At a certain voltage, the battery starts 'gassing'. Gassing, the production of hydrogen and oxygen, reduces the life expectancy as well as the

required maintenance periods and should be avoided. A controller avoids excessive gassing by maintaining the voltage at the end-of-charge level of about 14.0 V, depending on the battery type and temperature. The end-of-charge level may be adapted to the battery temperature by means of a temperature sensor.

In contrast to charging, the voltage of a battery being discharged declines. In a deep-discharged state, chemical processes occur which will reduce the battery lifetime quickly. To protect the battery, the loads have to be cut off at a certain voltage. High-quality batteries are less sensitive to deep discharging. Controllers are designed for maximum input and output currents. Safety measures may include protection against short circuit, overvoltage, overloads, reverse polarity, wrong connection and environmental hazards. Controllers should indicate the state of charge and a load cut-off.

The inverter

Solar modules and batteries operate with DC. The technical electricity sector, however, is based on AC. Many electrical appliances, devices and accessories are only available for AC. An inverter transforms low-voltage DC supplied by a solar system into high-voltage AC. The input of an inverter is designed for 12 V (24 V, 48 V, etc.), depending on the type. At the output it produces 110 or 240 V AC. Inverters are designed for stand-alone as well as for grid-connected systems.

Inverters are available with a wide variety of different qualities and features. Not surprisingly, low-price inverters are generally of poor quality and not very suitable for solar operation. Inverters are designed for a maximum load. They provide a certain overload capacity for a short-term demand arising e.g. from the starting current of a motor. High-quality inverters have an efficiency of more than 90% between 10% and 100% rated power.

Wiring

Many system failures are caused by unsuitable or improperly installed cables, connectors, switches or sockets. Their current, voltage and power ratings should be in line with the system characteristics. Unlike AC, in DC systems switching the poles (+) and (-) will cause problems. The application of special DC fittings is recommended. Otherwise, high-quality AC items should be applied. The current ratings of AC-switches can be applied only for 12 V/ 24 V DC systems. In high-voltage DC systems, they have to be higher. The current drawn by an appliance or circuit can be calculated by dividing the power rating by the system voltage.

Like other electric components, accessories (particularly cables) cause losses. Since energy is expensive, these should be minimized. The power loss (in Watt) within a copper wired cable depends on the current, I (in Amperes), the cable length L (back and forth in metres) and the cross-section A (in square millimetres) of the wire as follows:

$$\Delta P = \frac{0.018 \times I^2 \times L}{A} \qquad (10)$$

Example 4.3

What current will be drawn by a 1kW inverter in a 24V system?

Solution

$$P = IV$$

$$I = \frac{P}{V} = \frac{1000}{24} = \mathbf{42A}$$

Example 4.4

Calculate the power loss within a cable for a generator current of 5A and a cross-sectional area of 1.5mm^2 with a length of 10m (one-way).

Solution

$$Power\ loss, \Delta P = \frac{0.018 \times (5A)^2 \times (2 \times 10)}{1.5mm^2}$$

$$= \mathbf{6W}$$

NB: The dimension of the cross-sectional area should be in square millimetres.. L is multiplied by 2 because it was one-way.

Losses can be reduced by increasing the cross-section of wires or by higher voltage levels.

1-4.2.2 Applications of PV systems

Appliances that can be powered by solar energy include:

- Lighting (lamps)
- Audio and video devices (egs. TV, receivers or video recorders)
- Refrigerators
- Water pumps, etc.

Water pumps

Some tens of thousands of pumps powered by solar generators are currently in use worldwide. A variety of pumps for low water demand are available on the market. These include single-stage centrifugal pumps for lifting heads <10m and diaphragm pumps for larger heads, both DC- and AC-operated. They are surface-mounted, floating or submerged pumps. DC pumps with brushes need regular replacements.

The energy requirement (E) for a daily water demand (Q), a pumping head (h, water level to tank inlet or pipe outlet) and a pump efficiency (η = 0.2 – 0.4, depending on type and pumping head) is

$$E\ (kWh) = \frac{\eta \times h(m) \times\ Q\left(\frac{m^3}{d}\right)}{367} \quad (11)$$

Sizing the system

System sizing means adapting the supply to the demand. A system that is too small will not satisfy the demand, and a system that is too big will be a waste of money. Supply (insolation) and demand are both variable throughout the year. Matching both variables is only possible to a certain extent. With every solar system there will be times when there is energy in excess and times when it is scarce. This is no different from generator sets. Due to periods of maintenance, scarce fuel supply, breakdowns, etc., there are also shut-down times. With proper sizing and a battery autonomy of 3 days, energy shortages can be limited to about 1%; that corresponds to 4 days in a year. In sizing a system, the following steps are important.

1. **Calculating the demand -** A list must be made of all appliances needed with their respective power requirements. A decision is then made on how many hours they will be operated. The demand is then calculated as E_{LOAD}.
2. **Sizing the generator -** Insolation records are taken from the nearest meteorological station(s). Insolation levels are generally not subject to much local variation. Next a design insolation (G_{DESIGN}) which is the worst insolation during those times of the year that the building is in full-time operation is chosen.

The generator has to cover the energy consumption (E_{LOAD}) plus the system losses. The system losses are calculated by the efficiencies (η) of the components. The Energy required,

$$E_{GEN} \approx \frac{E_{LOAD}}{0.85 \times \eta_{BAT} \times \eta_{INV}} \quad (12)$$

The battery efficiency, η_{BAT}, depends on the battery type.

η_{INV} is the mean operating efficiency of the inverter (not the maximum or rated efficiency) and is applicable in AC systems only. It may be as good as 0.9 (high-quality, small power variations) or as poor as 0.5 (low-quality, continuous low partial load).

Example 4.5

An AC system has a solar battery of efficiency 0.8 and an inverter of efficiency, 0.85. If the energy demand is 4.0kWh/d, calculate the energy required to be supplied by the generator.

Solution

$$E_{GEN} = \frac{E_{LOAD}}{0.85 * \eta_{BAT} * \eta_{INV}} = \frac{4}{0.85 * 0.8 * 0.85}$$

$$= \mathbf{6.9kWh/d}$$

Now the required generator power can easily be determined:

$$P_{GEN} = \frac{E_{GEN}}{G_{DESIGN}} \tag{13}$$

Where G_{DESIGN} is the design insolation.

3. **Sizing the battery -** Battery sizing has to take into consideration that only part of the rated capacity is available. This is the allowable depth of discharge (DoD).

   The battery capacity is calculated as:

$$Battery\ capacity, C_{BAT} = \frac{E_{LOAD} \times Aut \times 1000}{\eta_{INV} \times DoD \times U_{BAT}} \tag{14}$$

Example 4.6

Calculate the capacity for a 60V (U_{BAT}) solar battery with a DoD of 80%, an energy demand of 4.6kWh/d, a good inverter ($\eta_{INV} = 0.85$) and an Autonomy, Aut of 3 days.

Solution

$$Battery\ capacity, C_{BAT} = \frac{E_{LOAD} \times Aut \times 1000}{\eta_{INV} \times DoD \times U_{BAT}}$$

$$= \frac{4.6 \times 1000 \times 3}{0.85 \times 0.8 \times 60} = \mathbf{338Ah}$$

4. **Sizing the inverter -** The inverter can be sized to operate all loads at the same time. It is very unlikely, however, that all loads are in operation at the same time. Generally it is adequate if the inverter output matches the sum of the power ratings of all high-priority loads. Furthermore, the overload capacity of the inverter should be sufficient to start every single appliance (motors, etc.).

Installation

Mounting the generator

- The output of the generator can be maximized by choosing an optimal orientation. As a rule of thumb, the module should be inclined at the degree of latitude towards the equator.
- To allow rain water to run off, the tilt angle should be at least 15°. Close to the equator, the module should be turned in the direction to face the sun at noon during the month with the lowest insolation.

- Output will obviously be lower if the generator is in the shade of buildings or trees. Even if just a small part is shaded, the whole generator will be affected. Therefore, avoid any shading especially around noon.
- An installation on the ground is simpler and allows for regular cleaning and control. If theft and vandalism are a problem, the generator should be installed on the roof or on long poles. Space should be left for air circulation on the underside to reduce the temperatures. The modules have to be fixed securely.
- The mounting structure should be made of weather-proofed materials such as galvanized steel. Stainless steel or aluminum are also suitable, but expensive.



Self Assessment 1-4

1. What you understand by the term solar energy?

2. What are solar cells made of ?

3. For a 12V, 60Ah battery at a 80% depth of discharge, calculate: (i). the remaining battery capacity (ii). the days of autonomy, Aut if the daily energy demand is 192Wh.

4. Calculate the power loss within a cable for a generator current of 4A and a cross-sectional area of 1.7mm^2 with a length of 15m (one-way).



Answer tips

1. Read introduction to session 1-4.1
2. Read session 1-4.2
3. (i). 12 Ah (ii). 3 days
4. 5W

SESSION 2-4: SOLAR THERMAL APPLICATIONS

Solar Thermal Collectors convert solar radiation into heat. The main uses are water heating and space heating for homes and businesses. There are many different types, but they can be categorized as flat plate collectors and concentrating collectors. The flat-plate collector is the most important type of solar collector because it is simple in design, has no moving parts and requires little maintenance. It can be used for a variety of applications in which temperatures ranging from 40˚C to about 100˚C are required.

A collector essentially consists of the following components.

- a coated flat plate which absorbs solar radiation and transforms it into thermal energy
- an insulated storage tank used to reduce thermal losses of heated water glass or plastic
- cover to reduce upward thermal losses of the collector
- bottom insulation to reduce downward thermal losses
- tubes and channels for circulating water to collect thermal energy
- wooden or metallic frame to house the collector assembly

In any collection device, the principle usually followed is to expose a dark surface to solar radiation so that the radiation is absorbed. A part of the absorbed radiation is then transferred to a fluid like air or water. Solar thermal applications include:

- Solar Water Heating
- Solar Cooking
- Solar Drying
- Solar Water Distillation, etc.

These applications are illustrated in Figures 22 and 23. Solar water heater is basically a flat-plate collector in which heat transfer fluid is water. Figure 22 shows the working principle of solar water heater.

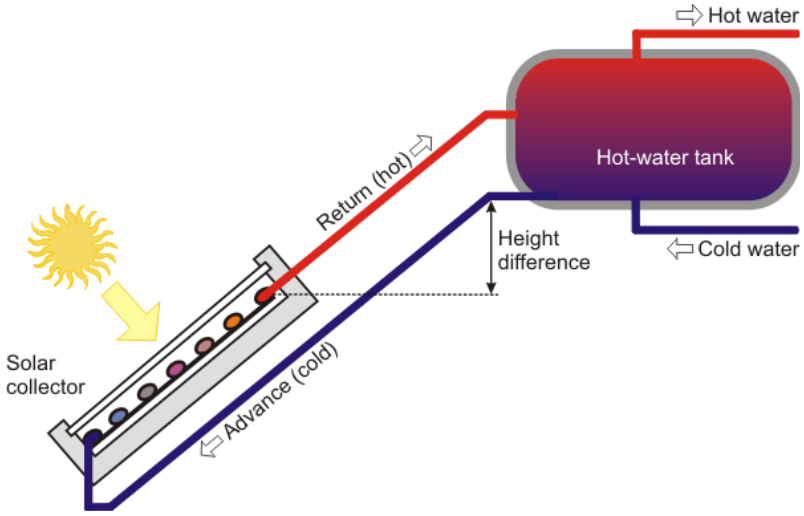


Figure 22: Solar water heating (Passive system)

The principle of thermosiphon is just like boiling the water. In a flat bed collector in cold water flows to the collector, it gets warm by sunshine and flows upward as it becomes lighter than cold water and stored in the tank which can be used directly.

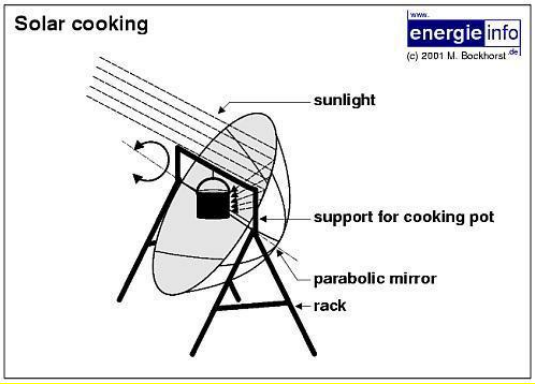


Figure 23: solar cooking

Self Assessment 2-4

1. What are the components of a solar collector?

2. List three applications of solar thermal power.

Answer tips

1. Read session 2-4
2. Read session 2-4

Learning Track Activities

Unit Summary

1. In this unit, we have discussed the conversion of energy from the sun into forms that can be applied for lighting and heating. We have discussed a stepwise approach to

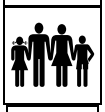
producing electricity from the sun and briefly at producing heat for cooking and other thermal application.

2. We have discussed the applications of solar energy in our everyday life.



Key terms/ New Words in Unit

1. Photovoltaic
2. Radiation
3. Insolation
4. Pyranometer
5. Charge controller
6. Depth of discharge
7. Inverter




(choose an appropriate activity and give detail instructions)

- **Review Question:** [insert details of review question]
- **Discussion Question:** Discuss the process of producing electricity from the sun given a standard solar panel.
- **Web Activity:** [insert details of web Activity or online v-classroom]
- **Reading:** [insert details of literature for reading]
- **Interactive CD:** [insert details of interactive CD]



Unit Assignments 4

What are the factors that determine the service life of batteries?

Calculate the capacity for a 60V (U_{BAT}) solar battery with a DoD of 75%, an energy demand of 4.5kWh/d, a good inverter $(\eta_{INV} = 0.85)$ and an Autonomy, Aut of 3 days.

An AC system has a solar battery of efficiency 0.8 and an inverter of efficiency, 0.85. If the energy demand is 5.0kWh/d, calculate the energy required to be supplied by the generator.

Unit 5

WIND ENERGY - FUNDAMENTAL THEORIES AND CONVERSION

Introduction

This unit discusses the production of energy from moving air. Global wind regimes are first presented, including land and sea breezes. The unit then discusses power in the wind, wind turbines, wind resource assessment and impacts of wind power development.



Learning Objectives

After reading this unit you should be able to:

1. Explain how wind is formed and the factors that are most important for wind energy production.
2. Describe the principal types of wind turbines.
3. Discuss wind resource assessment and impacts of wind power projects

Unit content

Session 1-5: Overview of wind energy

- 1-5.1 Introduction
- 1-5.2 Power in the wind

Session 2-5: Wind turbines and wind resource assessment

- 2-5.1 Description of wind turbines
- 2-5.2 Wind resource assessment

SESSION 1-5: OVERVIEW OF WIND ENERGY

1-5.1 Introduction

Wind is the flow of gases on a large scale and on earth, it consists of the bulk movement of air. Wind is caused by differences in pressure as a result of temperature differences. When a pressure gradient exists, air is accelerated from the point of higher pressure to that of lower pressure. Globally, the two major driving factors of large scale winds (atmospheric circulation) are the differential heating between the equator and the poles (difference in absorption of solar energy leading to buoyancy forces) and the rotation of the earth. The regions around the equator are heated more by the sun than the rest of the globe. Hot air is lighter than cold air and will rise into the sky at high altitude and will spread to the North and the South. If the globe did not rotate, the air would simply arrive at the North Pole and the South Pole, sink down, and return to the equator. This explains large scale wind (Global wind) circulation such as hurricanes illustrated in Figure 24.

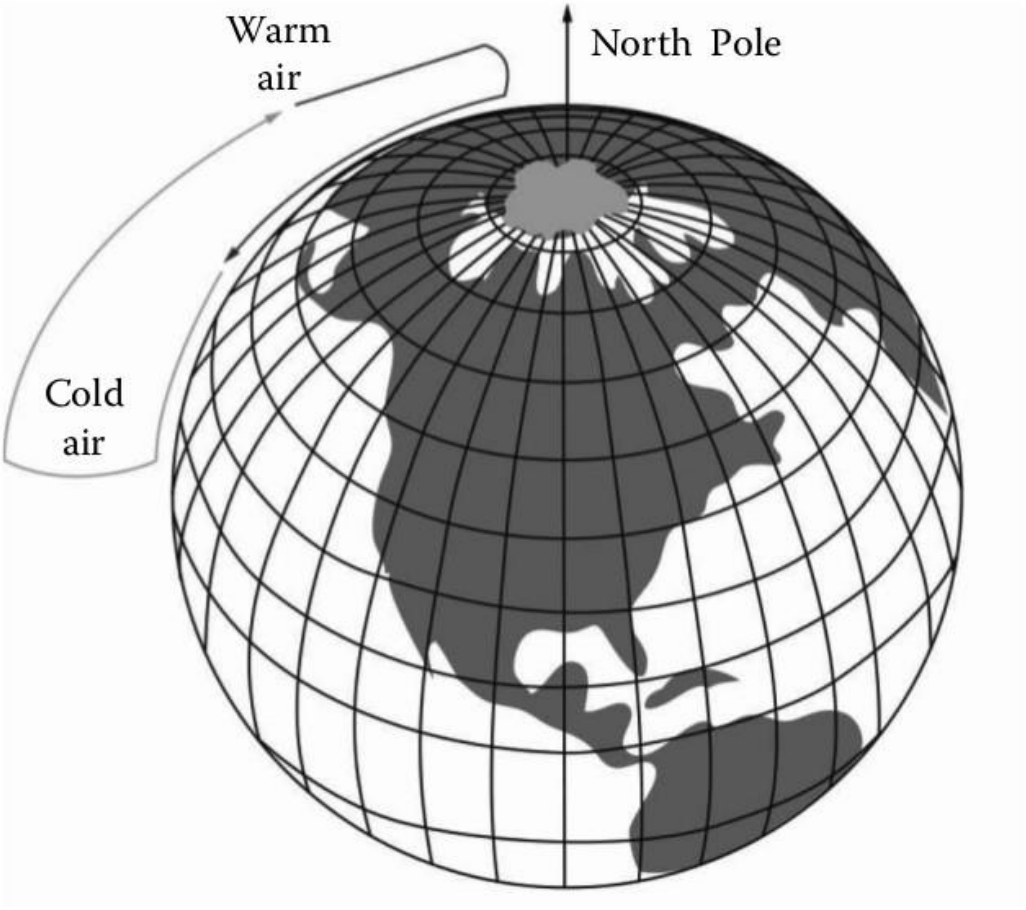


Figure 24: Global wind

For the purposes of utility, the interest has to do more with wind in much smaller areas called surface winds or local winds. Local winds are simply small-scale winds produced by a locally generated pressure gradient. They are caused either by topographic effects or variations in surface composition in the immediate area. The topographic effects may include the proportion of land and sea, the size of land masses and the presence of mountains or plains. The type of vegetation may also have a significant influence through its effects on the absorption or reflection of solar radiation, affecting temperatures and on humidity which in-turn affects pressure which is the main driving force of wind. Examples of local winds are sea, land, valley and mountain breezes.

1-5.1.1 Sea and Land Breezes

A sea breeze is a wind from the sea that develops over land near coasts. It is formed by increasing temperature differences between the land and water which create a pressure gradient over the land due to its relative warmth and forces higher pressure, cooler air from the sea to move inland. The sea is warmed by the sun more slowly because of water's greater specific heat compared to land. As the temperature of the surface of the land rises, the land heats the air above it by conduction. The warm air is less dense than the surrounding environment and so it rises (Figure 25). This causes a pressure gradient from the ocean to the land. The cooler air above the sea, now with higher sea level pressure, flows inland into the lower pressure, creating a cooler breeze near the coast.

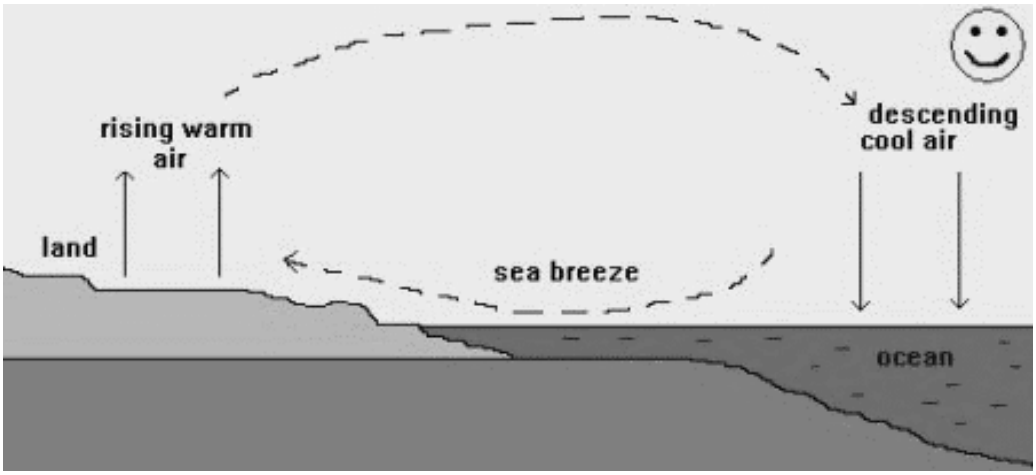


Figure 25: Sea breeze

At night, the land cools off quicker than the ocean due to differences in their specific heat values, which forces the dying of the daytime sea breeze. If the land cools below that of the adjacent sea surface temperature, the pressure over the water will be lower than that of the land, setting up a land breeze as long as the environmental surface wind pattern is not strong enough to oppose it (Figure 26). The land breeze at night generally has lower wind speeds, because the temperature

difference between land and sea is smaller at night. The land breeze will die once the land warms up again the next morning.

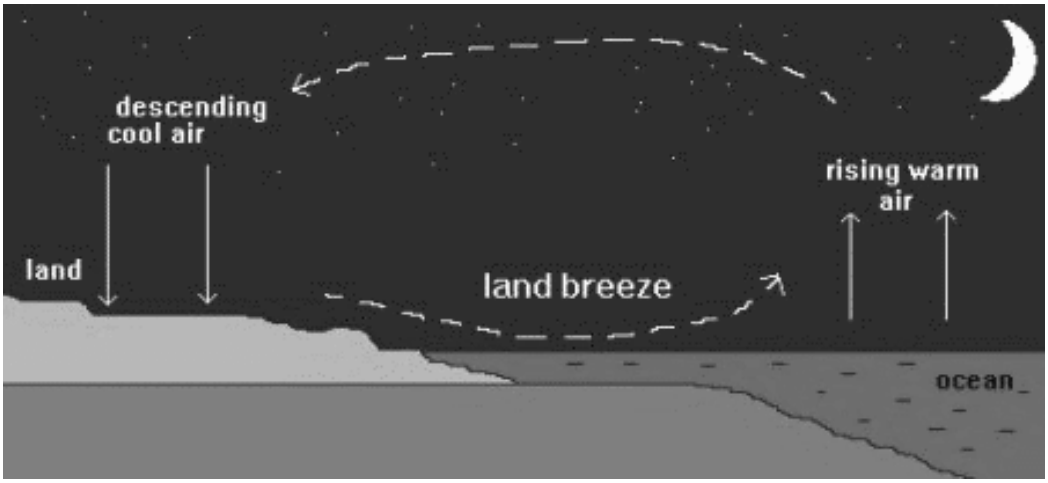


Figure 26: Land breeze

1-5.1.2 Valley and Mountain Breezes

Mountain and valley breezes form through a process similar to sea and land breezes. During daylight hours, the air along the slopes of the mountains is heated more intensely than the air over the valley floor. Because this warmer air is less dense, it glides up along the slope and generates a valley breeze. The occurrence of these daytime up-slope breezes can often be identified by the cumulus clouds that develop on adjacent mountain peaks. After sunset, the pattern may reverse. Rapid radiation cooling along the mountain slopes produces a layer of cooler air next to the ground. Because cool air is denser than warm air, it drains down-slope into the valley. This movement of air is called the mountain breeze.

1-5.2 Power in the wind

Wind energy is basically the kinetic energy of the air in motion. The moving molecules of air have kinetic energy, so locally the amount of air molecules moving across some area during some time period determines the power. This area is the area perpendicular to the wind flow. A wind turbine converts the kinetic energy of the wind into a torque which acts on the turbine blades. The torque is then transferred through the mechanisms of the turbine to produce power (typically electricity). The amount of energy which the wind transfers to the turbine rotor depends on the density of air, the rotor area and wind speed. The heavier the wind, the more energy can be extracted by the wind turbine. Mathematically, the power available in the wind to a turbine is given by the expressions in equation below.

$$P = \frac{1}{2}\dot{m}V^2 = \frac{1}{2}(\rho AV)V^2 = \frac{1}{2}\rho AV^3 \qquad (15)$$

Where:

P is power (W), ρ is density (kg/m^3), A is area (m^2) and V is velocity (m/s)

$$\frac{P}{A} = \frac{1}{2}\rho V^3 \text{ is referred to as the Wind Power Density} \quad (16)$$

Density

The kinetic energy of a moving body is proportional to its mass. The kinetic energy in the wind, thus, depends on the density of the air (it's mass per unit volume). In other words, the heavier the air, the more energy is received by the turbine.

Rotor Area

The rotor area determines how much energy a wind turbine is able to harvest from the wind. A typical 1 MW wind turbine has a rotor diameter of 54 m (a rotor area of some 2300 m^2)

Wind Speed

The wind speed is extremely important for the amount of energy a wind turbine can convert to electricity. The energy content of the wind varies with the cube of the average wind speed. A wind resource map of Ghana is shown in Appendix 4.

Example 5.1

A wind turbine of rotor diameter 30 m can only extract a maximum of 59% of the maximum available power in the wind. Assuming air density of 1.25 kg/m^3, calculate the maximum power the wind turbine can extract in an air stream of wind speed 15 m/s.

Solution

Maximum available power in the wind,

$$P = \frac{1}{2}\rho A V^3$$

ρ = 1.25 kg/m^3, V = 15 m/s, $A = \frac{\pi d^2}{4} = \frac{\pi \times 30^2}{4} = 225\pi$

$$P = \frac{1}{2} \times 1.25 \times 225\pi \times 15^3$$

$$P = 1491029.3\ W = 1.5\ MW$$

Maximum power that can be extracted = 59% of P

$$= \frac{59}{100} \times 1.5\ MW = 0.885 MW$$



Self Assessment 1-5

1. Differentiate between land breeze and sea breeze

2. Differentiate between valley breeze and mountain breeze

3. Define the wind power equation and explain the terms in the equation.

4. A wind turbine of rotor diameter 54 m can only extract a maximum of 55% of the maximum available power in the wind. Assuming air density of 1.25 kg/m^3, calculate the maximum power the wind turbine can extract in an air stream of wind speed 16 m/s.



Answer tips

1. Read session 1-5.1.1
2. Read session 1-5.1.2
3. Read 1-5.2
4. 3.2 MW

SESSION 2-5: WIND TURBINES AND WIND RESOURCE ASSESSMENT

2-5.1 Description of wind turbines

Wind turbines are the devices used to extract power from the wind. They depend on the same aerodynamic forces created by the wings of an aeroplane to cause rotation. If the mechanical energy is used directly by machinery, such as a pump, it is called a windmill. If the mechanical energy is converted to electricity, the machine is called wind turbine, wind generator, wind power unit, wind energy converter or aerogenerator. An anemometer that measures wind speed is usually part of most wind turbine control systems. Wind turbines can be divided into two

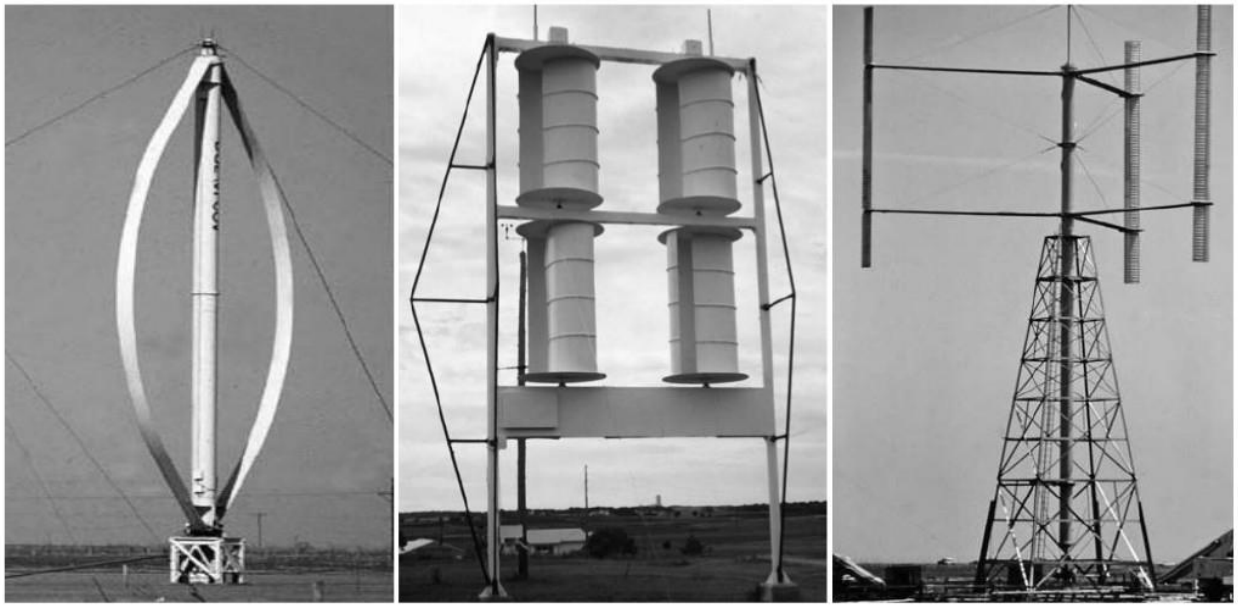
categories namely; vertical axis wind turbine (VAWT) and horizontal axis wind turbine (HAWT). The two categories are based on the axis around which the turbine blades rotate.

2-5.1.1 Vertical Axis Wind Turbine (VAWT)

A vertical axis wind turbine is one in which the axis of rotation of the shaft of the blade is in the vertical plane. These turbines have the main rotor and shaft arranged vertically. The generator and gearbox can be placed near the ground and this makes it more accessible for maintenance. Vertical axis wind turbines are of three types namely; Savonius type rotors, Darrieus type rotors and Giromill type rotors (Figure 27).

Advantages of VAWT

1. A massive tower structure is less frequently used, as the turbines are more frequently mounted with the lower bearing mounted near the ground.
2. They are omnidirectional.
3. They can be located near the ground, making it easier to maintain the moving parts.
4. They may be built at locations where taller structures are prohibited.



Darrieus type Savonius type Giromill type

Figure 27:Vertical axis wind turbines

Disadvantages of VAWT

1. They are usually not self-starting and often require an electric motor to get them starting.

2. They produce large torque ripples and cyclic stresses on the tower, which contributes to poor reliability.
3. Having rotors located close to the ground where wind speeds are lower due to wind shear, VAWT's may not produce as much energy at a given site as a horizontal axis wind turbine with the same height.
4. While vertical axis wind turbine parts are located close to the ground, they are also located under the weight of the structure above it, which can make changing parts nearly impossible without dismantling the structure if not properly designed.

2-5.1.2 Horizontal Axis Wind Turbine (HAWT)

A horizontal axis wind turbine has the axis of rotation of the shaft of the blade in the horizontal plane. This type of turbine has the main rotor shaft and electrical generator at the top of a tower and must be pointed into the wind. Small turbines are pointed by a simple wind vane, whiles large turbines generally use a wind sensor coupled with a servo motor. Since the tower produces turbulence behind it, the turbine is usually pointed upwind of the tower. The turbine blades are made stiff to prevent them from being pushed into the tower by high winds.

Advantages of HAWT

1. Most HAWT's have variable blade pitch which gives the turbine blades the optimum angle of attack. This makes it possible for the turbine to collect the maximum amount of wind energy for the time of the day and season.
2. They have high efficiency since the blades always move perpendicularly to the wind, receiving power through the whole rotation.
3. The tall tower base allows access to stronger winds due to wind shear.

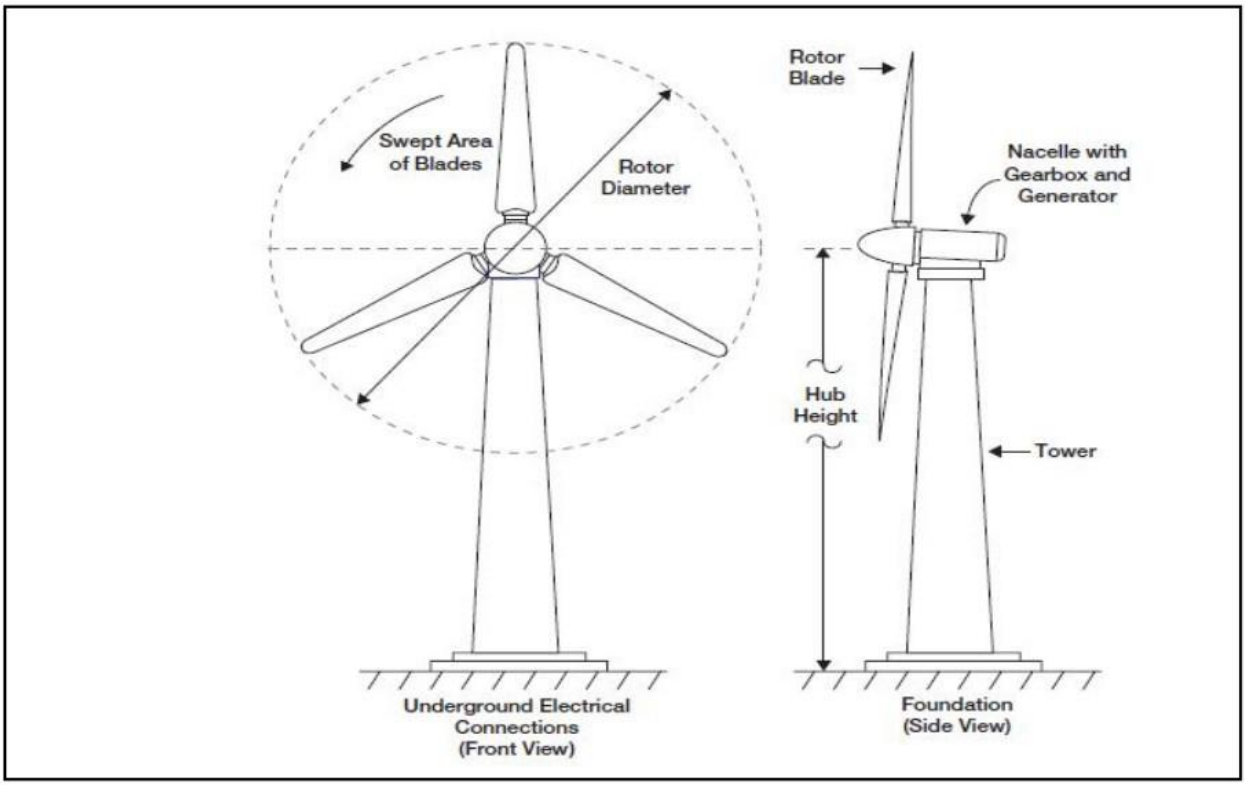


Figure 28: Schematic of a Horizontal Axis Wind Turbine

Disadvantages of HAWT

1. The tall towers and blades are difficult to transport.
2. Tall HAWT's are difficult to install, needing very tall and expensive cranes and skilled operators.
3. Massive tower construction is required to support the heavy blades, gearbox and generator.
4. HAWT's require an additional yaw control mechanism to turn the blades toward the wind.

Major Components of a HAWT

Horizontal axis wind turbines are mostly used for electricity applications. The major components of a typical modern horizontal axis wind turbine consist of the following (Figure 29);

- Rotor with three (3) blades which converts the energy in the wind into mechanical energy that is transferred to the rotor shaft
- Tall tower which supports the rotor high above the ground to capture the higher wind speeds.
- Gear box to match the slowly turning rotor shaft to the electric generator

- Solid foundation to prevent the wind turbine from blowing over in high winds and/or icy conditions
- Control system to start and stop the wind turbine and to monitor proper operation of the machinery

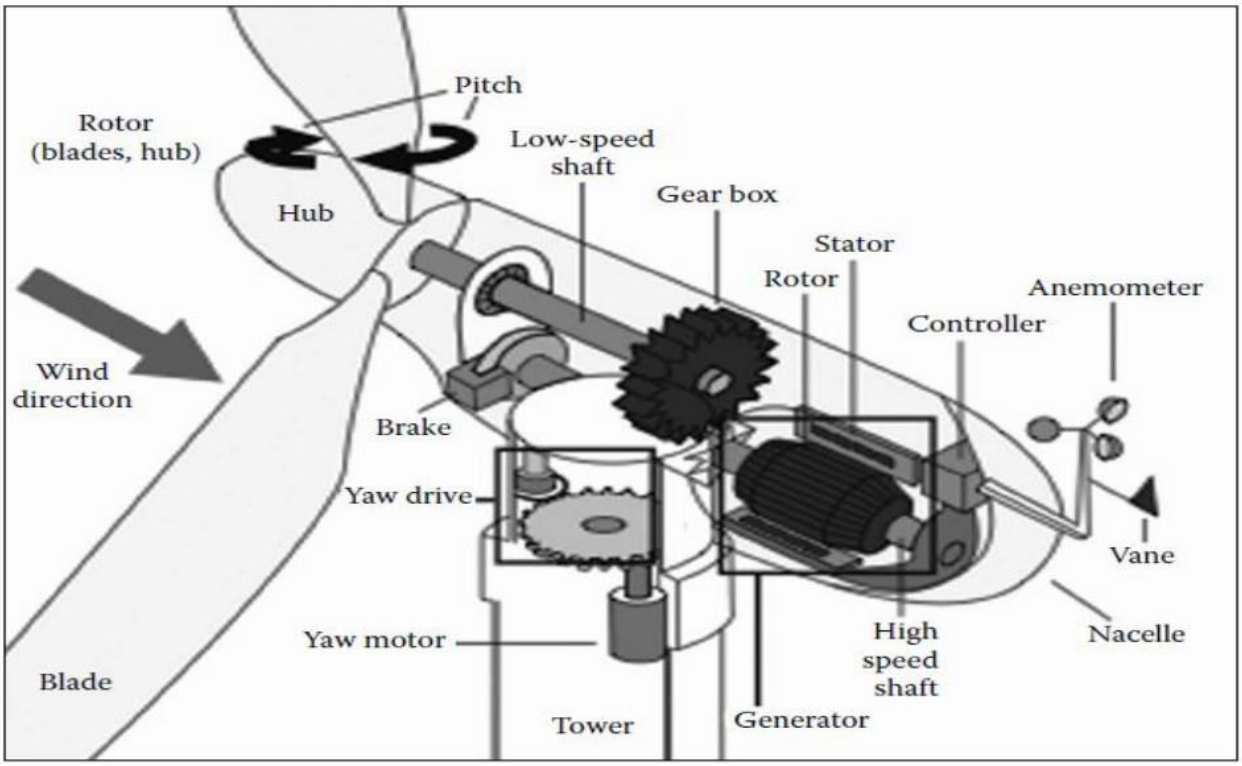


Figure 29: Components of a Horizontal Axis Wind Turbine

2-5.1.1 Technical Performance Characteristics of Wind Turbines

There are several technical parameters that characterize wind turbines, some of which are explained below;

Power Curve

A power curve, power versus wind speed, describes the normal operation of a wind turbine. It relates the power output of the turbine to its operational wind speeds as shown in Figure 30. The power curve of a wind turbine depicts the characteristics of a particular turbine such as the cut-in, cut-out and the rated wind speeds. A wind turbine produces its rated power output at its rated wind speed.

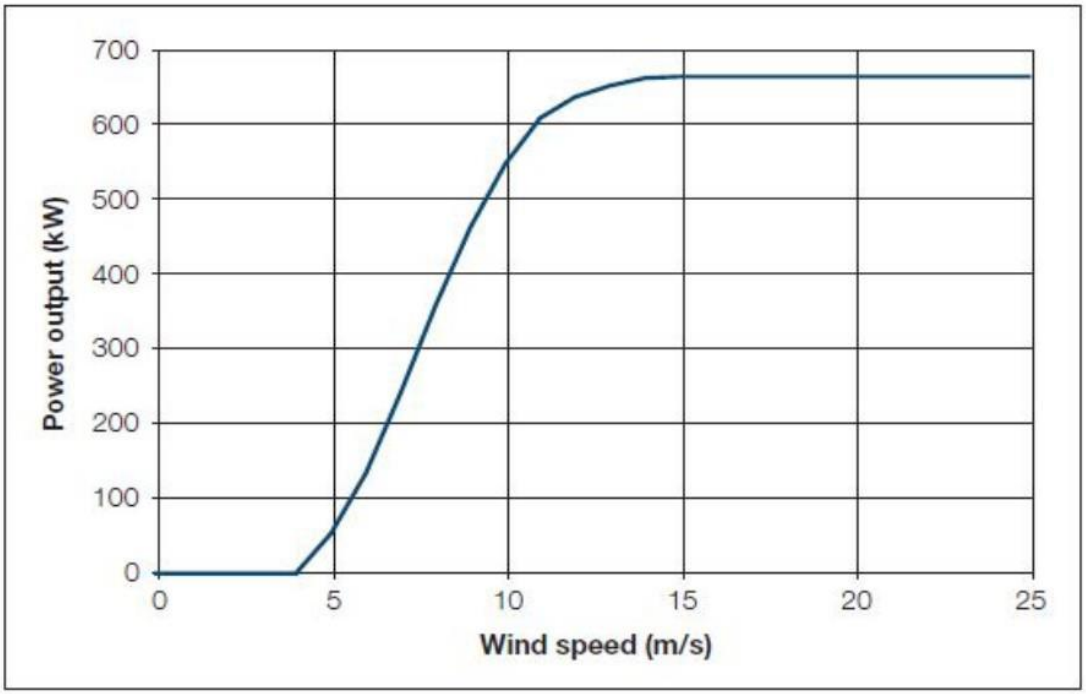


Figure 30: Vestas V47 – 650 kW Turbine Power Curve

Operational Speeds

The important operational wind speeds of a wind turbine are explained below:

- Start-up wind speed – the wind speed that will turn an unloaded rotor
- Cut-in wind speed – the minimum wind speed at which power is generated
- Rated wind speed – the wind speed at which the wind turbine is designed to run to produce rated power
- Cut-out wind speed – the wind speed at which the wind turbine is programmed to stop in order to prevent damage
- Maximum design wind speed – the wind speed above which damage could occur to the wind turbine

Power Rating

Wind turbines are typically defined by their power generating capacity. The rated power is produced at the rated wind speed of the turbine. The rated wind speed is usually a strong wind of about 15 m/s.

Capacity Factor

The capacity factor (C_p) of a wind turbine is not necessarily a technical performance characteristic as it depends also on the nature of the available wind resource. It relates the actual energy output to the maximum possible output if the machine were running at its rated power

during all of the 8760 hours of the year. It is defined as the ratio of the actual annual energy output to the maximum possible annual energy output.

$$C_p = \frac{Actual\ annual\ energy\ output}{Maximum\ possible\ energy\ output} \quad (17)$$

Example 5.2

A 50 MW wind farm produces 98 GWh of energy annually at 80 m hub height above ground level. Calculate the capacity factor.

Solution

Capacity factor,

$$C_p = \frac{Actual\ annual\ energy\ output}{Maximum\ possible\ energy\ output}$$

Actual annual energy output = 98 GWh

Maximum possible energy output = rated power * number of hours in the year

$$\text{Maximum possible energy output} = 50\text{MW} \times 8760$$

$$= 438000\ MWh = 438\ GWh$$

$$C_p = \frac{98\ GWh}{438\ Gwh} = 0.22$$

2-5.2 Wind resource assessment

The precise evaluation of the wind resource at a given site is one of the most important aspects in the development of a wind energy project as the available wind resource at the project site can dramatically impact the cost of wind energy production. Evaluation of wind resource data at a site is necessary since it is the inaugural step in assessing the wind power potential at that particular site of interest. There are two aspects of wind resource assessment namely:

1. Determination of the general wind power potential and
2. Determination of wind power potential and predicted energy production for wind farms. 10 m height above ground level (a.g.l.) is the standard for wind speed measurement for world meteorological data. Having a cubic relation with the power, the wind speed is the most critical data needed to appraise the power potential of a candidate site.

The wind is never steady at any site. It is influenced by the weather system, the local land terrain and the height above the ground surface. The wind speed varies by the minute, hour, day, season and year. Therefore, the annual mean wind speed needs to be averaged over ten or more years since such a long term average raises the confidence in assessing the energy capture potential of a site. However, long term measurements are expensive and most projects cannot wait that long. In such situations, about one year data is enough to predict the long term annual wind speed at the site under consideration. Since wind is driven by the sun and the seasons, the wind pattern generally repeats over the period of one year.

2-5.2.1 Wind Speed Measurement

The measurement of wind speeds is mostly done with a cup anemometer shown in Figure 31. The cup anemometer usually has three cups which rotate about the vertical axis to capture the wind. The number of revolutions per minute of the cups, being proportional to the wind speed is registered and processed electronically to give the value of the wind speed. Usually, the anemometer is fitted with a wind vane to determine the wind direction as well. Alternatively, an anemometer may be fitted with propellers instead of cups.



Figure 31: Cup Anemometer

The best way of measuring wind speed at a prospective wind turbine site is to fit an anemometer to the top of a mast which has the same height as the expected hub height of the wind turbine to be used. By this way, one avoids the uncertainty associated with extrapolating the wind speed to a different height. By fitting the anemometer to the top of a mast, one minimizes the disturbances of air flow from the mast itself.

2-5.2.2 Wind Shear

Wind speed increases with height above ground level due to the frictional drag of the ground, vegetation, and buildings. As such, any plans to harness the wind must take this variation into account. The power and log laws are used to estimate wind speeds typically between 20 – 120m height a.g.l. from that measured at about 10 m and below. The power law has been proven to give very good estimates of wind speeds so it is normally used.

2-5.2.3 Power Law

The power law for wind shear is defined as:

$$\frac{U_h}{U_{ho}} = \left(\frac{h}{h_o}\right)^{\alpha} \qquad (18)$$

Where:

h is the height at which speed U_h is to be estimated.

ho is the height at which speed U_{ho} was measured.

α is the wind shear exponent or the friction coefficient.

The wind shear exponent varies with height, time of day, season of the year, nature of the terrain, wind speed and temperature. Although several mathematical relationships have been developed for its estimate, the wind shear exponent is best determined empirically and is site specific. The average value of the wind shear exponent has been determined by many measurements around the world to be about one seventh (1/7).

2-5.2.4 Wind Power Density

The wind power density, measured in watts per squared metre (W/m^2), indicates how much energy is available at the site for conversion by a wind turbine. It is a useful way to evaluate the wind resource available at a potential site because it is convenient shorthand for how energetic the winds are during a period of time.

2-5.3 Potential environmental and social impacts of wind power development

Although Wind power is one of the prominent sources of energy globally, its development has environmental and social shortfalls, some of which are explained below.

2-5.3.1 Noise

The slicing of wind turbine blades through the air is the primary source of noise for wind farms. Although wind turbines built recently make substantially less noise than earlier models, noise from wind turbines is a potential problem if wind farms are sited too close to residences.

2-5.3.2 Bird Strikes

Since wind turbine blades generally work at high altitudes, birds can fly into fast-moving rotor blades and be killed.

2-5.3.3 Visual Impacts

The presence of wind turbines produces changes in views and skylines, and therefore, has a visual impact on the area in which they are cited. Visual impacts may be an especially important consideration if the turbines are to be located in pristine or wilderness areas. The access roads and power lines needed for grid-connected turbines can cause additional aesthetic impacts.

2-5.3.4 Interference with Telecommunications

Wind turbines normally interfere with television and other telecommunications signals, but these impacts seem to be typically localized to the vicinity of the wind farm.

2-5.3.5 Safety

Like any industry that includes moving machinery, safety is an issue with wind farms. Particular hazards from equipment failure include injury from equipment failures such as blades breaking off.



Self Assessment 2-5

1. If there is the need to install wind turbines in the middle (island) of dual carriage roads in Kumasi to supply power to traffic and street lights, which type of wind turbine will be most appropriate and why?

2. Differentiate between the cut-in and start up wind speed of a horizontal axis wind turbine.

3. Differentiate between the rated power and rated wind speed of a wind turbine.

4. Demonstrate with equations why wind speed is so important in wind resource assessment?



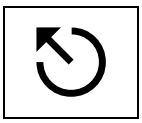
Answer tips

1. Read session 2-5.1.2

2. Read session 2-5.1.1
3. Read session 2-5.1.1
4. Read session 1-5.2.



Learning Track Activities

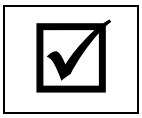


Unit Summary

1. In this unit, we have studies wind formation and discussed various wind regines such as land breeze, sea breeze, valley breeze and mountain breezes. We also discussed how wind power is determined from a given wind regime.

2. An overview of horizontal and vertical wind turbines was presented.

3. We have also discussed wind resource assessment and the impacts of wind energy.



Key terms/ New Words in Unit

1. Buoyancy forces
2. Land breeze
3. Sea breeze
4. Mountain breeze
5. Valley breeze
6. Rotor
7. Anemometer

(choose an appropriate activity and give detail instructions)

Review Question: 1). A wind turbine of rotor diameter 54 m can only extract a maximum of 55% of the maximum available power in the wind. Assuming air density of 1.25 kg/m^3, calculate the wind speed if the maximum power the wind turbine can extract in an air stream is 1.4 MW.

2). A 65 MW wind farm produces 150 GWh of energy annually at 80 m hub height above ground level. Calculate the capacity factor.

3). Assuming air density of 1.25 kg/m^3, calculate the power density of wind blowing at 10 m/s.

Discussion Question: Discuss why a community situated close to a wildlife reserve may reject the erection of wind turbines to produce electricity in the community.

Unit Assignments 5

The annual mean wind speeds measured at 20 m and 50 m above ground level at Anloga in the Volta Region of Ghana are 5.4 m/s and 6.7 m/s respectively.

i. Calculate the power law wind shear exponent to three decimal places.

ii. ii. Estimate the annual mean wind speed at 80 m above ground level with 20 m as the basis and using the calculated wind shear exponent value in "i" above.

Unit 6

APPLICATIONS OF RENEWABLE ENERGY TO AGRICULTURE IN GHANA

Introduction

[type here introduction of unit 6]



Learning Objectives

After reading this unit you should be able to:

1. Identify energy needs in agriculture that can be met by renewable energy
2. Discuss the applications of renewable energy specifically to agricultural activities

Unit content

Session 1-6: Overview of energy use in agriculture

- 1-6.1 Introduction
- 1-6.2 Agriculture in Ghana
- 1-6.3 Using energy to modernise agriculture in Ghana

Session 2-6: Renewable energy options for agriculture

- 2-6.1 Introduction
- 2-6.2 Solar dryers and pumps
- 2-6.3 Windmills
- 2-6.4 Biofuels and biogas

SESSION 1-6: OVERVIEW OF ENERGY USE IN AGRICULTURE

1-6.1 Introduction

Agriculture practices in many developing countries continue to be based to a large extent on animal and human energy. Insufficient mechanical and electrical energy is available for agriculture, and hence the potential gains in agricultural productivity through the deployment of modern energy services are not being realised. Agricultural energy demand can be divided into direct and indirect energy needs. The direct energy needs include energy required for land preparation, cultivation, irrigation, harvesting, post-harvest processing, food production, storage and the transport of agricultural inputs and outputs. Indirect energy needs are in the form of sequestered energy in fertilizers, herbicides, pesticides, and insecticides.

Mankind has adapted a variety of resources to provide for energy in agriculture. Animal draught power obtained through domestication of cattle, horses and other animals has existed for over 8,000 years, the water wheel is over 2,000 years old, and windmills were introduced over 1,000 years ago. Direct sun energy for drying and biomass fuels for heating have also been prominent as agricultural energy inputs for centuries.

The bulk of direct energy inputs in developing countries, particularly in the subsistence agriculture sector, is in the form of human and animal work. Human work has a limited output, but humans are versatile, dextrous and can make judgements as they work. This gives humans an advantage in skilled operations such as transplanting, weeding, harvesting of fruits and vegetables and working with fibres. Water lifting and soil preparation need less skill but more energy input. A sustainable rate at which a fit person can use up energy is around 250-300 W, depending on climate and needing 10-30 minutes/hour rest. The efficiency of energy conversion is only about 25%, with a maximum sustainable power output of 75 W.

1-6.2 Agriculture in Ghana

Ghana has an area of nearly 24 million hectares and a population of 21 million. The terrain varies from low-lying areas to highlands with the climate ranging from cool temperate localities, to tropical regions and to savannah conditions. The mainstay of the country's economy is agriculture – contributing about 37% to GDP and employing over 40% of the population – based on human and animal power using age-old farming tools. Inspite of efforts to mechanise Ghana'a agriculture, regrettably, greater part of energy for farming in Ghana is still manual. Human Power (using hoes and cutlasses) and to some extent animal power (cattle, horses and donkeys) are still predominantly used in some aspects of agriculture in all the farming districts of the country. There is a generally low energy output from human and animal power resulting in lower

productivity. Mechanically driven equipment and machinery are not very popular in the country's agriculture not only because of the investment cost, but also because the operation costs are very high. One of the most expensive operation costs is the cost of fuel for running these equipment and machines, due largely to the rising cost of fossil fuels.

The iron-age tool, the hand hoe, still dominates agriculture in Ghana. Animal ploughs are used to some extent in Northern Ghana where hand pushed ploughs, weeders and planters have also been adapted. Commercial farmers use inorganic fertilisers to some extent and these are mostly applied manually. Irrigation is not a common practice for subsistence farmers and in most places where some form of irrigation is done, this is largely manual. Manual collection of water, which is back breaking, is practised. Irrigation is an important pre-requisite for increased output since crop failure is often attributed to unreliable rainfall. Mechanised means of water pumping are required which would include solar, wind or biogas powered pumping systems. Almost all crops are harvested manually except in some few commercial farms.

Head loads are used to transport produce from the farm to the house or market. In some cases, bicycles are used. Commercial farmers use tractor-trailers and trucks. The use of wheel barrows and animal drawn carts is limited. Considering that an increase in productivity would result in an increase in harvesting and transportation operations, it is worthwhile investigating the renewable energy options for transporting different crops. Most crops are preserved by direct sun drying. As a result, some crops are often discoloured (especially fruit), infested with insects and micro-organisms, or contaminated with dust and small stones which have to be removed before further processing. Fish is preserved by refrigeration or smoking using wood fuels. There are limited options for fresh fruit and vegetable preservation resulting in great losses on farms and in the markets. Further processing on the farm would help increase rural incomes. Processing raw materials at source would reduce the amount of waste hauled over long distances and increase options for rural employment.

1-6.3 Using energy to modernise agriculture in Ghana

Agriculture is an important, but not dominant, user of energy in developing countries. Nevertheless, improving energy services should include increasing the energy input to agriculture so that gains in productivity, enhanced food security and rural economic development can be made. Even a small amount of additional energy, normally insignificant at the level of national energy balances, can make an important contribution to a local rural economy.

Trend in agricultural energy use in Ghana is similar to most of sub-Saharan Africa where mechanized agriculture and modern irrigation techniques are still largely under-developed. Availability of modern forms of energy is essential to increasing the productivity of the agricultural sector and opportunities exist to make use of clean and environmentally friendly energy sources. Due to its capacity for production of biomass, agriculture is also a potential source of renewable energy supply. Recognizing the dual role of agriculture as the first step in

mobilising its energy function is critical to any plans towards modernising agriculture in Ghana. Together with the challenges faced by the international community in responding to the local and global environmental impacts of energy use, there is now a major opportunity for agriculture to play an important extra role in sustainable energy development, a very important tool to modernising Ghana's agriculture.

Table 12: Conventional and renewable energy options for modernizing agriculture

Activity	Energy Source	Conventional Alternative	Renewable Alternative
Tilling, planting and weeding	Manual labour using the hand hoe	Tractors using diesel	Hand ploughs/planters, ox-ploughs and tractor using biofuels
Irrigation	Manual labour using head loads	Canals, irrigation using diesel powered water pumps	Canals, irrigation using solar, wind and biofuels
Application of fertiliser	Applied manually	Inorganic fertiliser	Compost and slurry from slurry from fertiliser manually
Harvesting	Manual labour using pangas, axes, hoes	Combine harvesters and power saws using petroleum fuels	Ox drawn harvesters, combine harvesters and power saws using biofuels
Transport	Manual labour using head loads	Tractors, lorries	Wheelbarrows, hand pushed carts, ox/donkey/horse drawn cart, bio-fuelled vehicles
Preservation	Direct sun drying, smoking	Electric or kerosene powered refrigerators, electric powered dryers (electricity generated using petroleum)	Solar and biomass kilns and dryers, solar/biomass/wind/micro - hydro powered refrigerators
Processing	Manual labour using pestle and mortar or grinding stones	Mechanised grinders using diesel generators	Paddle grater, mata mills, bio-fuelled generators

Globally modernising agriculture has meant shifting from the lower levels of energy intervention to the higher levels. For agriculture, there are three energy levels based on the "energy ladder" approach. The three-stage evolution can be considered as follows:

- basic human work for tilling, harvesting and processing, together with rain-fed irrigation, none of which involve an input from an external fuel source;
- then the use of animal work to provide various energy inputs;
- finally, the application of renewable energy technologies such as wind pumps, solar dryers and water wheels, together with modern renewable and fossil fuel based technologies for motive and stationary power applications, and for processing agricultural products.

In order to modernise Ghana's agriculture, there must be a shift from traditional energy (manual labour) to modern energy. However, this could result in increased environmental degradation, mainly as a result of pollution, de-vegetation and waste disposal. Ghana could concentrate on use of clean, environmentally sound energy sources. Some of the conventional and renewable energy options that would result in modernising Ghana's agriculture are given in Table 12.



Self Assessment 1-6

1. What is the principal source of farm energy in Ghana?

2. What renewable energy options exist for Ghana to modernise her agriculture and in which areas can these options be applied?



Answer tips

1. Read session 1-6.2
2. Read session 1-6.3

SESSION 2-6: RENEWABLE ENERGYOPTIONS FOR AGRICULTURE

2-6.1 Introduction

As shown in Table 13, renewable energy—such as solar, wind, and biofuels—can play a key role in creating a clean, reliable energy future for agriculture. The benefits are many and varied, including a cleaner environment. Photovoltaic (PV) can be used to power lighting, electric fencing, small motors, pumps, aeration fans, irrigation valve switches, automatic supplement feeders, etc. Solar dryers could replace direct drying and have advantages: designed to provide protection from insects, rodents, birds, as well as weather. Small wind electric systems can make significant contributions to farm energy needs and biofuels can be applied in engines for tilling, planting, weeding, irrigation, fertiliser application, harvesting, transport, preservation and processing.

Table 13: Renewable Energy Options for Agriculture

Technology	Application	Requirement	Benefits
Bioenergy/ Biofuels	Electricity, process heating and cooling, transport fuels, etc	Farm wastes, wood wastes, food processing wastes, waste water, agricultural crops	Reduced pollution from waste streams, control of odours
Mini-hydroelectric	Electricity (even water for irrigation)	Stream or river with reliable flow	Latest technologies do not require drop in water elevation
Solar PV	Electricity for lighting, electric fencing, pumps, aeration fans, etc.	Abundant sunshine	Can be incorporated into building structure as roofing (additional energy can be sold out)
Solar dryers	Drying of crops,etc	Abundant sunshine	Provide protection from insects, rodents, birds, as well as weather
Wind machine	Electricity	Good wind resource, space for tower	Small footprint

The type of renewable energy technology used in agriculture depends on the type of energy required, access to the renewable energy source and the design of the agricultural facilities and processes. Using renewable energy for agriculture in Ghana will create opportunities for decreased reliance on imported fossil fuels; cut pollution and greenhouse gas generation and enhance the country's economy by creating jobs. Canals and irrigation systems using solar, biogas and biofuel powered engines, combine harvesters using biofuels for their operation and solar/biomass/wind/micro-hydro powered refrigerators are just some of the few operations that can be carried out with renewable energy.

2-6.2 Solar dryers and pumps

A basic solar dryer consists of an enclosure or shed, screened drying trays or racks, and a solar collector. The design of a solar crop drying system need not be complicated – it can simply be a glossy box with a dark-coloured interior to collect solar energy, which heats the air inside the box (Figure 32). The heated air is then moved through the crop material either by natural convection or with a fan.

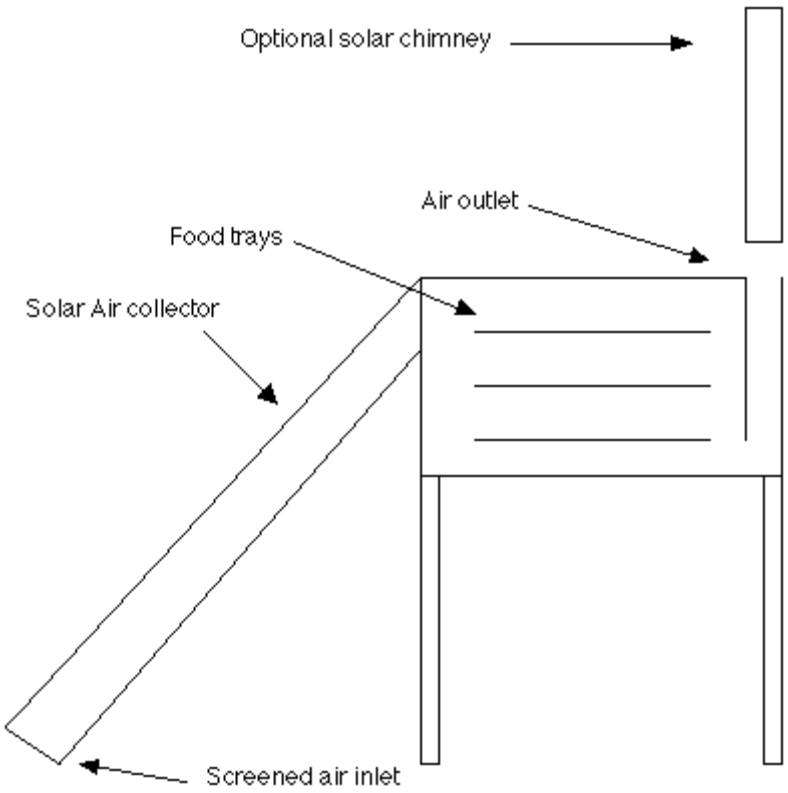


Figure 32: Features of a basic solar dryer

There are a variety of solar dryers. They can be made in different sizes and designs based on the quantity and type of food to be dried. The size and shape also depends on whether it is to be used

on domestic or commercial basis. The cost of a dryer varies with the design, size and materials used. The poor acceptance of solar dryers by farmers is generally related to the cost of the drying system that does not always match the income of the farmers. Direct mode natural convection solar dryers such as cabinet and tent dryers are relatively cheap, easy to maintain and easy to use. They can be produced from locally available materials, and by local carpenter. They are suitable for small-scale farmers whereas forced convection of any mode (direct, indirect or mixed mode) could be a good option for heavy-duty drying such as timber and cocoa drying. In general, solar drying is more appropriate when:

- the value per ton of products dried is high
- the proportion of the product currently spoiled in the open air is high
- the dryer is more often used.

Solar electric energy can even be used to move some sprinkler irrigation systems. PV systems are also extremely well-suited for pumping water for livestock in remote pastures, where electricity from power lines is unavailable. The sun's rays are received in a collector and transmitted to a heat engine, which converts the solar energy into mechanical power to run a water or irrigation pump (Figure 33). PV is often much less expensive than the alternative of extending power lines into these remote areas.

Most tropical countries are blessed with a plentiful sunshine all the year round. Ghana receives good amount of solar energy (especially in the Northern part) with an average annual solar radiation of 16-229MJ/m^2. There is thus plenty of potential for the development of solar energy for agricultural use.

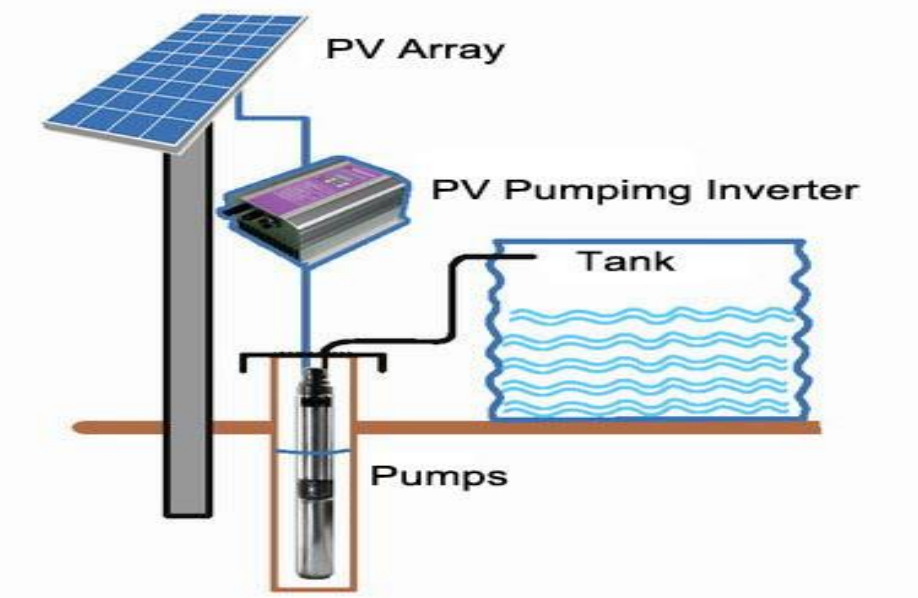


Figure 33: Typical arrangement of the components of a solar-powered pumping system

2-6.3 Windmills

Small wind electric systems can make a significant contribution to Ghana's agricultural energy needs. The energy of the wind, like that of flowing water, is more or less limited for farm use, chiefly because it cannot be controlled and is seldom available when needed. Consequently, the use of wind power on the farm is confined largely to water pumping, because whenever the wind blows, even if but once or twice a week, enough water can be pumped and conveniently stored to last several days or until the wind blows again. The power of the wind is made available by means of the common windmill (Figure 34). The power developed by this device depends primarily upon the size of wheel and the wind velocity. However, a number of other factors such as the type of wheel, design of wheel and mill and height of tower affect the performance of a windmill

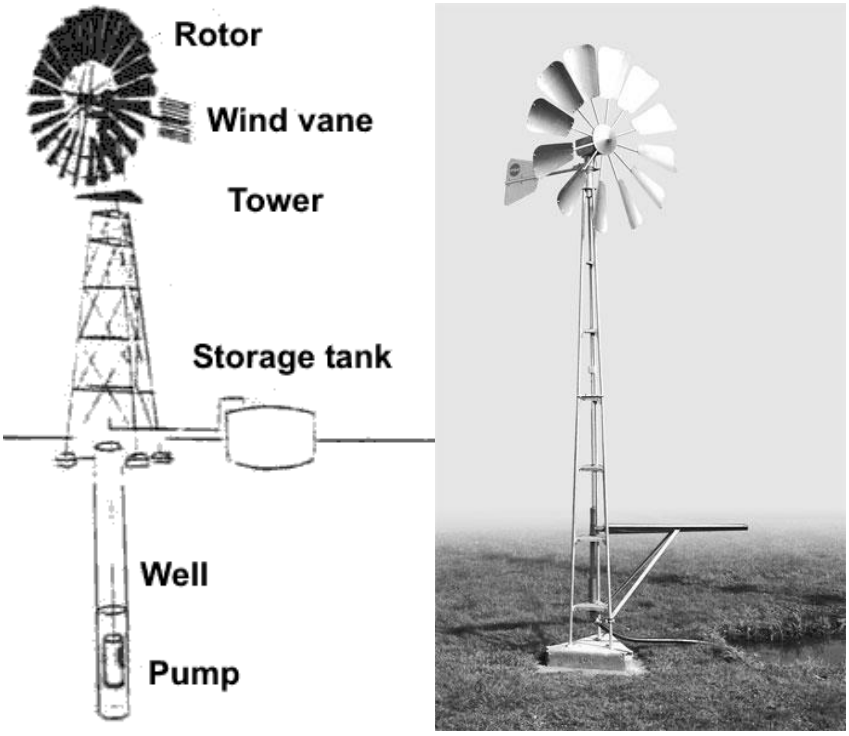


Figure 34: Schematic and photo of a windmill application for water pumping

2-6.4 Biofuels and biogas

One of the most exciting things about biodiesel for farmers is that it can provide a local or home-grown fuel supply. Diesel engines of most agricultural machines can run on pure of blended biodiesel without any form of adjustments, which is particularly good news for farmers, who often have pickup trucks, tractors and even generators that run on diesel.

Production of biogas through anaerobic digestion of animal manure and slurries as well as of a wide range of digestible organic wastes, converts these substrates into renewable energy and offers a natural fertiliser for agriculture. Biogas plants, while providing a source of gas to farmer households and digestate to fertilise agricultural crops, also provide a good alternative to managing waste from agriculture. Biogas plants which process feedstock of agricultural origin are usually referred to as agricultural biogas plants. The most common feedstock types for this kind of plants are animal manure and slurries, vegetable residues and vegetable by products, dedicated energy crops (DEC), but also various residues from food and fishing industries etc. Animal manure and slurries, from cattle and pig production, are the basic feedstock for most agricultural biogas. Anaerobic digestion of animal manure and slurries is considered to improve their fertiliser value for the reasons listed below:

- Manure and slurries from different animals (cattle, pig, poultry etc.) are mixed and codigested, providing a more balanced content of nutrients
- Anaerobic digestion breaks down complex organic material such as organic nitrogen compounds,increasing the amount of plant-available nutrients
- Co-digestion of manure with other substrates adds various amounts of nutrients to the feedstock mixture.



Self Assessment 2-6

1. Identify two ways in which solar energy can be applied to agriculture in Ghana

2. Discuss the use of windmills to pump water for irrigation.

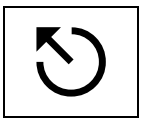


Answer tips

1. Read session 2-6.2
2. Read session 2-6.3



Learning Track Activities



Unit Summary

1. In this unit, we have identified a range of agricultural energy needs that can be supplied with the various renewable energy forms.

2. We have also discussed how renewable energy options can be appropriately applied to meet the energy needs in Ghana's agriculture.

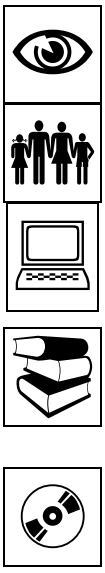


Key terms/ New Words in Unit

1. Tilling

2. windmill

3. solar dryers



- **Discussion Question:** Identify a range of farm activities and discuss how renewable energy can be used to meet the energy needs f these activities.



Unit Assignments 6

How can solar energy meet the needs of farm activities? Discuss the issue in depth, giving an overview that begins with the delivery of the panels on the farm.

Course Summary

In this course, we have defined and explained renewable energy and their potentials for replacing energy produced from conventional sources. We have found out that the share of renewable energy in the global energy mix has increased significantly over the last decade or so. The production process/conversion of biofuels, biogas, solar energy and wind energy has been discussed, including basic economics for some of the technologies. The use of renewable energy in everyday life and especially in agriculture has also been discussed.

Appendices

Appendix 1: Biodiesel feedstock classification

Edible oils

A large group of edible oils are used for biodiesel production. Throughout the world, the typical lipid feedstocks for biodiesel production are refined vegetable oils. Within this group, the oil of choice varies with location according to availability; the most abundant lipid is generally the most common feedstock. The reasons for this are not only the desire to have an ample supply of product fuel, but also because of the inverse relation between supply and cost. Refined oils can be relatively expensive under the best of conditions, compared with petroleum products, and the choice of oil for biodiesel production depends on local availability and corresponding affordability. Thus, rapeseed and sunflower oils are used in the European Union, palm oil predominates in biodiesel production in tropical countries such as Malaysia and Indonesia, and soybean oil and animal fats are the major feedstocks in the United States. Other edible oil feedstocks which have been successfully tried for biodiesel production (some of which are non-traditional) include palm kernel oil, peanut oil, oils from Ethiopian mustard (Brassica carinata) – which is closely related to rapeseed, cardoon (Cynara cardunculus) - closely related to sunflower and safflower, gold-of-pleasure (Camelina sativa), linseed (Linum usitatissimum) and tiger nut (Cyperus esculentus). Among the edible oils, palm oil and coconut oil which are produced from crops that give outstandingly high hectare yields (see Table 2) and do very well in the tropics good choices in some African countries for biodiesel production. The major problem with all these plant oils is their use as food sources. In a world where hunger is becoming endemic in some areas, especially in the least developed and developing countries, the competition between the use of food-based raw materials for energy production and as sources of food has become a major criticism for biofuel programmes. This is particularly true of many African countries where due to drought, war and other factors, they are unable to produce enough food to feed their people and hence are always at the mercy of charitable support.

Non-edible oils

There are two main factors underpinning the search for alternative and non-edible oils for biodiesel production either in place of or to supplement the production from edible oils. These are (i) the use of the edible oils for food and in industrial applications such as production of soap and cosmetics and (ii) the high prices of such oils as a result of the competition between their use for food and other purposes and their use for biodiesel production. The price of biodiesel is very much affected by the price of the lipid feedstock. It is estimated that about 88% of the total cost of biodiesel production comes from the refined lipid feedstock used. Finding alternative feedstocks that are cheaper can reduce the price of biodiesel and make their use more

economically attractive, (even without subsidies) than petro-diesel. Non-edible oils that tend to be cheaper can fill this gap. Many countries have a large number of non-traditional lipid sources which have the potential of being converted to biodiesel. A number of non-edible oils have been identified for their possible use for biodiesel production. These include oils from the physic nut (Jatropha curcas Linn), neem tree (Azadirachta indica), jojoba (Simmondsia chinensis Link Schneider), castor oil (Ricinus communis), tall oil, rubber (Hebea brasiliensis) seed oil, sal oil, rice bran, mahua oil, palas (Butea monosperma) oil, kusum (Schlelchera trijuga) oil and karanja (Pongamia glabra) oil. Jatropha curcas Linn is one the high yielding oil plants with a future for biodiesel production in Africa. It has been identified as a sustainable source of biodiesel and many countries are already researching into its exploitation as an abundant biodiesel source. Another major advantage with this plant is that a large portion of the tropics has been identified as a Jatropha zone. The zone covers a much bigger area than that for oil palm. Jatropha is a tough, drought-resistant and frost hardy shrub that bears oil-rich seeds under optimum growing conditions. It starts bearing fruits from the first year, and develops production after three years. Jatropha grows well in Zimbabwe and Zambia and is widely used in India and West African countries (e.g. Mali, Ghana, Nigeria) for biodiesel production. It can successfully be grown on sub-optimal land as long as it receives above 300 mm of summer rainfall, but it produces a much higher yield under sub-tropical conditions.

Animal fats

Four main types of animal fats have been employed for biodiesel production. These are pork lard, beef tallow, fish fat and poultry fat. Animal fats are typically less expensive than refined vegetable oils for two main reasons: (i) they are a by-product rather than a primary product of animal agriculture and (ii) the demand for these fats is lower than for the more common vegetable oils. A major disadvantage of animal fats in as far as biodiesel production is concerned is that they contain a higher content of saturated fatty acids (SFA) than do vegetable oils. This property leads to relatively high melting points, which can result in precipitation and hence poor engine performance in cold weather conditions. On the positive side, animal fat derived biodiesel fuels, because of their higher saturated fatty ester content, generally have higher cetane values than vegetable oil-derived biodiesel.

Waste frying oils

The increasing production of waste frying oils (WFOs) from household and industrial sources is a growing problem all around the world. This residue is regularly poured down the drain, resulting in problems for wastewater treatment plants and energy loss, or is integrated into the food chain through animal feeding, thus becoming a potential cause of human health problems. There are many possible end-uses for this waste, such as the production of soaps or of energy by anaerobic digestion and thermal cracking. Recently there is an interest in using such oil for the production of biodiesel.

There are two main attractions for the utilization of waste or used oils for biodiesel production:

2. These oils are relatively cheap and hence their use can reduce the cost of biodiesel production.

3. Their disposal poses environmental problems and, therefore, their recycling into biodiesel is an environmentally benign process which also makes economic sense.

A large variety of waste oils are available for biodiesel production. The sources of these oils can be categorized into the following:

- Waste oil from households and restaurants

- Waste oil from food industry

- Waste oil from non-food industry

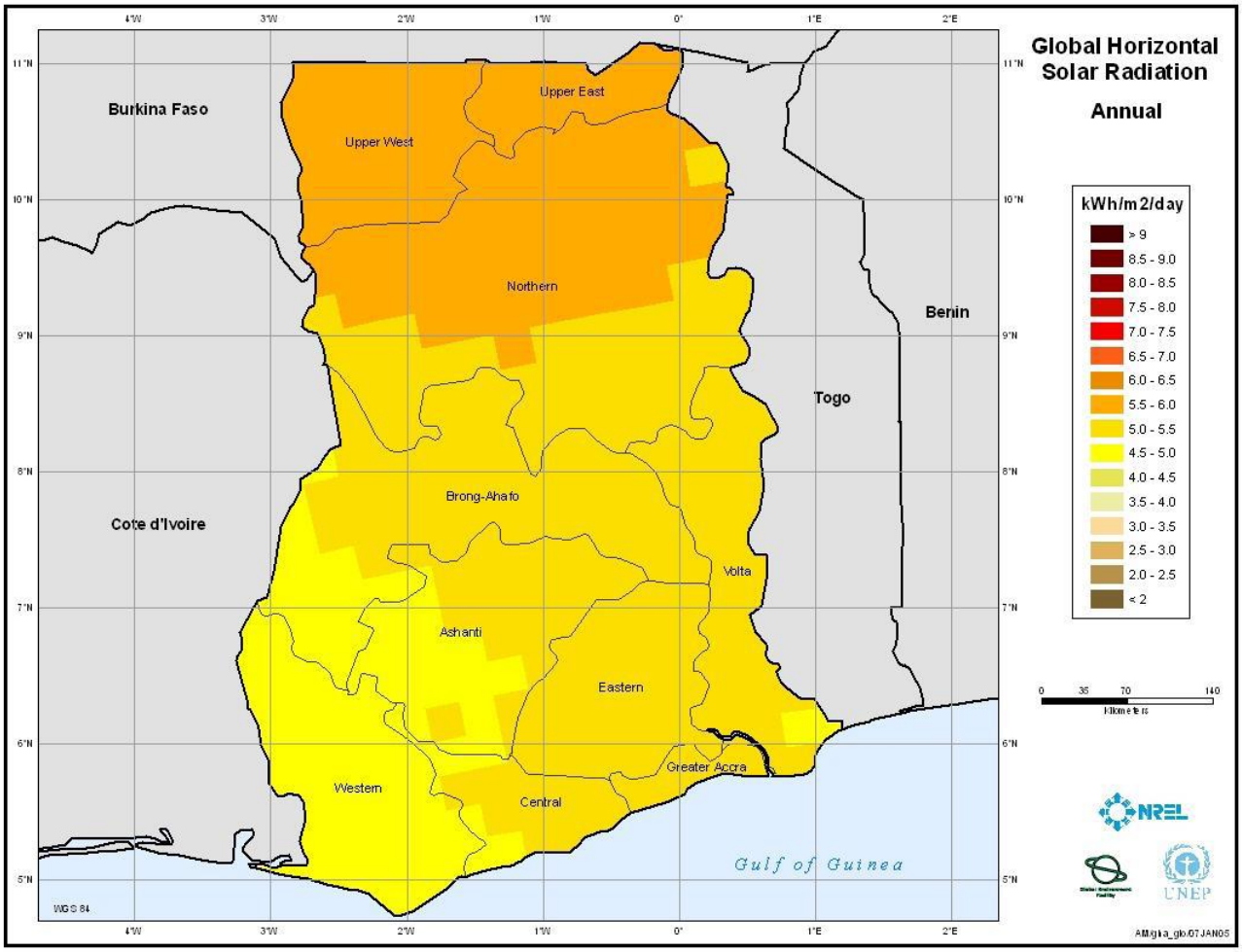
Rape, soybean, palm and coconut oils are the waste oils most frequently used in biodiesel production. The use of these waste vegetable oils (WVOs) requires additional processing to filter out residues and to handle the free fatty acids produced by high temperatures.

Microalgae

Microalgae have been suggested as very good candidates for fuel production because of their advantages of higher photosynthetic efficiency, higher biomass production and faster growth compared to other energy crops. Algae can grow practically in every place where there is enough sunshine. Some algae can grow in saline water. The most significant aspect of algae oil production is the yield and hence the yield of the biodiesel. It has been estimated that the yield (per acre) of oil from algae is over 200 times the yield from the best performing plant or vegetable oil source. Microalgae are the fastest growing photosynthetic organisms. They can complete an entire growing cycle every few days Approximately 46 tons of oil/hectare/year can be produced from diatom algae. Different algae species produce different amounts of oil. Some algae produce up to 50% of oil by weight.

Recently the potential of cultivating algae near power plants has been recognized as the algae can be fed by CO2 emissions. This is possible because carbon dioxide and nitrogen oxides are the primary nutrients for the growth of microalgae. The fact that oil-rich microalgae can feed on the emissions of coal, petroleum, and natural gas power plants in its growth process makes the production of biodiesel from microalgae a promising technology. The production figures given earlier also suggest that there is a huge potential for the production of biodiesel using microalgae. There is, however, the need for extensive research on the exploitation of microalgae for biodiesel production.

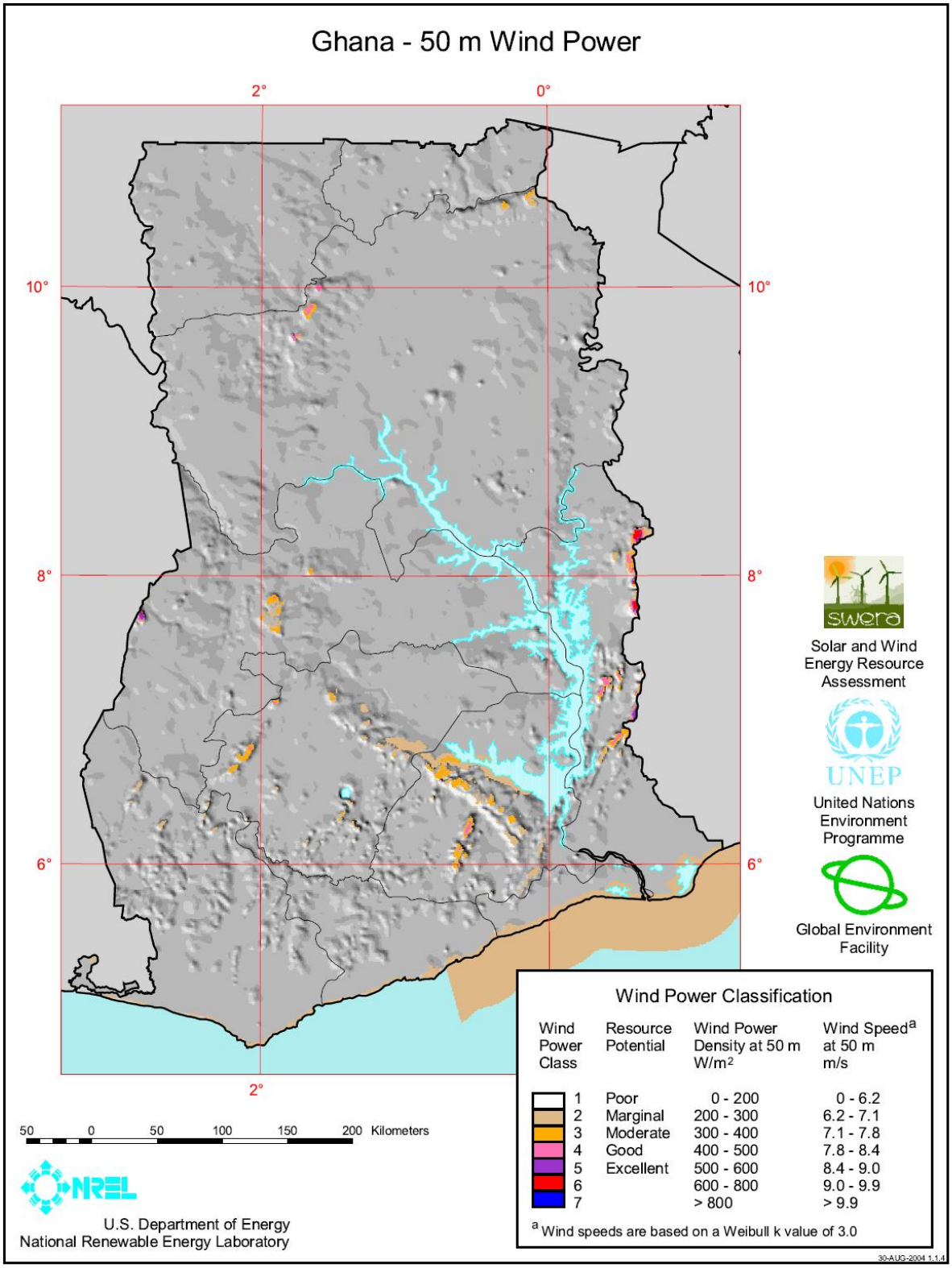
Appendix 2: Global Solar Radiation Map of Ghana



Appendix 3: Monthly Averages of Solar Irradiation (kWh/m2-day) at 9 Synoptic Stations

MONTH	KUMASI	ACCRA	AXIM	NAV'GO	HO	ADA	K'DUA	WENCHI	TAMALE
JAN	4.818	4.660	4.882	5.391	4.872	4.995	4.711	5.193	5.124
FEB	5.313	5.206	5.399	5.400	5.224	5.381	5.139	5.495	5.479
MAR	5.305	5.256	5.569	5.783	5.509	5.649	5.260	5.483	5.613
APR	5.356	5.665	5.605	5.958	5.716	5.937	5.434	5.711	5.890
MAY	4.709	5.416	5.051	5.934	5.576	5.570	5.287	5.507	5.869
JUN	4.029	4.613	3.936	5.719	4.916	4.978	4.641	4.972	5.510
JUL	4.036	4.189	4.242	5.339	4.601	5.064	4.074	4.356	4.954
AUG	3.783	4.527	4.230	5.098	4.187	5.065	3.842	4.120	4.841
SEP	3.992	5.107	4.382	5.324	4.663	5.510	4.437	4.405	5.004
OCT	4.707	5.623	5.178	5.677	5.500	5.872	5.174	4.927	5.472
NOV	5.000	5.510	5.466	5.616	5.624	5.480	5.241	5.127	5.695
DEC	4.552	4.930	4.986	4.824	5.074	5.359	4.857	4.905	5.213
Av'ge	4.633	5.059	4.911	5.505	5.122	5.409	4.841	5.017	5.389

Appendix 4: Wind Resource Map for Ghana - 50 m Wind Power



Selected Answers to Unit Assignments

[Supply selected answers to Unit Assignments]

Course Quiz/Exams

[Supply course quiz of this course here for the attention of the Institute's examinations officer]

RESEARCH/PROJECT AREAS AND RELATED TOPICS IN THIS COURSE

1. Design of solar dryers
2. Production and testing of biofuels from different feedstock
3. Design of windmills for rural water pumping and irrigation
4. Mapping of jatropha farms in Ghana

SOME CASE STUDIES TO THIS COURSE

[Supply some case studies in this course for your students]

MY PAGE

Name: _____ Learning Centre: _____

Contact: Tel. _____ Email: _____ Emergency Name/Phone: _____

Important numbers: Student number _____ Examination number _____

Program: _____ Year: _____ Course code/title: _____

Course objectives: _____

_____

Course dates/Semester No ( ): Starts_____ Ends _____

FFFS schedule/Dates: _____

Quiz dates: _____

Assignments hand in dates: _____

Revision dates: _____

Group discussion/work members (names and contacts): _____

_____

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End of course Self Evaluation:
I have completed all Units & interactive sessions□, mastered all learning objectives, completed all self Assessments, unit summary, key words and terms, discussion questions, review questions, reading activity, web activity, unit assignments, and submitted all CA scoring assignments, learner feedback on this course and submitted my comments and course focus contributory questions to facilitator for discussion.

Self-grading: self assessment questions score _____ % Unit Assignments scored _____ %

My course conclusion remarks: _____

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